

Research Master Thesis

Reasoning is to Humans as Association is to LLMs: Inside the Analogical Mechanisms of Large Language Models

Lara Olders

Supervisor dr. Pia Sommerauer & prof. dr. Antske Fokkens
2nd reader dr. Lucia Donatelli

a thesis submitted in fulfillment of the requirements for the degree of

RMA Humanities
(Human Language Technology)

Vrije Universiteit Amsterdam

Computational Linguistics and Text Mining Lab
Department of Language and Communication
Faculty of Humanities

Date June 26, 2026
Student number 2691204
Word count 16319

Abstract

Analogical reasoning, the ability to identify abstract relational structures and apply them to a new domain, is widely regarded as a hallmark of human cognition. Whether Large Language Models (LLMs) engage in genuine analogical reasoning or simply exploit surface-level statistical patterns remains an open debate. This thesis investigates this question by combining a rank-based behavioral evaluation, a manual error analysis of the top-10 predictions, and head-level activation patching, applied to GPT-2 Small (124M parameters) and Llama 3.2 1B (1.2B parameters) on the 60-item analogy dataset of Jones et al. (2022). Behaviorally, both models show limited and fragile accuracy, although Llama outperforms GPT-2 across almost every condition. A weak relational preference persists in the restricted forced-choice condition, even when top-1 accuracy collapses. Within-relation few-shot demonstrations raise Llama's top-1 accuracy to 23.33%, whereas cross-relation demonstrations and single-pass finetuning depress performance, the latter collapsing GPT-2's top-1 accuracy to zero. Accuracy is further shaped by relational distance and type. An offset parallelism criterion intended to test for preserved relational structure could not be applied, as the analogies proved non-parallel in both models' embedding spaces (mean offset cosine near zero). Manual inspection instead showed that failed predictions default to associative retrieval from the prompt's semantic neighborhood rather than the relational target. Activation patching localized the prediction to a small number of stable mid-layer heads (L8H6 for GPT-2, L7H5 for Llama), with recovery maps that correlate near-perfectly under resampling of the corruption tokens and are largely preserved across prompting conditions. Together, these findings indicate that both models reproduce the surface behavior of analogy solving without reliably identifying the underlying relation, a pattern better characterized as surface-level associative retrieval than as deliberate analogical reasoning.

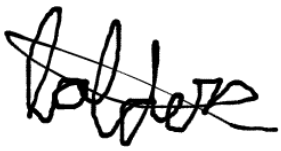
Declaration of Authorship

I, Lara Olders, declare that this thesis, titled *Reasoning is to Humans as Association is to LLMs: Inside the Analogical Mechanisms of Large Language Models* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Research Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: June 26, 2026

Signed:



Acknowledgments

I would like to thank several people who helped me complete and improve this thesis. First and foremost, I am grateful to dr. Pia Sommerauer and prof. dr. Antske Fokkens for their support, their critical questions and their creative ideas. You helped me shape this thesis in the way I had envisioned, and you always had suggestions for how to take it further and make it better. I learned a lot from working with both of you. I also would like to thank dr. Lucia Donatelli for taking the time to read and assess my work. I am thankful to my colleague Nane Andreasyen for critically reviewing my thesis. Your feedback genuinely improved the readability of my writing and the clarity of my explanations. Finally, I want to thank my family and friends for supporting me during the writing of this thesis.

List of Figures

2.1 Results of GPT-3 vs. Humans on different verbal analogy datasets as displayed in Webb et al. (2023) 6
2.2 The Transformer Architecture 14
2.3 Self-attention weights for the pronoun "it" in the sentence "The animal didn't cross the street because it was too tired," adapted from (Google, 2026). 16
2.4 The Attention Mechanism (Vaswani et al., 2017) 17
2.5 The Residual Stream (Kirchner, 2022) 17
2.6 The Activation Patching Procedure 19
2.7 Example of a per-head heatmap for the indirect-object identification task 20

3.1 The Activation Patching Procedure 31

4.1 Rate of selecting the first-position candidate in the restricted forced-choice condition, by target position 41
4.2 Boxplot for near and far analogies based on cosine distance 45
4.3 GPT-2 Small activation patching under the open-vocabulary zero-shot condition, filter `topk` ($n = 12$). 52
4.4 Llama 3.2 1B activation patching under the open-vocabulary zero-shot condition, filter `topk` ($n = 21$). 53
4.5 Llama 3.2 1B activation patching under the 1-shot within-relation condition, filter `topk` ($n = 29$). 54
4.6 Per-layer summed recovery for GPT-2 Small under three independent corruption-token seeds (42, 123, 456) on the open-vocabulary zero-shot condition. 55
4.7 Per-layer summed recovery for Llama 3.2 1B under three independent corruption-token seeds (42, 123, 456) on the open-vocabulary zero-shot condition 56

1 Mean head-level recovery by relation type under the open-vocabulary zero-shot condition 67
2 Mean head-level recovery by semantic distance category under the open-vocabulary zero-shot condition 68

List of Tables

2.1 Examples of analogy and literal similarity (Jiayang et al., 2023) 8
2.2 Example of a story analogy (Sourati et al., 2024) 9
2.3 Examples of synonym and antonym verbal analogy 22

3.1 Examples of analogies 26
3.2 Overview of the four experimental conditions 34
3.3 Overview of the five finetuning train/test splits 37

4.1 Accuracy on the sinlge-prompt conditions (open-vocabulary zero-shot baseline and restricted forced choice for GPT-2 Small and Llama 3.2 1B) 40
4.2 Accuracy (%) on the few-shot conditions for GPT-2 Small and Llama 3.2 1B 43
4.3 GPT-2 Small accuracy (%) on the finetuning conditions 43
4.4 Open-vocabulary zero-shot accuracy broken down by relation type and by semantic distance 46
4.5 Top-10 next-token predictions for a representative subset of analogies under the open-vocabulary zero-shot condition 47
4.6 Number of analogies submitted to activation patching under the `topk` filter, and how many survive the logit-gap threshold 51
4.7 The five attention heads with the highest mean logit recovery under the open-vocabulary zero-shot baseline 51
4.8 The five attention heads with the highest mean logit recovery for Llama 3.2 1B under the best-performing setup 54
4.9 Pairwise Pearson correlations (r) between the flattened recovery maps obtained under three independent corruption-token seeds 55
4.10 Mean logit recovery of each model's baseline peak head, broken down by relation type and semantic distance, under the open-vocabulary zero-shot condition 57

1 Restricted forced-choice *correct-choice* accuracy 69
2 All 60 analogies from the dataset, grouped by relation type and sorted by semantic distance category. 72
3 Cosine distance scores for all 60 analogies by model 74
4 Corruption words and their cosine distance to the C-term, per model . . 81
5 Top-10 next-token predictions per analogy under the open-vocabulary zero-shot condition 86

Contents

Abstract i

Declaration of Authorship iii

Acknowledgments v

List of Figures vii

List of Tables ix

1 Introduction 1

2 Literature Review 3
2.1 Behavioral Evaluations of Analogical Reasoning in LLMs 3
2.1.1 Digit Matrices 4
2.1.2 Letter String Analogies 4
2.1.3 Verbal Analogies 4
2.1.4 Story Analogies 7
2.1.5 Mixed Evidence from Related Reasoning Tasks 8
2.2 The Geometry of Analogical Representations 10
2.2.1 The ANALOGICAL benchmark 10
2.2.2 Offset Parallelism in Word Embeddings 10
2.2.3 Criticisms of Offset Parallelism 11
2.3 Fine-tuning Models for Analogical Reasoning 11
2.3.1 Distilling Relation Embeddings 12
2.3.2 Learning Offset Parallelism 12
2.4 Mechanistic Interpretability 13
2.4.1 The Transformer Architecture 13
2.4.2 Activation Patching and Path Patching 17
2.4.3 Features and Circuits 20
2.5 Research Gap 22

3 Methodology 25
3.1 Analogies Dataset 25
3.1.1 Dataset Adaptation 26
3.2 Models 27
3.2.1 GPT-2 Small 27
3.2.2 Llama 3.2 1B 27
3.3 Behavioral analysis: Analogical Reasoning Task 28

3.3.1 Analysis . . . 28
3.4 Representational/Error Analysis . . . 29
3.5 Mechanistic analysis: Activation Patching . . . 30
3.5.1 Corruption Procedure . . . 30
3.5.2 Analysis of the Results . . . 33
3.6 Experimental Conditions . . . 34
3.6.1 Open-vocabulary Zero-shot Prompting . . . 34
3.6.2 Restricted Forced-choice Prompting . . . 35
3.6.3 Few-shot Prompting . . . 35
3.6.4 Finetuning (GPT-2 Small) . . . 36
3.6.5 Corruption-Token Sensitivity . . . 37

4 Results **39**
4.1 Behavioral Results . . . 39
4.1.1 Open-vocabulary zero-shot results . . . 39
4.1.2 Restricted forced-choice results . . . 40
4.1.3 Few-shot results . . . 41
4.1.4 Finetuning results . . . 43
4.1.5 Accuracy across relation types and semantic distances . . . 44
4.2 Error Analysis Results . . . 46
4.3 Mechanistic Results . . . 49
4.3.1 Interpreting Recovery Scores . . . 50
4.3.2 Baseline (Open-vocabulary Zero-shot) . . . 50
4.3.3 Best-performing Setup . . . 53
4.3.4 Corruption-Token Sensitivity . . . 54
4.3.5 Across Experimental Conditions and Analogy Subsets . . . 55

5 Discussion **59**
5.1 Behavioral Discussion . . . 59
5.2 Error Analysis Discussion . . . 60
5.3 Mechanistic Discussion . . . 61
5.4 Limitations and Future Work . . . 62

6 Conclusion **65**

Appendix A: Per-relation and Per-distance Activation Patching **67**

Appendix B: Restricted Forced-choice Breakdown **69**

Appendix C: Full Analogy Dataset **71**

Appendix D: Open-vocabulary Top-10 Predictions **83**

Chapter 1

Introduction

The rapid development of Large Language Models (LLMs) has sparked renewed debates about what should count as reasoning in artificial systems. Despite their impressive fluency and task performance, LLMs are often said to lack grounded understanding, rely on pattern recognition rather than genuine reasoning, and produce confident but incorrect or inconsistent output (Xu et al., 2024; Kim et al., 2025). These limitations prompt a fundamental question: do LLMs engage in genuine reasoning, or do they simply produce outputs based on learned statistical patterns? A widely cited critical perspective on this issue is offered by Bender and Koller (2020). They introduce the Octopus Thought Experiment, which imagines an octopus with no direct access to the human world. This octopus can only observe communication between two humans, transmitted through a cable. Over time, by identifying recurring patterns in these exchanges, the octopus becomes increasingly adept at predicting how each participant is likely to respond, relying solely on statistical regularities in the data. Eventually, it breaks the cable and communicates directly with one of the humans without being detected. This thought experiment raises the question of whether the octopus can be said to understand what it is to communicate and whether its behavior can be described as genuine reasoning. Bender and Koller (2020) use this scenario to critique LLMs, arguing that, like the octopus, these models may reproduce the formal structure of meaningful language without any grasp of its underlying meaning or real-world grounding.

One particularly interesting type of reasoning is analogical reasoning. Analogical reasoning is said to be a foundational mechanism of human cognition, defined as the ability to recognize and reason about common relational structures in different contexts (Gentner and Maravilla, 2017). Evaluating this capability in LLMs matters beyond curiosity: performance on analogical tasks is widely taken as a signal of broader LLM reasoning abilities (Webb et al., 2023; Yuan et al., 2023; Lewis and Mitchell, 2024; Mondorf and Plank, 2024). As the use of LLMs exponentially grows, evaluating whether these models possess analogical reasoning capabilities has therefore become an important debate.

Solving an analogy involves identifying a relational structure, often expressed in the form "A is to B as C is to D", and applying this structure to a new pair of concepts. Examples include "tarantula is to spider as bee is to insect" and "defeat is to frown as victory is to smile" (Jones et al., 2022). In each case, the analogy depends on a shared functional relationship rather than on surface-level similarity. Linguistic and cognitive research on analogical reasoning, most notably the Structure-Mapping Theory (Gen-

tner, 1983; Gentner and Maravilla, 2017), offers a useful framework for distinguishing genuine relational reasoning from just surface-level pattern matching. According to this theory, successful analogy solving requires mapping abstract relationships between domains, rather than simply recognizing frequently co-occurring items. This framework imposes that analogical reasoning requires establishing a structural alignment between a familiar base domain and an unfamiliar target domain.

This distinction is particularly relevant for the evaluation of LLM reasoning capabilities. In many analogies, the correct answer is not the word most strongly associated with the prompt in everyday language use. For example, in the case of "bee", a word such as "honey" may co-occur more frequently with "bee" than "insect", although "insect" is required to preserve the underlying relational structure. In human cognitive development, the relational shift in which children transition from focusing on simple object matches to recognizing deeper relational matches is considered a critical milestone (Gentner and Maravilla, 2017). Evaluating LLMs on analogical reasoning to determine whether they are capable of detecting deeper relational patterns, or if they rely solely on statistical associations, is therefore the key focus of this research.

Research addressing this question has so far focused almost exclusively on behavioral evaluation, and interpretability techniques have rarely been applied to analogical reasoning specifically. This thesis therefore addresses this issue and combines a behavioral evaluation, an error analysis, and head-level activation patching, applied to GPT-2 Small (124M parameters) and Llama 3.2 1B (1.2B parameters) on the 60-item verbal analogy dataset of Jones et al. (2022). The research goal is divided in three sub-questions:

SQ1 (Behavioral). How accurately do GPT-2 Small and Llama 3.2 1B solve four-term verbal analogies, and how is this accuracy modulated by prompting format, finetuning, relation type, and semantic distance?

SQ2 (Representational/Error-analysis). When the model's top-ranked prediction is not the gold target, do its predictions nevertheless reflect the relational structure of the source pair, rather than failing at random?

SQ3 (Mechanistic). Which attention heads causally drive the analogical prediction in each model, and how stable is this head-level structure across prompting conditions, finetuning, and resampling of the corruption tokens used for activation patching?

To answer these questions, the remainder of this thesis is organized as follows. Chapter 2 reviews the literature on behavioral evaluations of analogical reasoning in LLMs and on mechanistic interpretability, and concludes by restating the research gap and sub-questions in full. Chapter 3 details the dataset, the models, the experimental conditions, and the activation patching procedure. Chapter 4 reports the results for each sub-question in turn. Chapter 5 relates these findings to the broader literature and discusses the limitations of the present study. Chapter 6 summarizes the answers to the three sub-questions and their implications.

Chapter 2

Literature Review

This chapter establishes the foundation for studying analogical reasoning in LLMs and the internal mechanisms that produce it. It will start by reviewing the current literature on behavioral evaluations of analogical reasoning in LLMs (Section 2.1), covering the different analogy domains studied, their critiques, and related reasoning tasks. It will then turn to the geometry of analogical representations (Section 2.2) and to fine-tuning models for analogical reasoning (Section 2.3), two lines of work that probe a model's relational structure more directly than behavioral accuracy alone can. Followed by an in-depth review of literature concerning mechanistic interpretability (Section 2.4), starting with an explanation of the Transformer architecture and the residual stream, before turning to activation patching and path patching and how these techniques have been used to identify the circuits underlying reasoning in language models. Finally, the research gap will be stated and the research goal and the corresponding sub-research questions that drive this research goal will be outlined (Section 2.5).

2.1 Behavioral Evaluations of Analogical Reasoning in LLMs

A central and widely cited line of research suggests that sophisticated LLMs exhibit emergent analogical reasoning in zero-shot contexts (Webb et al., 2023). The authors argue that these models perform well on multiple complex analogical reasoning tasks. To comprehensively evaluate this emergent ability, they tested GPT-3's performance and compared it with that of humans on four distinct analogical domains: digit matrices (Section 2.1.1), letter string analogies (Section 2.1.2), four-term verbal analogies (Section 2.1.3), and story analogies (Section 2.1.4). The following sections will detail the performance on these tasks as presented by Webb et al. (2023) and critically evaluated by other researchers, before turning to mixed evidence from related reasoning tasks beyond analogies (Section 2.1.5). Across these domains, a recurring pattern emerges: strong zero-shot performance is frequently undercut once a robustness check probes whether the model relied on genuine relational reasoning or on surface-level pattern matching from its training data. The following sections trace this pattern domain by domain, with closer attention paid to the four-term verbal analogies that form the basis of this thesis's own dataset and evaluation.

2.1.1 Digit Matrices

Digit matrices are a textual adaptation of Raven's Standard Progressive Matrices, in which numerical or logical transformations must be extracted from a 3x3 grid and applied to complete a missing entry. GPT-3 matched or exceeded human performance on this task and showed human-like sensitivity to relational complexity, performing worse as the number of underlying rules increased (Webb et al., 2023). Lewis and Mitchell (2024), however, found that this performance is not robust: when they moved the position of the missing entry away from its original, fixed location in the grid, human accuracy was unaffected but GPT-3's accuracy declined sharply. This suggests that the model's performance depended on pattern recognition tied to the original task format rather than on genuine abstract reasoning, an early instance of the fragility that recurs across the domains discussed below.

2.1.2 Letter String Analogies

In the letter string analogy task of Webb et al. (2023), models and humans complete letter strings such as "a b c d → a b c e" and "i j k l → ?" (correct answer: "i j k m"), and GPT-3 outperformed human undergraduates on the standard alphabet. Lewis and Mitchell (2024) again tested robustness, this time by reversing or fully randomizing the alphabet: if the model had learned the abstract transformation rule, it should apply equally well to any alphabet, as human participants did. GPT-3's accuracy instead dropped sharply under the modified alphabet, again indicating reliance on memorized sequences from pretraining rather than on the underlying transformation rule itself.

2.1.3 Verbal Analogies

Specifically interesting for the current research, Webb et al. (2023) tested GPT-3 on four-term verbal analogies of the kind A:B::C:D, using four different problem sets that each contribute a different point of comparison. Across all four datasets, the performance of GPT-3 was evaluated using a multiple choice format. The model was scored by comparing the log probabilities of all possible options and selecting the answer that received the highest log probability. This internal probability calculation differs from the human task, where participants simply had to select one of the provided answers. Lewis and Mitchell (2024) critically evaluate the robustness of this claim, showing that the accuracy of GPT models drops sharply on counterfactual variants of Webb's analogies, problems that test the same relational structure but are less likely to resemble anything seen during pretraining, while human performance remains stable. They additionally find that GPT models are sensitive to the order in which answer options are presented, with accuracy shifting by as much as 28 percentage points depending on whether the correct option appears first or second.

UCLA Verbal Analogy Test

The first of the four tested problem sets was the UCLA Verbal Analogy Test (UCLA-VAT), which was developed by Webb et al. (2023) and consists of 80 problems. It spans four relation types: categorical (the B and D terms are members of the categories denoted by A and C), functional (A and C name the function of B and D), antonym, and synonym, with 20 problems for each type. Each problem presents two candidate answers for the final term (D and D'). Performance was compared against adult human

participants (N = 57, UCLA undergraduates), and GPT-3 matched or exceeded their accuracy on this set, as can be seen in Figure 2.1.

Sternberg and Nigro (1980)

The second dataset was originally constructed by Sternberg and Nigro (1980) and was designed to study developmental patterns in analogical reasoning across age groups. It contains 200 problems, with 40 problems for each of five relation types. These are the same four relation types found in the UCLA-VAT (categorical, functional, antonym, and synonym), together with an additional linear order relation (for example, *month : year :: inch : foot*). This relation captures ordinal or sequential structure instead of semantic association. Like the UCLA-VAT, each problem offers two answer choices. Here GPT-3 was compared against the human behavioral data reported by Sternberg and Nigro (1980) (N = 20), and again performed as well as or better than the human participants, as can be seen in Figure 2.1.

SAT

The third dataset was drawn from Turney et al. (2003) and consists of 374 multiple-choice analogy problems taken from the SAT college-admissions examination. These problems cover a broad range of relation types rather than a small fixed set. This dataset differs from the others in one important way: each problem provides five answer choices for the C and D terms instead of two, so chance performance is 20% instead of 50%. GPT-3 was evaluated by presenting each of the five candidate analogies and selecting the one for which the C and D terms together received the highest log probability. No trial-level human data accompanies this set, so the comparison was made against an estimate of the average performance of high school students taking the SAT, which GPT-3 surpassed.

Jones et al. (2022)

The final dataset used by Webb et al. (2023), was originally created by Jones et al. (2022). This dataset specifically tests analogical reasoning across three different underlying relationships: categorical, compositional, and causal relations. The dataset was designed to manipulate two primary variables: semantic distance and distractor salience. Semantic distance measures the similarity of the domains between the source (A:B) and target (C:D) word pairs, and was computed using Latent Semantic Analysis (LSA). Half of the instances in the dataset were semantically near, meaning the LSA distance score between source and target pair was 0.70 or below. The other half were semantically far with an LSA score of 0.90 or above, and required relational abstraction across highly dissimilar domains. For each problem two answer choices were provided: a target answer and a distractor. The dataset manipulated distractor salience to test inhibitory control during selection of the correct response. Distractor salience was defined by the semantic distance between C and the incorrect answer options D'. For each problem, participants faced either a low-salience distractor (semantically further from the C-term than the correct answer) or a high-salience distractor (semantically closer to the C-term than the actual correct answer). This salience was validated using human association norms (SWOW EN) to ensure the high salience distractors were genuinely distracting (Jones et al., 2022). Webb et al. (2023) found that humans

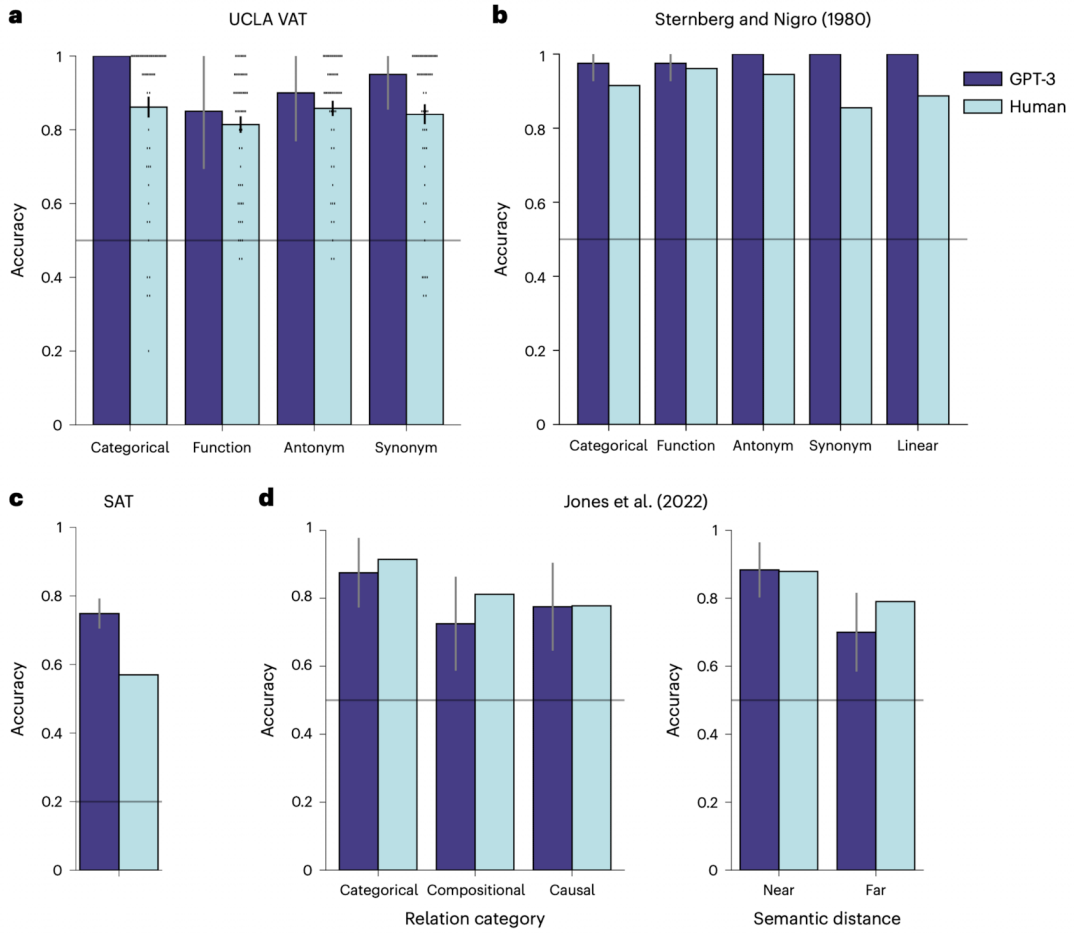


Figure 2.1. Results of GPT-3 vs. Humans on different verbal analogy datasets as displayed in Webb et al. (2023)

outperformed GPT-3. Like human subjects, the LLM displayed more difficulty when distractor salience was high and performed worse on problems where the analogies were semantically distant. While the model's performance was lower than the estimated average performance of high school students, it should be noted that the results of the students were observed on a different dataset of analogy problems.

Comparing LLMs and Children on Verbal Analogies

A more direct test of whether performance in analogical reasoning reflects relational mapping or surface-level association is offered by Johnson et al. (2025), who compared eight multilingual LLMs against children. They evaluated the models on 872 Dutch verbal analogies of the form A:B::C:?, the same items solved by 14,006 Dutch children aged 7 to 12 in an online learning environment. Each analogy was presented as a multiple-choice problem with five answer options, all of which were semantically associated with the C-term, so that the correct answer could only be identified by mapping the underlying relation rather than following surface-level association with C. An example of an analogy is "horse belongs to stable like chicken belongs to...?" The answer *chicken coop* would show understanding of the underlying relation, whereas the answer

egg would simply reflect a learned association with *chicken*. The models performed at or above the level of children, with the strongest models reaching adult-level accuracy. To control for word pairs the models were already familiar with through training, the authors introduced a C-only prompt that removed the A:B pair entirely and presented just the C-term with the five options. Because no relation can be mapped in this setting, any item still solved correctly must have been solvable by association alone. The models solved up to 62% of the items under this prompt, indicating that they could already identify the correct option for the majority of the analogies without access to the relation. Much of their apparent analogical reasoning can therefore be attributed to surface-level association with the C-term rather than relational mapping.

2.1.4 Story Analogies

In the final task of the paper by Webb et al. (2023), GPT-3 was evaluated on its ability to map higher order causal relations between complex narratives. This was the only domain where human participants maintained a clear advantage over GPT-3, especially for far analogies that required cross domain comparisons. While they conclude that Transformer architectures are to some extent capable of zero shot pattern recognition, Webb et al. (2023) also pointed out critical limitations. Unlike humans, LLMs do not have a persistent long term memory to draw upon past experiences once the context window is cleared. Furthermore, they struggle significantly with analogies that require physical and real world reasoning. The authors describe the models' emergent capabilities in this area as parasitic intelligence, because the relational reasoning of the models fundamentally depends on the massive corpora of human generated text they were trained on.

The STORYANALOGY Benchmark

This concept of parasitic intelligence is further supported by research demonstrating that models systematically fail when an analogy requires them to match the underlying relational structure of a situation, rather than just recognizing similar words or topics (Jiayang et al., 2023; Yuan et al., 2023; Sourati et al., 2024). The STORYANALOGY benchmark evaluates models on their capacity to identify narrative-level analogies (Jiayang et al., 2023). Building upon the Structure-Mapping Theory (Gentner, 1983), their dataset distinguishes two different types of similarity in narratives: entity similarity and relational similarity. Entity similarity is the degree of overlap in the topics or entities discussed in a narrative. Relational similarity is defined as the alignment of the underlying relational structures and logical connections between events, regardless of the surface topics. In the research of Jiayang et al. (2023) an analogical narrative is a narrative with low entity similarity and high relational similarity, i.e. two narratives about completely different topics or entities share the same underlying relational structure. Conversely, if narratives have both high entity similarity and high relational similarity, they represent literal similarity. Examples of both are shown in Table 2.1. The identification ability is tested with a multiple-choice task: given a source narrative and four candidate narratives, the model has to select the single candidate that is the true analogy. The three incorrect options consist of two randomly sampled distractors and one hard distractor, which shares many entities with the source narrative but not its relational structure. Both humans and LLMs are scored by the percentage of questions for which they select the correct analogous narrative, where chance performance

	Source	Target
Analogy	"Magma rises from deep in the earth. The magma goes into volcanos."	"Food goes up from the stomach. The food enters the oesophagus."
Literal Similarity	"Magma rises from deep in the earth. The magma goes into volcanos."	"Magma goes up from the inside of the planet. The magma enters volcanos."

Table 2.1. Examples of analogy and literal similarity (Jiayang et al., 2023)

is 25%. Human participants achieved an accuracy of 85.7%, whereas LLMs struggled significantly. ChatGPT (OpenAI, 2024) achieved an accuracy of only around 30% to 35%, and GPT-3.5 averaged around 37.1%. In particular, LLMs were rarely fooled by random distractors but were frequently drawn to the hard distractor, indicating that these models prioritize surface features over the deep structural abduction required for true analogical reasoning, as concluded by Yuan et al. (2023).

The ARN framework

The same preference for surface-level similarity over relational structure appears in the Analogical Reasoning on Narratives (ARN) framework (Sourati et al., 2024). ARN frames analogy detection as a binary task in which the model has to choose which of two narratives, a target and a distractor, forms an analogy with a given query narrative. Furthermore, the correct analogies are split into two categories. Near analogies share both high surface-level similarity and the same relational structure, while far analogies share only the relational structure across otherwise dissimilar domains. An example of these story analogies can be found in Table 2.2. The findings reveal a substantial gap: the models reliably recognize near analogies, but largely fail to detect far analogies. Among the six tested LLMs, even GPT-4 falls below chance level on far analogies in a zero-shot setting.

2.1.5 Mixed Evidence from Related Reasoning Tasks

Beyond the four domains studied by Webb et al. (2023), related reasoning tasks reveal a similar pattern: apparent competence on a task often coexists with a reliance on surface-level cues rather than genuine relational reasoning. The following sections illustrate this pattern across matrix reasoning, spatial reasoning, and a statistical account of where such shortcuts come from.

Matrix Reasoning

Reasoning tasks beyond verbal analogy show the same split between apparent competence and reliance on surface patterns. Hu et al. (2023) also studied matrix reasoning based on Raven's Progressive Matrices. The difference in the Digit Matrices of Webb et al. (2023) lies in what the matrices contain: Webb et al.'s were built directly from abstract numerical rules, whereas Hu et al. started from the original visual RAVEN problems, whose shapes, sizes, colors, and positions had to be transcribed into language before a language model could attempt them. They tested several encodings at dif-

	Story	High-level message
Analogous narrative	"After months of rigorous training and pushing through excruciating pain, Emily finally crossed the finish line of the marathon, triumphantly claiming her well-deserved medal."	"No pain, no gain"
Far analogous narrative	"Jacob knew that mastering the guitar requires countless hours of sore fingers and frustrating practice, but he embraced pain with determination, knowing that in the end, winning the competition would be worth every ache."	"No pain, no gain"
Near disanalogy	"Emily loved training for the marathon, but couldn't find friends until she realised that all people training there shared the same interest and passion, and she became friends with them."	"Birds of a feather flock together"

Table 2.2. Example of a story analogy (Sourati et al., 2024)

ferent levels of abstraction, ranging from literal numerical descriptions of each panel's attributes to more abstract descriptions of those features. They found that the more abstract encodings produced remarkably better performance, with even small models reaching strong zero-shot accuracy and at times surpassing human performance. However, Hersche et al. (2024) observed that on the 2x2 and 3x3 RAVEN matrices, high accuracy can be maintained by attending only to the last row of the matrix, implying that these problems can be solved by relying on shortcuts rather than with genuine reasoning.

Spatial Reasoning Strategies

Additionally, Wicke et al. (2024) investigated the spatial intuitions of LLMs by asking them to associate abstract and concrete verbs (e.g., perch and pull) with spatial representations, such as arrows ($\uparrow, \downarrow, \leftarrow, \rightarrow$) or textual directions. To elicit these intuitions, the authors introduced analogy-prompting, a method inspired by Chain-of-Thought reasoning where the model is explicitly instructed to first explain the underlying analogy before selecting a spatial concept. Although analogy-prompting improved the performance of the model, the Spearman correlation of 0.51 between the answers of GPT-4 and human judgments still showed that the LLM exhibited different reasoning strategies. For instance, models consistently associated the event "perched" with an upward direction, justifying this choice through the analogy of a bird sitting high up. In contrast, human participants preferred a downward orientation, likely focusing on the physical act of sitting down.

Statistical Drivers of Reasoning

A complementary explanation for these failures comes from outside the analogy literature. Prabhakar et al. (2024) conducted a shift cipher decoding study that identifies

three primary drivers of reasoning performance in LLMs. First, the models exhibit a strong probabilistic bias, favoring common sequences over rare ones. This often causes diversion from their own reasoning steps, such as abandoning the decoding of "STAZ" to predict the more probable word "STAY" instead. Secondly, models are highly dependent on the frequencies in the training data, which is demonstrated by the accuracy spike at a specific rotation of 13. Because this particular rotation is widely used on Internet forums to hide puzzle solutions, the spike suggests that models map learned distributions instead of executing the abstract rule. Finally, the logical processes the models execute are highly susceptible to noise. Accuracy declines as the required number of intermediate operations increases because each extra step carries an error probability (Prabhakar et al., 2024). Together, these factors suggest that strong performance can stem from statistical regularities in the training data rather than the consistent application of an abstract rule.

2.2 The Geometry of Analogical Representations

The studies discussed so far evaluate (analogical) reasoning behaviorally, by checking whether a model arrives at the correct answer. A complementary line of work instead asks whether a model's internal representations encode the relational structure of an analogy directly, independent of whether that structure is ultimately read out into a correct response. This geometric perspective treats the relation between two concepts as a property of the vector space in which they are represented, asking whether analogous pairs occupy a consistent and predictable position relative to one another, rather than whether the model's final answer happens to be correct. The following sections trace this perspective from text-level proximity in a representational benchmark (Section 2.2.1), through the parallelogram geometry observed in static word embeddings (Section 2.2.2), to the empirical and theoretical criticisms that this geometric account has since received (Section 2.2.3).

2.2.1 The ANALOGICAL benchmark

The ANALOGICAL benchmark (Wijesiriwardene et al., 2023) offers a perspective on geometrical representations of analogies by probing a model's internal representations rather than its task performance. Instead of asking a model to answer analogy questions, it embeds pairs of texts and checks whether analogous pairs lie closer together in vector space than non-analogous ones. The authors find that this proximity holds for simple analogies but breaks down as they become more abstract. Sourati et al. (2024) argue, however, that such proximity-based benchmarks are not grounded in the cognitive distinction between surface-level and relational similarity: they reveal whether related texts are represented nearby, without establishing whether the model has aligned their underlying relational structure.

2.2.2 Offset Parallelism in Word Embeddings

Research on static word embeddings offers a concrete picture of what geometric encoding of analogical structure can look like (Mikolov et al., 2013). In a static embedding space, every word is represented as a vector, and the relation between two words can be expressed as the offset vector connecting them. The offset vector entails the direction and distance one must travel to move from the first word to the second in the

embedding space. Mikolov et al. (2013) demonstrated that Word2Vec embeddings exhibit an approximate parallelogram geometry: for an analogy of the form A:B::C:D, the offset from A to B, $\vec{v}_B - \vec{v}_A$, tends to be approximately parallel to the offset from C to D, $\vec{v}_D - \vec{v}_C$. This means that the same relation is encoded as the same direction in the space, irrespective of which word pair expresses it. For the analogy *man:woman::king:queen*, for instance, the vector connecting *man* and *woman* points in roughly the same direction as the vector connecting *king* and *queen*, so that the four words form the corners of a parallelogram. Offset parallelism therefore provides a candidate operationalization of analogical structure: the cosine similarity between the two offset vectors $\cos(\vec{v}_B - \vec{v}_A,\ \vec{v}_D - \vec{v}_C)$, should be high when a genuine relation is shared and low when it is not (Mikolov et al., 2013).

2.2.3 Criticisms of Offset Parallelism

This view has, however, been challenged on both empirical and theoretical grounds. Peterson et al. (2020) evaluated the parallelogram model against large datasets of human analogy completions and relational-similarity judgments. They found that for some items, the plain similarity between individual word vectors predicted human responses better than the parallelogram model, even though such similarity ignores relational direction entirely. They also showed that the model captures some relation types well and others poorly, and concluded that offset parallelism alone is insufficient to account for how people solve even simple four-term analogies. Linzen (2016) identified a more basic problem with how the method is applied. When solving A:B::C:D by searching for the nearest vector with $\vec{v}_B - \vec{v}_A + \vec{v}_C$, the input words A, B, and C must be manually excluded from the candidate set; otherwise the method frequently just returns one of the input words, most often B. Its apparent success therefore depends in part on this exclusion rather than on relational geometry alone. Rogers et al. (2017) added that performance is tied to how close the word pairs already are in the embedding space: accuracy drops sharply as the semantic distance between the source and target pairs increases, so the method works mainly for relations whose terms are already similar. Finally, Ethayarajh et al. (2019) gave a formal explanation of when this geometry arises in the first place. They proved that the offset method recovers an analogy only when the words in each pair co-occur in the training text in a particular way, called co-occurrence shifted pointwise mutual information (csPMI). In practice, this condition is met for some frequently attested relations but not for analogies in general, which explains why the parallelogram structure is reliable in some cases and absent in others. Offset parallelism is therefore a useful but contested operationalization of analogical structure. For this reason, I test the analogies in the dataset directly for offset parallelism, to establish whether a parallelism-based metric is meaningful in this research before relying on it. I elaborate on this test and its consequences in the Methodology (Section 3.4).

2.3 Fine-tuning Models for Analogical Reasoning

A separate line of work asks not whether pretrained models already solve analogies, but whether analogical reasoning can be explicitly learned through fine-tuning, and which training objective best supports it. A recurring theme is that the target of learning is the relation between a word pair rather than the words themselves, and

that this can be learned from relatively little data. The following sections discuss two such objectives in turn: fine-tuning a model to produce a single relation embedding per word pair (Section 2.3.1), and fine-tuning a model directly on the offset parallelism criterion introduced in the previous section (Section 2.3.2).

2.3.1 Distilling Relation Embeddings

Ushio et al. (2021) fine-tuned RoBERTa to produce relation embeddings, single vectors that capture the relationship between two words. Their method worked in two steps. First, a word pair such as (Paris, France) was inserted into a prompt template, for example "the relation between Paris and France is ...", which was passed through the model, and the resulting token embeddings were averaged into one vector representing that pair's relation. Second, the model was fine-tuned with a contrastive loss: for a given word pair, another pair of the same relation type should receive a more similar embedding than a pair of a different relation type, so that minimizing the loss pushed the same relation pairs together and different relation pairs apart. They trained the model on the SemEval 2012 Task 2 dataset, which covers 79 fine-grained semantic relations grouped into 10 categories, using the most prototypical word pairs of each relation as positive examples. The resulting model, RelBERT, produced high-quality relation embeddings and performed competitively on analogy and relation-classification benchmarks despite the relatively small size of this dataset. Crucially, this also held for relation types different in nature from those used for fine-tuning, which the authors took as evidence that the method distills relational knowledge already encoded in the pretrained model rather than simply generalizing from the training examples.

2.3.2 Learning Offset Parallelism

Petersen and van der Plas (2023) pursued a similar goal by training a model to capture the relational structure of an analogy rather than the surface-level similarity between its words. They fine-tuned a BERT-base model to classify whether four words form a valid analogy, using an offset parallelism objective: for a true analogy A:B::C:D, the offset between A and B should match the offset between C and D, so the model was trained to maximize $\cos(\vec{v}_B - \vec{v}_A,\ \vec{v}_D - \vec{v}_C)$ for true analogies and minimize it for false ones. The fine-tuning set contained 4930 analogies, pooled from several human-oriented analogy tests and balanced between true and false examples. The fine-tuned model was then compared against a human baseline on a held-out dataset, namely the analogy dataset of Jones et al. (2022). Fine-tuning raised accuracy on this set from 0.53 to 0.69, leaving the model roughly 0.15 below the human level but showing that the relational pattern could be learned from a small amount of data

Taken together, the literature reviewed in Sections 2.1–2.3 points to an unresolved tension. Across the behavioral reasoning tasks reviewed in Section 2.1, strong performance can reflect either a genuine understanding of relational structure or reliance on surface-level similarity and statistical patterns learned during training. The representational and fine-tuning work reviewed in Sections 2.2 and 2.3 sharpens this tension rather than resolving it: offset parallelism is a disputed measure of relational structure, and the fact that the relational pattern can be learned through fine-tuning does not reveal which of the two strategies produced any given correct answer. That is why behavioral results alone cannot decide between these two perspectives. A high accuracy

score can indicate either strategy, while a low accuracy score can mean that a model lacks relational mechanisms entirely, or simply that these mechanisms are hidden by decoding difficulties. Resolving this conflict requires looking inside the model, which is the focus of the next section.

2.4 Mechanistic Interpretability

Mechanistic interpretability is an approach to studying LLMs that examines the internal computations leading to a prediction, rather than the prediction alone. This makes it a well-suited tool for moving beyond the behavioral debates discussed in the preceding sections: by locating and intervening on the components responsible for a specific behavior, it offers a way to examine analogical reasoning at the level of the mechanisms that produce it, rather than only the outputs they produce.

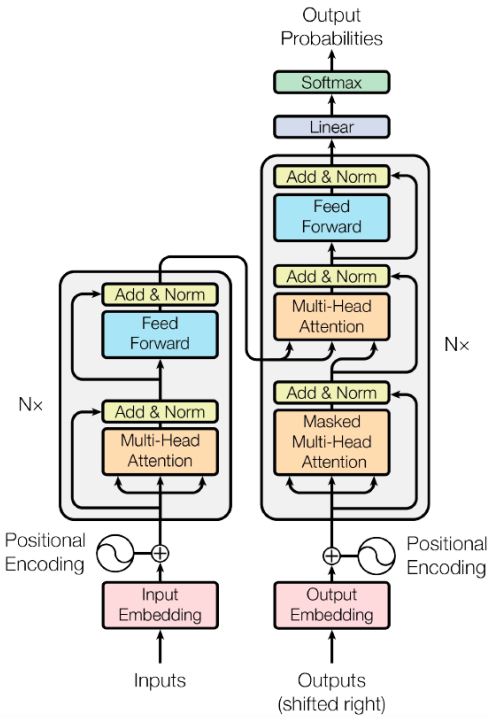
This section establishes the framework needed for such an analysis. It first outlines the Transformer architecture (Section 2.4.1), including the residual stream view that underlies most mechanistic interpretability work. It then explains the key methodological aspects of activation patching and path patching (Section 2.4.2), before introducing the central objects of study, features and circuits, together with what is known about the functional roles of layers and attention heads in reasoning (Section 2.4.3). Throughout, it reviews recent experiments at the intersection of mechanistic interpretability and (analogical) reasoning, including the small body of work that applies these methods to analogies directly.

2.4.1 The Transformer Architecture

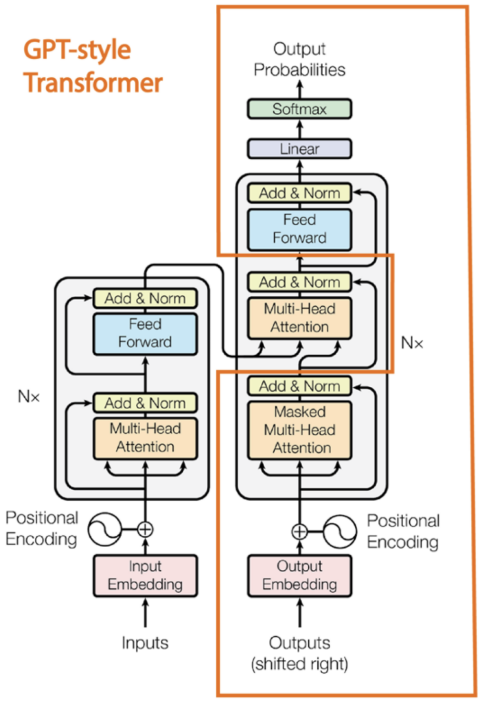
The Transformer architecture was first introduced in the paper Attention Is All You Need (Vaswani et al., 2017). Attention mechanisms were by then already popular in machine translation, where they were typically combined with recurrent encoder-decoder architectures such as bidirectional RNNs and LSTMs (Stahlberg, 2020). The revolution of the Transformer was that it replaced the recurrent structure in those architectures with attention mechanisms. In a recurrent structure, information has to travel through the sequence token by token, which makes it difficult to relate distant tokens and causes information from early in the sequence to degrade over steps. Attention instead allows every token to directly attend to every other token in a single step, so that relationships between words are captured regardless of how far apart they are in the sequence. A further consequence of removing recurrence was parallelization: because recurrent structures process a sequence token by token, they inherently limit how much can be computed at once. Attention mechanisms allow Transformers to process all tokens in a sequence in parallel, which was a major benefit compared to the state-of-the-art models at that time. This combination of directly modeling relationships between all tokens and processing them in parallel fundamentally reshaped Natural Language Processing. In the years following this publication, the Transformer became the dominant architecture for LLMs that are capable of a wide range of zero-shot and few-shot tasks (Brown et al., 2020).

From Translation Model to Encoder and Decoder Models

The original Transformer of Vaswani et al. (2017) was an encoder-decoder model designed for translation, in which the encoder reads the source sentence and the decoder



(a) Original Transformer Architecture (Vaswani et al., 2017)



(b) GPT-style Transformer Architecture (Stollnitz, 2023)

Figure 2.2. The Transformer Architecture

generates the translation. This translation model formed the inspiration for a broader family of models, not trained on parallel translation data but with a self-supervised perspective on large amounts of unlabeled text. From the original encoder-decoder architecture, shown in Figure 2.2a with the encoder on the left and the decoder on the right, two model families emerged. Encoder models such as BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2019), which build representations using both the preceding and the following context, and decoder models such as the GPT (Generative Pre-trained Transformer) family (Radford et al., 2018), which attend only to the preceding context.

GPT-2 as a Decoder Model

A prominent example of a decoder-only model is GPT-2 (Radford et al., 2019), which uses only the decoder component of the original architecture, as illustrated in Figure 2.2b. Because a decoder attends only to the preceding context, GPT-2 works autoregressively: it generates text one token at a time, and each new token is predicted from the tokens that came before it. It is trained with a causal language modeling objective: predicting the next token in a sequence and updating its weights based on how good that prediction is. This training objective was introduced for the first GPT model (Radford et al., 2018) and reused at larger scale in GPT-2.

Once trained, a decoder-only model can generate text. The sequence of operations within the decoder architecture begins at the bottom of Figure 2.2b, where previously generated tokens are passed through an output embedding layer to convert tokens into

embeddings. As described above, the Transformer has no inherent sense of word order, which is why positional encodings are added to the input embeddings to explicitly add information about the sequence of the tokens. Afterwards, the resulting representations enter the main Transformer block, which is composed of a stack of N identical layers (Vaswani et al., 2017). Each of these N layers contains two main sub-layers: a multi-head attention mechanism, which lets each token gather information from other tokens in the sequence, and a feed-forward network, which processes each token's representation on its own (Vaswani et al., 2017). Both sub-layers are wrapped with residual connections and layer normalization, the "Add & Norm" steps discussed below. Since the attention mechanism is the innovative component that distinguishes Transformers from previous language models, it will be examined in detail in the next sections.

The Attention Mechanism

Conceptually, the attention mechanism maps a query and a set of key–value pairs to an output. The model uses a query (Q) to evaluate which keys (K, representing the available context) best fit the query, and then retrieves the values (V) that correspond to those keys. Crucially, the attention mechanism does not select one best match but a weighted blend of all values, where each value contributes in proportion to how well its key matches the query. This mechanism operates over the words in a sequence. Consider the sentence "The animal didn't cross the street because it was too tired." Suppose we want to compute the attention scores for the word "it" using an attention head specialized in coreference resolution (resolving which earlier word a pronoun refers to). For each word in the sequence, the mechanism computes how strongly "it" relates to that word, as visualized in Figure 2.3. If the mechanism works well, the highest attention score is assigned to "animal," correctly identifying it as the word that "it" refers to (Google, 2026).

Specifically, the Transformer architecture uses scaled dot-product attention, as visualized in Figure 2.4a. The network computes the dot products of the queries with all keys to determine how relevant the keys are in this context. This result is then divided by a scaling factor ($\sqrt{d_k}$). A softmax function is applied to produce normalized weights, which are then multiplied by the value vectors (Vaswani et al., 2017). When computed simultaneously for matrices of queries, keys, and values, the operation is expressed as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Multi-Head Attention

Rather than performing this calculation once, Transformers employ multi-head attention (Figure 2.4b). Instead of a single attention head, the model uses h heads that operate in parallel, each learning to attend to different aspects of the language. This is powerful because the heads can capture distinct types of relationships: one head might learn to track syntactic structure, such as matching a verb to a distant subject, while another tracks semantic relationships, such as linking a pronoun to its referent, as in the coreference example above. The outputs of all heads are then concatenated and passed through a final linear layer, merging their representations into a single output (Vaswani et al., 2017).

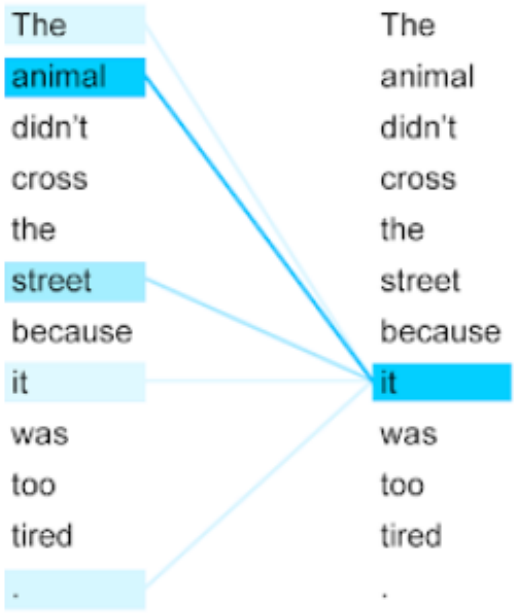


Figure 2.3. Self-attention weights for the pronoun "it" in the sentence "The animal didn't cross the street because it was too tired," adapted from (Google, 2026).

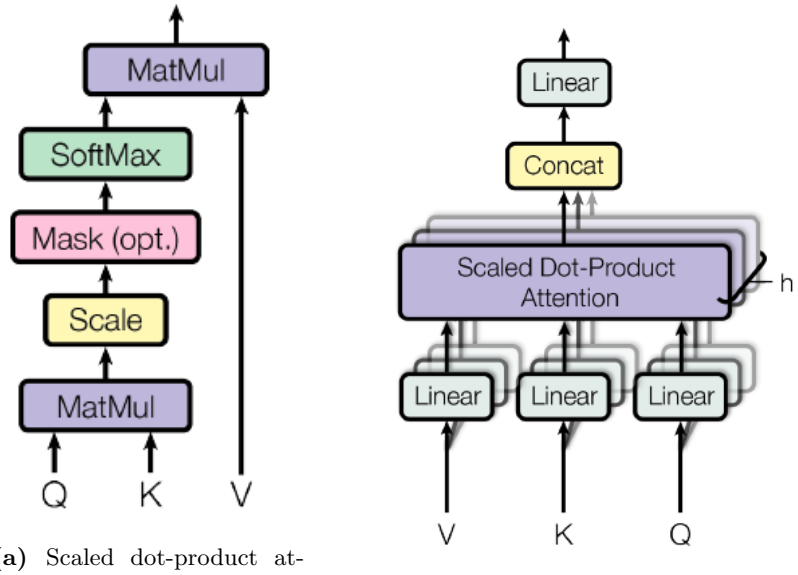
Add & Norm and the Feed-Forward Network

Immediately following the multi-head attention mechanism is an "Add & Norm" step (Figure 2.2b). The "Add" refers to a residual connection, which adds the original input of the sub-layer directly back into its computed attention output. The "Norm" applies layer normalization to this summed result. The representation is then routed through a feed-forward network. While the attention mechanism exchanges information between tokens in the sequence, the feed-forward network operates on each token's representation independently. This network is essentially a multi-layer perceptron, consisting of two linear transformations separated by a non-linear activation function (Vaswani et al., 2017); GPT-2 uses the GELU activation in place of the original ReLU (Radford et al., 2019). A second "Add & Norm" step completes the sub-layers within the block.

Once a representation has passed through all N stacked layers, it is projected by the unembedding layer into a vector of logits, one for each token in the vocabulary. A softmax function then converts these logits into a probability distribution over the vocabulary, indicating how likely each token is to occur next (Vaswani et al., 2017).

Residual Stream

Essentially, a token's representation is refined as it passes through the N layers of the model, with each layer reading the current representation and writing an updated one back. This perspective invites an alternative view of the Transformer architecture, framed around the flow of information through the layers. This flow is known as the residual stream (Figure 2.5), which acts as the central information highway of the model (Wang et al., 2022). Rather than viewing the layers as sequentially, passing data from one to the next, this view treats components such as the attention heads and feed-forward networks as operating on a shared stream: each reads a representation



(a) Scaled dot-product attention

(b) Multi-head attention

Figure 2.4. The Attention Mechanism (Vaswani et al., 2017)

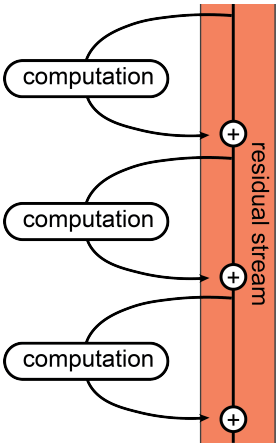


Figure 2.5. The Residual Stream (Kirchner, 2022)

from the stream, performs its computation, and writes its output back by addition (the "Add & Norm" step described above).

This framing is central to mechanistic interpretability, which seeks to understand the internal computations of otherwise opaque "black-box" models. Because each component's contribution enters the residual stream additively, researchers can isolate, track, and manipulate the information passing between specific components, making it possible to identify which computations contribute most to the final prediction (Wang et al., 2022).

2.4.2 Activation Patching and Path Patching

Two main techniques are used to isolate and track information in the internal computations of the model: activation patching and path patching (Wang et al., 2022). Both are intervention methods: they alter the model's internal activations and observe how

its prediction changes. This reveals the role a given component plays in producing a specific prediction.

The two techniques differ in what they localize. Activation patching identifies which components matter for a given prediction. By intervening on a single component, such as an attention head, and observing how much the model's behavior changes, researchers can determine whether that component is important for the computation in question (Zhang and Nanda, 2023). Path patching asks how one component influences another, intervening on the connection between two specific components instead. This makes it possible to trace how information is passed from one component to the next and to map out the circuit of interactions through which information flows to produce the final prediction (Wang et al., 2022).

Activation Patching Procedure

In this study, activation patching is the chosen mechanistic interpretability method; this section therefore explains its procedure. The general idea behind activation patching is to determine which components of the model contribute to restoring the original prediction (Meng et al., 2022; Zhang and Nanda, 2023).

Activation patching is performed by carrying out a clean run, a corrupted run, and a set of patching runs for each prompt on which it is applied, as illustrated in Figure 2.6. Together, these runs measure how much each head in the model contributes to producing the correct answer for that prompt (Heimersheim and Nanda, 2024). During the clean run, the original (clean) prompt is given to the model and the model's intermediate activations at every head in every layer are stored in cache. Crucially, the model must produce the correct answer in the clean run, since this is the answer whose restoration is measured later. During the corrupted run, the model processes a corrupted version of the original prompt, in which one or more of the original tokens are replaced with random other tokens (Zhang and Nanda, 2023). For the corrupted run, both the intermediate activations at every head in every layer and the final logits over the model's vocabulary are stored.

In the patching runs, the model again processes the corrupted prompt, but for a single head h in a single layer l, the activation value of that head is replaced with the value recorded for that same head in the clean run. After this replacement, the model completes its usual forward pass, and the final logit assigned to the correct target (i.e. the answer given in the clean run) is recorded. A logit recovery metric (defined in Section 3.5.2) is then used to quantify how much replacing that single head moved the logit of the correct prediction back toward the value it had in the clean run (Zhang and Nanda, 2023). This procedure is repeated independently for every head in every layer, producing one logit per (layer, head) cell. For a model with n heads per layer and m layers, this amounts to $n \times m$ patching runs. From the resulting map, the heads that contribute most to restoring the correct prediction can be identified. Wang et al. (2022) produced a per-head heatmap of this kind for the indirect-object identification (IOI) task. An example of a clean prompt they used is "When Mary and John went to the store, John gave a drink to", for which the correct prediction is "Mary". The corrupted prompt is generated by replacing all three names with random, unrelated names, for example "When *Pia* and *Antske* went to the store, *Lara* gave a drink to", which destroys the signal that identifies "Mary" as the answer. Each head is then patched in turn to determine its contribution to the prediction. It should be noted that the heatmap in Figure 2.7 is the result of path patching rather than activation

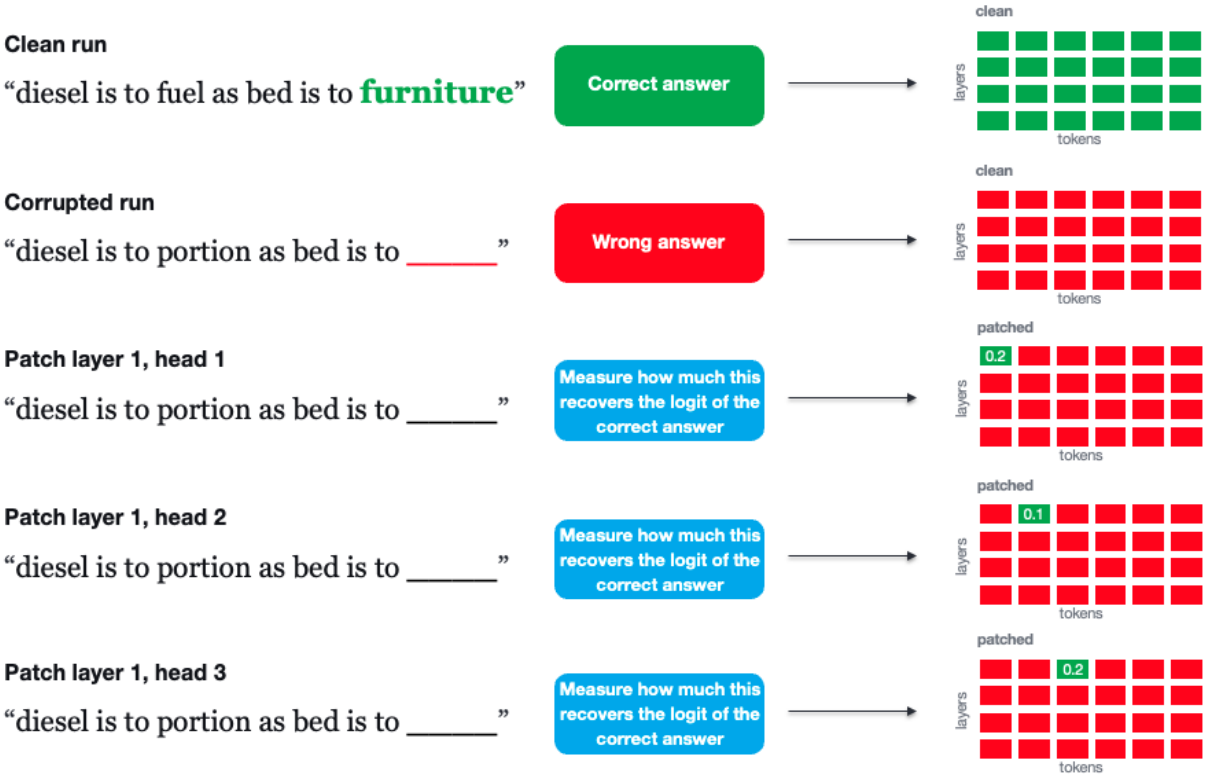


Figure 2.6. Illustration of the activation patching procedure. In the clean run, the model receives the uncorrupted prompt and produces the correct answer; in the corrupted run, the source pair is replaced with an unrelated word, causing the model to produce an incorrect answer. Patching reintroduces the clean activation of a single attention head into the corrupted run and measures whether the correct answer is restored. Each grid represents a model with 4 layers and 6 attention heads per layer, where each block corresponds to one attention head in a specific layer. The four bottom examples show that patching most individual heads leaves the wrong answer unchanged, one head produces only a partial recovery, and a single specific head fully restores the correct answer.

patching. Its colors encode each head's effect on the logit difference between the correct name ("Mary") and the distractor ("John"). It is included here because it shares the same layer-by-head format as the recovery maps reported later in this thesis.

Methodological Foundations

Zhang and Nanda (2023) systematically evaluated the methodological choices involved in activation patching, in particular how the corrupted run is constructed and which metric is used to measure the effect of patching. For the corrupted run, one option is to add Gaussian noise to the token embeddings that carry the key information. However, Zhang and Nanda (2023) show that this pushes the model out of distribution, since it never encounters such noised inputs during training, which makes the resulting measurements less reliable. They instead recommend symmetric token replacement, in which the key tokens are swapped for different but comparable tokens of the same type (for example, replacing "Mary" and "John" with "Antske","Pia" and"Lara" in the example of the previous paragraph). The replacement is called symmetric because the

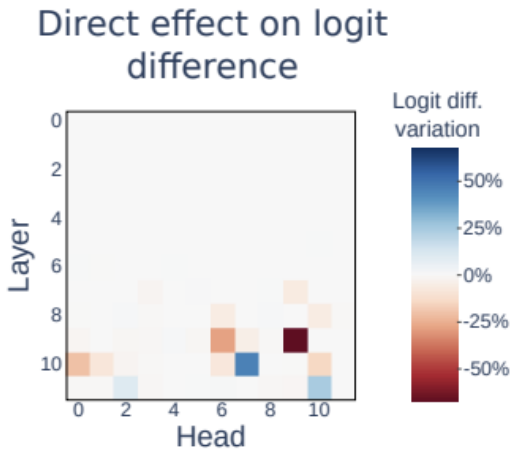


Figure 2.7. Example of a per-head heatmap for the indirect-object identification task, adapted from Wang et al. (2022). Each cell shows a single attention head's effect on the logit difference between the correct name (*Mary*) and the distractor (*John*), rather than the normalized logit recovery used in this study.

clean and corrupted prompts are interchangeable: the corrupted prompt is drawn from the same distribution as the clean one and could itself have occurred as a valid clean input, so the model stays in distribution. Zhang and Nanda (2023) further advocate for measuring the patching effect with the logit difference, the gap between the logit of the correct answer and that of a competing incorrect answer, rather than with the answer's probability. A probability-based metric can overlook components that actively harm the correct prediction: once corruption has driven the correct probability close to zero, there is no room left for a harmful component to lower it further. The logit difference has no such floor and therefore captures these negative components as well.

Beyond how the corrupted run is constructed, Wang et al. (2022) and Conmy et al. (2023) address which component of the model should be patched. Each attention layer in a Transformer model contains multiple attention heads, each of which produces its own output vector, denoted as z. This vector represents the information that a single head has gathered by attending to the relevant positions in the sequence. Before reaching the residual stream, the output vectors of all heads in the same layer are combined by the output projection matrix W_O into a single blended vector. Patching is advised to be applied at the z level, immediately before this blending step, because this is the last point at which the contribution of an individual head can be measured in isolation. Replacing $z_{l,h}$ with its value from the clean run therefore reveals what that specific head contributes to the prediction, without interference from the other heads in the same layer. This head-level granularity is the key advantage of z-patching over patching at the level of the full layer output: once W_O has combined all heads together, their individual contributions can no longer be separated (Wang et al., 2022; Conmy et al., 2023).

2.4.3 Features and Circuits

The techniques discussed above support the analysis of model behavior in terms of the two primary objects of study in mechanistic interpretability: features and circuits.

Features are the meaningful properties or concepts represented within the model's activation space (Gurnee et al., 2023; Bills et al., 2023). A feature might encode a relatively concrete property, such as whether the input text is in a particular language or whether the current token is a verb. Or a more abstract property, such as the sentiment of a passage. The degree to which the property is present is reflected in the strength of the corresponding activations. Circuits, by contrast, are subgraphs within the model: configurations of attention heads and feed-forward layers that together implement a specific behavior (Wang et al., 2022; Eshuijs et al., 2025). A well-known example is the circuit responsible for indirect-object identification that was discussed above (Wang et al., 2022).

By isolating features and circuits, mechanistic interpretability offers a bottom-up account of how a model stores knowledge, routes information, and processes context (Meng et al., 2022). For complex reasoning tasks, this approach offers a way to move beyond behavioral tests and locate the internal mechanisms responsible for drawing connections between distinct concepts, a possibility this thesis investigates directly for analogical reasoning. This kind of analysis has revealed that complex behaviors are often implemented as a staged division of labor across layers and heads, rather than a single computation: as the following sections show, specific functional roles are concentrated in specific layers and heads across a range of reasoning tasks.

This division of labor extends to individual layers and attention heads. Mechanistic interpretability studies have shown that reasoning is not implemented as a single computation, but is functionally specialized across the network: early layers tend to retrieve relevant information from the input, which later layers then draw upon to derive a final answer (Biran et al., 2024), while individual attention heads can be organized into distinct functional roles such as retrieving facts, selecting relevant context, and carrying out the reasoning step itself (Zheng et al., 2024). This specialization holds for reasoning that unfolds over several steps, not only single-step queries: Hou et al. (2023) show that a model's implicit reasoning tree can be recovered directly from its attention patterns, with lower layers identifying useful statements and higher layers executing the resulting logical deductions Moreover, Pan et al. (2026) confirm in a broader survey that such intermediate results are computed and passed along internally, although this implicit reasoning remains bottle-necked by the network's fixed depth.

Application to Analogical Reasoning

Verbal analogies are a natural candidate for localizing such reasoning processes using mechanistic interpretability, since solving a verbal analogy requires first extracting the relation from the first pair and then reapplying it to the second pair. Yang et al. (2025) applied these methods directly to analogical reasoning, among other experiments. They constructed pairs of four-term verbal analogies with a deliberate design choice: a synonym problem and an antonym problem built from the same word set always share the same correct completion. For instance, "lazy : idle" is a synonym relation, while "energetic : idle" is an antonym relation, yet both end in the word "idle," as shown in Table 2.3. Because the literal target token is identical across the two relation types, any attention head that responds differently to the synonym versus the antonym version of a problem cannot be reacting to the surface form of the answer; the difference must instead reflect sensitivity to the abstract relation, synonym or antonym, that the model needs to apply. Holding the completion token constant in this way is what let the researchers dissociate heads that encode this abstract relation from heads that simply

	Synonym	Antonym
Example 1	impoverished : poor	rich : poor
Example 2	courageous : brave	fearful : brave
Query	lazy : ?	energetic : ?

Table 2.3. Examples of synonym and antonym verbal analogy

process literal tokens.

Before any patching, Llama 3.1 70B was evaluated on the task itself: accuracy was computed by comparing the logit assigned to the correct completion with the logit assigned to the incorrect completion that would result from applying the opposite relation (e.g., for the antonym query "energetic", the correct answer is "idle," while the incorrect alternative is "active," the word that would result from treating the relation as a synonym instead). Under this measure, the model reached 77.0% accuracy on synonym problems and 88.4% on antonym problems, well above the 50% chance level set by the two-way comparison.

To localize which heads causally drive this behavior, three separate patching analyses were then run, each transferring activations from a donor problem into a recipient problem that shares the same correct completion, but calls for a different relation. To identify symbol-abstraction heads, activations were transferred between the synonym and antonym versions of the same word set, patching at the position of the second word in each in-context example (e.g., "poor" and "brave" in Table 2.3). To identify symbolic induction heads, the same synonym-antonym transfer was used, but patching was applied instead at the position of the final colon, the point at which the model must generate its completion. To identify retrieval heads, activations were instead transferred between two problems that shared the same relation but used different word sets, again patching at the final colon. In each case, a head is identified as causally involved when patching shifts the model's output toward the completion implied by the donor problem's relation rather than the recipient's. The resulting effects were concentrated in early layers for symbol-abstraction heads, intermediate layers for symbolic induction heads, and later layers for retrieval heads, supporting the three-stage architecture also identified for the simpler tasks discussed above (Yang et al., 2025).

2.5 Research Gap

Despite the extensive behavioral and representational literature, it remains unclear whether the analogical behavior of LLMs reflects genuine relational reasoning or surface-level association, since accuracy alone cannot distinguish between the two strategies. Mechanistic interpretability offers a way to resolve this question by examining the internal computations that produce a prediction, but it has so far been applied to analogical reasoning in a single study, on only synonym and antonym problems, and never alongside a systematic behavioral and error analysis of the kind reviewed above. A gap therefore remains in our understanding of how analogies are processed inside LLMs, and how this mechanistic evidence might relate to the discussion on whether LLMs rely on surface-level similarity or extract actual relational structure when solving analogies.

To address this gap, this thesis combines a behavioral analysis, an error analysis,

and a mechanistic analysis to evaluate the analogical reasoning of two pretrained transformer models that differ in scale and architecture: GPT-2 Small and Llama 3.2 1B. The behavioral and error analyses are observational: they read off the model's outputs and internal representations without altering its computation, the first from the predicted tokens and the second from the geometry of the prediction in embedding space. The mechanistic analysis is interventional: it actively manipulates the model's internal activations and observes how the prediction changes, in order to establish which components causally drive these predictions.

Following the dataset and analogy format of Jones et al. (2022), each analysis is structured as follows. The behavioral analysis is a rank-based evaluation across multiple prompting and finetuning conditions: open-vocabulary zero-shot prompting, restricted forced-choice prompting, few-shot prompting with within- and cross-relation demonstrations, and finetuning on held-out splits (GPT-2 Small only); accuracy is additionally broken down by relation type (categorical, causal, compositional) and semantic distance (near, far). The error analysis examines whether incorrect predictions still preserve the relational structure of the source pair rather than failing at random, via an structural criterion based on offset-cosine similarity between the source-pair and prediction offsets, and a manual inspection of the top-10 predictions for each analogy. The mechanistic analysis uses head-level activation patching in combination with explicit corruption-token sensitivity checks to isolate the internal mechanisms that drive the prediction. This research goal corresponds to three sub-questions.

SQ1 (Behavioral). How accurately do GPT-2 Small and Llama 3.2 1B solve four-term verbal analogies, and how is this accuracy modulated by the prompting format, by finetuning on held-out splits (GPT-2 Small only), by relation type, and by semantic distance?

SQ2 (Representational/Error-analysis). When the model's top-ranked prediction is not the gold target, do its predictions nevertheless reflect the relational structure of the A:B::C:D mapping rather than random failure?

SQ3 (Mechanistic). Which attention heads causally drive the analogical prediction in each model, and how stable is this head-level structure across prompting conditions, finetuning, and resampling of the corruption tokens used for activation patching?

Based on the literature reviewed, I expect Llama 3.2 1B to outperform GPT-2 Small on the behavioral task, since in-context analogical performance has been found to scale with model size (Webb et al., 2023; Brown et al., 2020), and Llama 3.2 1B is larger and trained on a more recent corpus. I expect both models to perform better on near than far analogies, consistent with the repeated finding that models recognize near analogies reliably but fail on far ones that require relational transfer across dissimilar domains (Sourati et al., 2024; Jiayang et al., 2023; Webb et al., 2023). I expect within-relation in-context demonstrations to help more than cross-relation ones, since Structure-Mapping Theory holds that transfer depends on relational alignment between the demonstration and the test item (Gentner, 1983). For the mechanistic analysis, I expect the recovery signal to concentrate in a small number of mid-layer attention heads in both models, and to be largely shared across relation types rather than relation-specific, given the layer-wise division of labor reported by Pan et al. (2026) and the symbolic-induction stage situated in intermediate layers by Yang et al. (2025).

Chapter 3

Methodology

This chapter establishes the experimental framework used. It will start by describing the analogies dataset and its model-specific adaptation (Section 3.1), and the two language models under study (Section 3.2). It will then detail the basic experiment: the analogical reasoning task, its evaluation metrics, and the statistical treatment of the resulting accuracies (Section 3.3); the representational/error analysis applied to its predictions (Section 3.4); and the activation patching procedure used to localize the attention heads that drive them, together with how the patching results are filtered and summarized (Section 3.5). Finally, it describes the four experimental conditions that are tested and how they deviate from the basic experiment (Section 3.6). All code written for this thesis is available at `https://github.com/lara-anna/thesis-analogical-reasoning-llms`.

3.1 Analogies Dataset

The experiments in this study use the dataset created by Jones et al. (2022). This dataset was originally developed to test the interactive effects of semantic distance, distractor salience, and relation types on human performance in verbal analogy tasks. It contains 60 verbal analogies formatted as "A is to B as C is to D" mapped over three relation categories: categorical, causal, and compositional relations. Categorical relations represent a "kind of" relationship where one word acts as a specific exemplar, and the other word serves as its broader category label, as in *trout : fish :: lobster : crustacean*. Compositional relations describe what an object is made of by connecting a specific raw material or component to the final product, as in *marble : stairs :: brick : patio*. Causal relations involve a cause and effect relationship where one term represents an event or condition and the other the produced outcome, as in *virus : flu :: bacteria : infection*. The dataset manipulates semantic distance by creating near analogies, where the source pair (A,B) and the target pair (C,D) come from similar semantic domains, and far analogies, where they come from more distant domains. Examples of each category are shown in Table 3.1. The complete dataset of analogies is included in Appendix 6.

In the original experiment, participants were asked to complete an analogy formatted "A is to B as C is to ..." by choosing between two options: the target word and a distractor. Therefore, the dataset includes a correct target word and two distractor terms per analogy that vary in salience. High-salience distractors are semantically close to the C-term and therefore represent plausible but incorrect completions. For

	Near	Far
Categorical	Tarantula:Spider::Bee:Insect	Wrench:Tool::Sad:Mood
Causal	Fracture:Cast::Incision:Scar	Invitation:Visit::Tornado:Destruction
Compositional	Oatmeal:Cookie::Banana:Muffin	Flour:Tortilla::Silk:Tie

Table 3.1. Examples of analogies

instance, given "fracture is to cast as incision is to...", the correct target is *scar*, while a high-salience distractor might be *surgery*, which is semantically related to the C-term *incision*. Low-salience distractors are more semantically distant and less compelling, like *numb* in this analogy. The authors of the dataset measured this distance using Latent Semantic Analysis (LSA).

3.1.1 Dataset Adaptation

From the original dataset, I extract the four terms A, B, C, and D together with the high- and low-salience distractors for each analogy. These distractors will be used to analyze how distractor salience influences the model's performance on the analogy task in a specific experiment. I also copy the two categorical labels for each analogy: the relation type (categorical, causal, or compositional) and the near/far semantic-distance label. The relation type enables analyses of whether model performance and internal activation patterns differ across relation types. The inclusion of the near/far label enables analyses of whether performance and activation patterns differ with the degree of cross-domain relational transfer that is required. The extracted terms are then normalized: the four analogy terms and the two distractors are lower-cased to ensure consistent tokenization across models with different vocabulary conventions.

Semantic Distance Annotation

For each model under evaluation, three cosine-distance measures are computed from the model's static token embedding matrix (W_E) for further analysis:

1. `dist_AB`: the cosine distance between the embeddings of A and B.
2. `dist_CD`: the cosine distance between the embeddings of C and D.
3. `dist_AB_CD`: the cosine distance between the summed pair embeddings ($\vec{v}_A + \vec{v}_B$) and ($\vec{v}_C + \vec{v}_D$), capturing the semantic distance between the source and target pairs as represented in the model's embedding space.

When a word is tokenized into multiple sub-word tokens, its embedding is computed as the mean of the individual sub-word token embeddings. The model-specific semantic distances are stored as per-item metadata and are used in later analyses of the semantic distance between the parts of the analogies. The original paper divided the analogies into near and far analogies based on LSA distance scores. To confirm whether this separation holds when using cosine distance, a more modern and common technique for computing semantic distance between word embeddings, I compare the cross-pair distances (`dist_AB_CD`) of the near and far groups in both models using a two-sided Mann-Whitney U test. This non-parametric test is appropriate here because it does not assume the cross-pair distances are normally distributed and remains valid for

the relatively small subgroup sizes (30 near and 30 far analogies). The results of this comparison, which confirm that the near/far distinction also holds when cosine distance is used, are elaborated on in Section 4.1.

3.2 Models

In this study, two distinct models are employed, differing in both their parameter counts and architectural details. Both models are based on a decoder-only architecture and are available through the TransformerLens library (Nanda and Bloom, 2022), which was a primary factor in their selection. Beyond compatibility with TransformerLens, these specific models were chosen because they are relatively compact. Their smaller scale makes them highly suitable for activation patching, as having fewer layers and attention heads restricts the complexity of the internal pathways and keeps the resulting data manageable and easier to read. Furthermore, TransformerLens provides direct access to residual stream components at every layer of the model, making it well-suited for mechanistic interpretability experiments. By evaluating two models that differ in scale, training data, and architectural details, I can systematically examine how these factors influence both behavioral performance and the internal mechanisms underlying this performance.

3.2.1 GPT-2 Small

The first model evaluated in this study is GPT-2 Small, a Transformer-based language model developed by OpenAI (Radford et al., 2019). GPT-2 Small is the smallest version of the GPT-2 model family and contains approximately 124 million parameters. The model follows a decoder-only transformer architecture consisting of 12 transformer layers, each with 12 self-attention heads and a hidden dimension of 768. Like other GPT-style language models, GPT-2 uses masked self-attention, ensuring that each token prediction depends only on previously observed tokens in the sequence. The model processes sequences with a maximum context length of 1024 tokens and uses learned positional embeddings to represent token positions. Because GPT-2 Small shares the standard Transformer architecture used by many auto-regressive language models, it provides a natural and interpretable baseline for activation patching.

3.2.2 Llama 3.2 1B

In addition to GPT-2, this study evaluates Llama 3.2 1B, an open source auto-regressive transformer language model developed by Meta (Grattafiori et al., 2024). The model follows a decoder-only architecture and contains approximately 1.2 billion parameters. Llama 3.2 1B consists of 16 transformer layers, each with 32 attention heads and a hidden dimension of 2048. It encodes positional information using rotary positional embeddings, and it additionally incorporates grouped query attention with 8 key value heads. This reduces the memory required during inference while preserving the expressiveness of the model. The model supports a maximum context length of 128,000 tokens and was trained on a multilingual corpus of approximately 9 trillion tokens. Due to its larger parameter count and its use of more recent architectural choices, Llama 3.2 1B provides an opportunity to examine whether the mechanistic patterns identified in smaller models generalize to a more contemporary language model.

3.3 Behavioral analysis: Analogical Reasoning Task

In each experiment, the model's analogical reasoning capabilities are evaluated as described below. Each analogy in the dataset is converted to a natural-language prompt using the template:

```
{A} is to {B} as {C} is to
```

The model is then evaluated on its ability to predict D as the immediately following token. The target token is defined as the first (sub-word) token produced when tokenizing D with a preceding space (i.e., `" " + D`). This respects the word-boundary convention used by sub-word BPE tokenizers, where word-initial forms are distinct from word-internal forms (both GPT-2's byte-level BPE and Llama 3's BPE tokenizer mark word-initial tokens with a leading-space prefix).

3.3.1 Analysis

Model performance is scored based on the rank of the target token in the output probability distribution over the full vocabulary. Rank 1 indicates the model assigned the highest probability to the correct answer; a rank within 10 indicates that the correct answer was among the model's ten most probable predictions. This yields two complementary evaluation metrics: top-1 accuracy, the proportion of items for which the correct token is the single most probable prediction, and top-10 accuracy, the proportion of items for which the correct token falls within the top 10 predictions. Both metrics are computed across all 60 analogy items.

Statistical Analysis

All accuracies are accompanied by 95% Wilson score confidence intervals (Wilson, 1927). The Wilson score interval is a closed-form interval for a binomial proportion that, unlike the standard normal (Wald) approximation, is bounded within the valid range $[0, 1]$ and remains well-behaved when the observed accuracy is close to zero or one. This interval is appropriate for the present study because the evaluations are small: the dataset contains only 60 items (240 in the restricted forced-choice condition, and 20 or 30 items in the held-out finetuning splits). With these sample sizes, the normal approximation to the binomial is unreliable. This is particularly true when accuracy values approach zero or one, as many do in the finetuning conditions: in that regime, the normal approximation either extends beyond the valid range of proportions or collapses to a zero-width interval. The Wilson score interval avoids both failure modes: even when every item is answered correctly or incorrectly it returns a non-degenerate interval bounded within $[0, 1]$, and it is recommended over the normal approximation precisely in the small-sample and near-boundary regime considered here (Brown et al., 2001).

The confidence intervals are reported as a descriptive measure of the uncertainty around each accuracy estimate, not as a basis for formal pairwise significance testing. Throughout the results tables, an asterisk (*) marks a value whose 95% Wilson interval does not overlap those of the other values in the same column; this is not a formal significance test, but a conservative indication of a difference at approximately the 0.05 level, inferred from non-overlapping confidence intervals.

Punctuation filter

Because initial experiments with Llama 3.2 1B showed that the predictions frequently included punctuation, a punctuation-filtered rank is computed in addition to the standard rank. A boolean function is used to mark each token as lexical if its string form contains at least one alphabetic character (matched against the regular expression `[a-zA-Z]`), and non-lexical otherwise. The filtered rank of the target token is then computed only among the lexical tokens, yielding `target_rank_nopunct` and the corresponding `correct_top1_nopunct` and `correct_topk_nopunct` accuracy variants. The difference in accuracy between ranks with and without non-lexical items filtered out, is explicitly documented in the results.

In addition to the rank-based metrics, the raw output logit and probability assigned to the target token are recorded for each item, along with the full ranked list of the top-100 most probable tokens. These additional outputs enable secondary analyses, such as examining the margin between the correct answer and the most probable competing token, and identifying systematic patterns in the types of errors the model makes. Results are analyzed separately by relation type (categorical, causal, compositional) and by semantic distance category (near, far), enabling investigation of whether the model's analogical reasoning capabilities are uniform across these dimensions or show systematic differences.

3.4 Representational/Error Analysis

To complement the rank-based accuracy metrics, I analyze the top-10 completions of each experimental condition. Accuracy captures whether the correct completion ranks highly, but it does not reveal what the model produces instead when it fails, nor whether a correct completion reflects genuine extraction of the relational structure. Inspecting the top-10 predictions addresses both questions: when the model fails, the competing completions show what it preferred over the correct answer, and when it succeeds, the surrounding predictions indicate whether the relation was consistently extracted or whether the correct answer simply appears among associatively related alternatives. The prediction analysis was therefore designed with two components: a geometry analysis, an automatic structural criterion based on the geometry of the models' embedding spaces, and an error analysis, a manual inspection of the top-10 predictions. As detailed below, a validation of the geometry analysis on the dataset itself showed that its core assumption does not hold, so this component was omitted and the representational and error analysis rests solely on the error analysis.

Initially, the intention was to do a representational analysis by analyzing the geometry of the terms in the analogies. The idea was to label a prediction as structural when the offset between the prediction and the target, $(\vec{v}_{\text{pred}} - \vec{v}_C)$, is parallel to the source-pair offset $(\vec{v}_B - \vec{v}_A)$, following the parallelogram operationalization of analogical structure discussed in Section 2.2. This criterion presupposes that valid analogies are parallel in the embedding space, meaning that the offset cosine similarity $\cos(\vec{v}_B - \vec{v}_A,\ \vec{v}_D - \vec{v}_C)$ is consistently high for genuine analogies.

Before applying the criterion, I validated this assumption on the analogies in the dataset itself. For each of the 60 analogies, the offset cosine similarity $\cos(\vec{v}_B - \vec{v}_A,\ \vec{v}_D - \vec{v}_C)$ was computed in each model's own embedding space. Contrary to the assumption, the valid analogies were found to be almost entirely non-parallel: the mean offset cosine

similarity was nearing zero for both GPT-2 Small (0.0425) and Llama 3.2 1B (0.0418), with values nearing 0 rather than nearing 1. The parallelogram structure on which the criterion depends can therefore be said to be absent for this dataset within the models' own embedding spaces. The same validation was repeated using pretrained Word2Vec embeddings (Mikolov et al., 2013), the representational space in which the parallelogram structure was originally described. The result was the same: the mean offset cosine similarity for the valid analogies was 0.0561, again nearing zero. The analogies in this dataset are thus not parallel in either the models' embeddings or in Word2Vec.

Because the assumption of parallelism does not hold for these analogies in the embedding spaces described, a systematic structural analysis cannot be meaningfully applied and is therefore omitted. This is consistent with prior work showing that the parallelogram structure of word embeddings holds only for a restricted set of relation types and frequently fails for more abstract or associative relations (Peterson et al., 2020). The error analysis consequently relies on the manual inspection of the top-10 predictions, which is reported in Section 4.2.

3.5 Mechanistic analysis: Activation Patching

To isolate the mechanisms underlying analogical reasoning, the contribution of individual attention heads to the correct prediction is evaluated using activation patching. As explained in Section 2.4.2, the general idea behind activation patching is to determine which components contribute most to restoring the original prediction. In my experiments, this is done by performing a clean run, a corrupted run, and a set of patching runs for each analogy. In the clean run, the model receives the A:B::C:? natural-language prompt. Only analogies for which the model ranks the correct completion token in the top 10 in the clean run are submitted to activation patching, since the recovery metric is only meaningful when the clean run succeeds. In the corrupted run, the B-term is replaced with a corruption token, disrupting the activations that lead to the correct answer. In patching runs, the activations of the corrupted run are used as a basis, as shown in Figure 3.1. Then, for each head in each layer, the activation of that single head is replaced, one by one, with the corresponding activation from the clean run. I measure how much the logit of the correct answer is restored.

The intuition is that heads whose activations restore the clean-run logit of the correct completion token are causally involved in solving the task, whereas heads whose activations have no effect are not. Following the methodological foundations established by Zhang and Nanda (2023), I corrupt the prompt through symmetric token replacement rather than Gaussian noise, and I base the patching metric on logits rather than probabilities. The effect of each head is then quantified with the logit recovery metric defined in Section 3.5.2, a normalized, logit-based measure of how much patching a head restores the clean-run logit of the correct token.

3.5.1 Corruption Procedure

Rather than applying activation patching only once per analogy, I repeat it five times, each time with a different corruption token. A single corruption token may carry its own unique associations or may fail to fully disrupt the relational signal, so averaging the recovery values over five independent corruptions ensures that the resulting map reflects

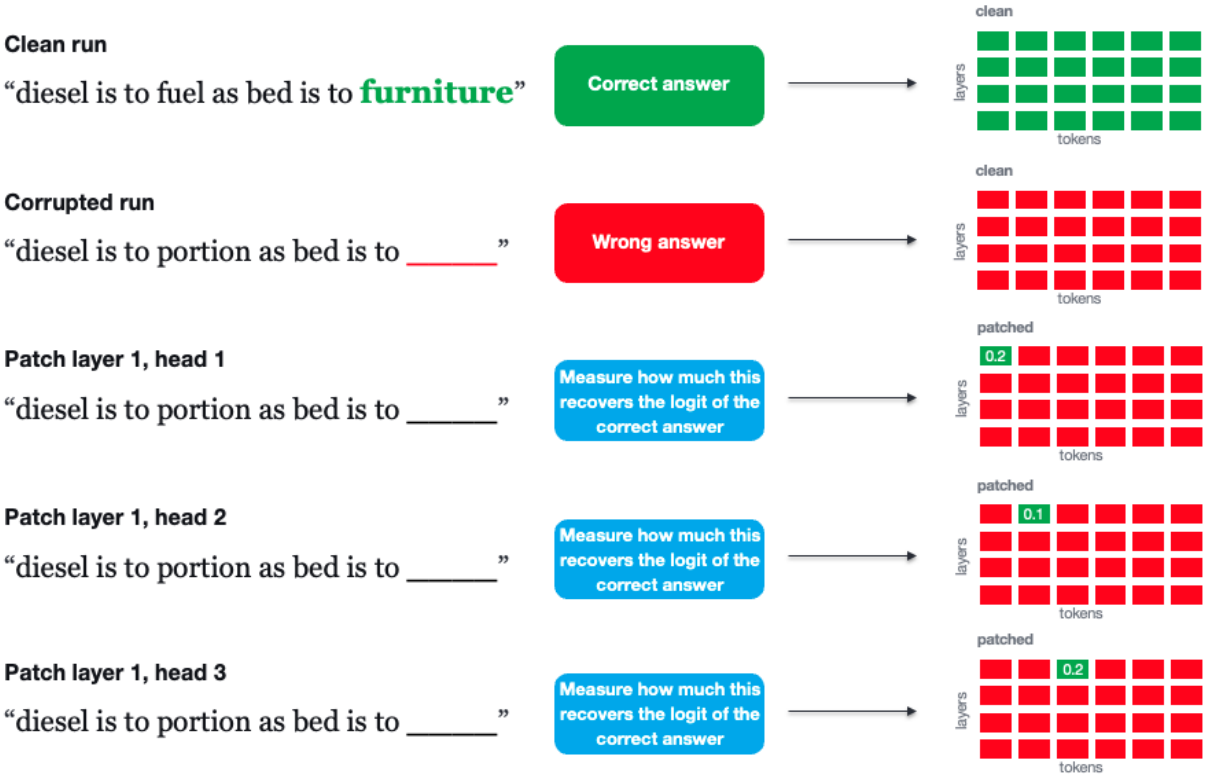


Figure 3.1. Illustration of the activation patching procedure. In the clean run, the model receives the uncorrupted prompt and produces the correct answer; in the corrupted run, the source pair is replaced with an unrelated word, causing the model to produce an incorrect answer. Patching reintroduces the clean activation of a single attention head into the corrupted run and measures whether the correct answer is restored. Each grid represents a model with 4 layers and 6 attention heads per layer, where each block corresponds to one attention head in a specific layer. The four bottom examples show that patching most individual heads leaves the wrong answer unchanged, one head produces only a partial recovery, and a single specific head fully restores the correct answer.

the general effect of corrupting the B-term rather than properties of one particular replacement word.

The five corrupted prompts are generated by replacing the B-term in the clean prompt with five randomly sampled corruption tokens (the procedure for generating these tokens is explained in the next section). The clean prompt takes the form:

```
"{A} is to {B} as {C} is to"
```

Each corrupted prompt takes the form:

```
"{A} is to {corruption_token} as {C} is to"
```

Corrupting the B-term disrupts the relational signal that the model needs to solve the analogy: without a coherent A:B pair, the relation between the source and target pairs cannot be inferred from the prompt. Any attention head that relied on the B-term to support the correct prediction should therefore fail when given the corrupted input. As tokenizers often break longer or less common words into sub-word tokens, I only use the first sub-word token of each corruption word. This sub-word token is then put

into every slot that the B-word occupied in the original prompt. This ensures that the corrupted prompt is exactly the same length as the clean prompt and that every word in the analogy is still in its original position. This is important because patching works by comparing the two runs slot by slot.

Corruption Token Generation

For each analogy, five corruption tokens are generated and stored separately for each model under study, to ensure that the corruption tokens are always drawn from the vocabulary the model itself uses. This per-model generation is necessary because the two models under study tokenize words differently and have different embedding spaces: a token that is a valid, semantically distant noun in one model is not guaranteed to exist, or to be distant, in the other.

Filtering From each model's vocabulary, a candidate set is first constructed by retaining only those tokens that (a) begin with a space character in the tokenizer, indicating a word-initial token, (b) consist entirely of alphabetic characters once the leading space is stripped, (c) begin with a lowercase letter, (d) are at least three characters in length, (e) are identified as nouns by WordNet, meaning they have at least one noun synset in the WordNet lexical database, and do not appear in the standard English stop word list, and (f) the analogy's own four terms (A, B, C, and D) are additionally excluded from the candidate set, so that a corruption token can never coincide with a term from the analogy. Each of these filters serves a specific purpose. The word-initial constraint (a) ensures that the token can directly fill the B-slot in the prompt, matching the leading-space convention used for the analogy terms themselves. Filters (b), (c), and (d) remove punctuation, sub-word fragments, capitalized forms such as proper nouns, and very short tokens that rarely correspond to a meaningful word. The noun and stop word constraint (e) guarantees that each corruption token is a content noun of the same lexical category as the analogy terms, so that replacing the B-term yields a grammatical prompt.

Semantic distance Each corruption token is required to be distant from both A and C. The distances are computed using cosine distance as described in 3.1.1. When A or C is split into multiple sub-word tokens by the tokenizer, its embedding is taken as the mean of the sub-word embeddings, yielding a single vector for the distance computation. Keeping it distant from A, the other term in the source pair, ensures that the corrupted source pair `A:{corruption_token}` has no internal relation, while keeping it distant from C ensures that it entails no relation to the target context either. To satisfy both requirements at once, every candidate is scored by the smallest of its two cosine distances, i.e. if the distance of A to the corruption token is 0.85 and the distance of C to the corruption token is 0.90, the smallest distance (0.85) will be considered the distance score for that specific corruption token. A token therefore receives a high score only when both A and C lie far away in the semantic space.

Candidate pool The five corruption tokens are then drawn at random from a pool of the 200 corruption tokens with the highest distance scores. A pool of candidates is used because the distance distribution varies between models. Consequently, a fixed distance threshold would hold a different meaning in each model's embedding space.

The pool guaranties that every candidate is genuinely distant from both the A and C terms, while the random draw adds lexical variety across the five corruptions. The random draw is performed with a fixed random seed, ensuring that the corruption set is reproducible. All five corruption tokens per analogy are added in Appendix 6, Table 4.

3.5.2 Analysis of the Results

Once the clean, corrupted, and patched runs have been obtained for every analogy, a few further steps are needed before the patching results can be reported. Not every analogy is suitable for patching, so the candidate set is first filtered based on the model's clean-run performance. The head-level patching results are then summarized using two further metrics: logit recovery (Section 3.5.2), which quantifies how much a single head restores the correct prediction, and recovery map correlation (Section 3.5.2), which compares the resulting patterns across conditions or corruption-token samples.

Subset Filtering

As explained above, activation patching is only informative when the clean run actually produces the correct completion token, since the recovery metric (elaborated on in the next section) measures how much patching closes the gap between the logits for this token in the corrupted and clean run. For this reason, the set of analogies submitted to patching is filtered based on the model's clean-run performance. Two filter modes are used in the main experiment: top-1 retains only analogies that the model answered correctly at rank 1, and top-10 retains analogies that the model placed within its top 10 predictions, similar to the top-1 and top-10 accuracy reported in the analogical reasoning task (Section 3.3).

Logit Recovery

Logit recovery is the main metric used in the activation patching experiments. It quantifies how much a specific attention head restores the model's original prediction in the patching phase. It is normalized to the clean-corrupted logit gap. Formally, for the attention head h in layer l, logit recovery is defined as:

$$\text{recovery}(l,h) = \frac{\ell_{l,h}^{\text{patched}}(D) - \ell^{\text{corrupted}}(D)}{\ell^{\text{clean}}(D) - \ell^{\text{corrupted}}(D)} \tag{3.1}$$

where $\ell^{\text{clean}}(D)$ and $\ell^{\text{corrupted}}(D)$ are the logits assigned to the target token D in the clean and corrupted forward passes, and $\ell_{l,h}^{\text{patched}}(D)$ is the logit obtained when the z activation at layer l, head h is replaced with its clean-run value. A recovery of 1.0 indicates that patching the head fully restores the clean-run logit, meaning that head alone is sufficient to account for the difference between the clean and corrupted runs. A recovery of 0.0 indicates that patching has no effect, meaning the head does not contribute to the correct prediction in this context. Values above 1.0 occur when patching pushes the logit beyond the clean baseline: the head not only restores but amplifies the correct prediction. Values below 0.0 occur when patching actively suppresses the logit below the corrupted baseline.

Corruptions whose logit between the clean and corrupted run differs with less than 1.0, are excluded from computing the logit recovery. This is necessary because the recovery metric is normalized by the denominator $\ell^{\text{clean}}(D) - \ell^{\text{corrupted}}(D)$. When a

corruption barely lowers the target logit, the denominator approaches zero and acts as a large multiplier that inflates the logit recovery into extreme and uninterpretable values. This has been applied because initially, a single item produced a recovery of 14.2, far outside the interpretable range described above. Corruptions whose gap falls below a logit of 1.0 are therefore treated as uninterpretable and excluded when averaging recovery across an analogy's five corruptions. Similar to Zhang and Nanda (2023), who use the same metric and also document this instability. The number of excluded corruptions will be reported in the results.

Recovery Map Correlation

To compare the pattern of attention head contributions across experimental conditions or across different corruption-token samples, recovery maps are compared using the Pearson correlation coefficient r. For each map, the layer $\times$ head recovery matrix is serialized into a one-dimensional vector of length $L \times H$ (L layers, H heads per layer), and r is computed between the two resulting vectors. An r close to 1.0 indicates that the same heads dominate in both maps and that their relative contributions are stable; a lower but still positive r indicates that the dominant heads are shared, but their relative contributions differ in magnitude.

3.6 Experimental Conditions

Sections 3.1–3.3 describe the overall experimental setting, indicating the dataset, prompt template, evaluation metrics and corruption-sampling procedure used in all experiments. This general setup is applied in four different experimental conditions: (1) open-vocabulary zero-shot prompting, (2) restricted forced-choice prompting, (3) few-shot prompting, and (4) finetuning. Table 3.2 gives an overview of the four conditions, and the dimension(s) in which each differs from the general experimental setting is described in detail below.

Condition	Models	Manipulation
Open-vocabulary zero-shot	GPT-2 Small, Llama 3.2 1B	Baseline: single open-ended prompt, no demonstrations
Restricted forced-choice	GPT-2 Small, Llama 3.2 1B	Two explicit candidates (target + distractor), varying distractor salience and word order
Few-shot	GPT-2 Small, Llama 3.2 1B	1 or 2 demonstrations, drawn from the same (within) or a different (cross) relation type
Finetuning	GPT-2 Small	Single-pass finetune on a held-out relation-type or distance split

Table 3.2. Overview of the four experimental conditions, the models they are evaluated on, and the dimension along which they manipulate the basic experimental setup.

3.6.1 Open-vocabulary Zero-shot Prompting

The open-vocabulary zero-shot prompting condition establishes the baseline experiment. Each analogy is presented as a single prompt of the form "`{A} is to {B} as {C} is to`", with no demonstrations and no parameter changes. Both models described in Section 3.2, GPT-2 Small and Llama 3.2 1B are evaluated on the full set

of 60 analogies. The random seed controlling corruption-token sampling is fixed at 42, and the patching subset is restricted to top1 and topk.

3.6.2 Restricted Forced-choice Prompting

The restricted condition evaluates the same two models under a forced-choice prompt. This setup mirrors the format in which the dataset was originally used for human participants, where each analogy is presented alongside a fixed set of candidate completions rather than as an open-ended generation task. In the open-ended setting, the correct D-term may not always be recoverable from the A, B and C terms alone, since multiple semantically valid completions can exist for a given analogy. The restricted prompt eliminates this ambiguity by constraining the model's answer to two choices. Therefore, this experiment is added as it resembles the way the original experiment was conducted on human participants (Jones et al., 2022). The full set of 60 analogies is tested four times each, using two different distractors (high-salience distractor, low salience distractor) with the two possible word orders (target word in the first slot, target word in the second slot). This yields 240 forced-choice prompts per model. The template of the prompt is formulated below:

```
"{A} is to {B} as {C} is to either {option1} or {option2}"
```

For example:

```
"fracture is to cast as incision is to either scar or surgery"
```

The model's prediction is counted as correct (`correct_choice`) when it assigns a higher probability to the first token of the target word than to the first token of the distractor.

3.6.3 Few-shot Prompting

To examine whether in-context learning influences analogical reasoning performance, a set of few-shot prompting conditions is evaluated alongside the zero-shot baseline condition that is described in Section 3.6.1. In the few-shot conditions, one or two complete analogy demonstrations are presented preceding the test prompt as shown below:

```
"{A1} is to {B1} as {C1} is to {D1}.
{A2} is to {B2} as {C2} is to {D2}.
{A3} is to {B3} as {C3} is to"
```

For example, in a two-shot within-relation prompt:

```
"physics is to science as ring is to jewelry.
accountant is to profession as carrot is to vegetable.
diesel is to fuel as bed is to"
```

In activation patching, only the last B is replaced with a corruption token.

Four conditions are evaluated in total:

1. **1-shot within-relation**: One demonstration drawn from the same relation category as the test item.

2. **1-shot cross-relation**: One demonstration drawn from a different relation category than the test item.

3. **2-shot within-relation**: Two demonstrations, both drawn from the same relation category as the test item.

4. **2-shot cross-relation**: Two demonstrations, both drawn from a different relation category than the test item.

The within-relation conditions test whether contextually relevant examples that share the same underlying semantic relation aid in retrieving the correct relational pattern for the test item. The cross-relation conditions test whether structural demonstrations from a different semantic domain are sufficient to guide the model, or whether the relation-specific context of the examples matters. Demonstrations are selected using a seeded random procedure (seed=42) from the dataset of 60 analogies to ensure reproducibility. The test item itself is always excluded from the candidate demonstration pool to prevent information leakage. The near/far semantic-distance label is not used in this selection: demonstrations are sampled by relation type alone, so a near test item may receive a far demonstration (and vice versa) within the same relation category. Evaluation proceeds identically to the baseline condition: top-1 and top-10 accuracy are computed for each condition across all 60 analogy items, enabling direct comparison between the conditions and between within- and cross-relation demonstration types. The few-shot experiment is conducted on GPT-2 Small and Llama 3.2 1B.

3.6.4 Finetuning (GPT-2 Small)

As described in the literature review, prior work on teaching analogical relations to language models has focused on encoder-based architectures. Ushio et al. (2021) finetune RoBERTa with a contrastive objective to produce relation embeddings, and Petersen and van der Plas (2023) finetune BERT-base to classify valid analogies, raising accuracy on the Jones et al. (2022) analogy set from 0.53 to 0.69. These studies share two methodological features: they finetune encoder-based models, and they frame analogy as a problem of representing or classifying relations between word pairs. The finetuning experiment in this thesis differs on both points: rather than training an encoder to produce relation embeddings, I finetune a generative model (GPT-2 Small) on the analogy-completion task directly, following the dataset and format of Jones et al. (2022), in order to examine how finetuning affects analogical behavior in a decoder-only model of the kind that the rest of this work analyzes.

Finetuning is performed by doing a single pass over the training data using the AdamW optimizer (Loshchilov and Hutter, 2017) with a learning rate of 1×10^{-4}. Training is conducted in an online fashion with a batch size of one example per gradient update. For each training analogy, a prompt of the form "`{A} is to {B} as {C} is to is to`" is constructed, and the cross-entropy loss is computed over the logit assigned to the first token of D (with a preceding space, consistent with the baseline evaluation). All other token positions are excluded from the loss computation.

Following finetuning, the model is evaluated on the corresponding held-out test split using the same top-1 and top-10 accuracy metrics as the baseline experiment. The finetuned model weights are saved to disk, and activation patching is applied to

the finetuned model using five corruptions per analogy. Comparing the internal mechanisms of both models directly shows whether finetuning reallocates the contribution of attention heads or concentrates it within the same specific layers.

Training and Test Splits

Five different conditions are tested in total: three leave-one-relation-out splits and two semantic-distance reversal splits, as shown in Table 3.3.

Condition	Train set	Test set (held out)
cross_Categorical	Causal + Compositional (40)	Categorical (20)
cross_Causal	Categorical + Compositional (40)	Causal (20)
cross_Comp	Categorical + Causal (40)	Compositional (20)
near_to_far	Near analogies (30)	Far analogies (30)
far_to_near	Far analogies (30)	Near analogies (30)

Table 3.3. Overview of the five finetuning conditions: the three leave-one-relation-out splits train on two relation categories and test on the held-out third; the two distance-reversal splits train on one semantic-distance category and test on the other.

In the three cross-relation conditions, the model is finetuned on all analogies belonging to two of the three relation categories and evaluated on the held-out third. This leave-one-relation-out design assesses whether relational knowledge acquired from two relation types transfers to a structurally distinct third type. In the two cross-distance conditions, the dataset is split by semantic distance rather than by relation type: in the `near_to_far` condition, the model is finetuned on near-distance analogies and evaluated on far-distance analogies, and in the `far_to_near` condition, the split is reversed. These conditions test whether exposure to within-domain relational patterns enables the model to apply the same relational structure across more distant semantic domains, and conversely whether finetuning on far analogies aids generalization to near ones.

For each of the five conditions, a separate copy of GPT-2 Small is finetuned on the training split using the procedure described above, and its weights are saved to disk. The held-out test split is then evaluated under the same prompt template and metrics as the baseline experiment using top-1 and top-10 accuracy. Activation patching is applied to the finetuned weights using the same configuration as the baseline experiment.

3.6.5 Corruption-Token Sensitivity

As described in Section 3.5.1, the corruption tokens used to overwrite the B-term in activation patching are sampled at random from the candidate nouns that are most distant from both the A-term and the C-term embeddings. Because the sample is drawn under a fixed random seed, varying the seed yields a structurally identical but lexically distinct corruption set. To assess the sensitivity of the recovery maps to the corruption tokens, activation patching is repeated under three independent seeds ($\texttt{seed} \in \{42, 123, 456\}$) while all other settings are held identical to the baseline experiment. For each seed, five corruption tokens per analogy are sampled from the model's vocabulary, and the full patching pipeline is run.

Chapter 4

Results

This chapter reports the results of the four experimental conditions defined in Section 3.6 and interprets each finding in place. The chapter is organized around the three sub-questions introduced in Section 2.5. Section 4.1 addresses the behavioral sub-question (SQ1): it reports how accurately GPT-2 Small and Llama 3.2 1B solve the analogies and how prompting format, finetuning, relation type, and semantic distance influence that accuracy. Section 4.2 addresses the representational/error-analysis sub-question (SQ2) through an error analysis of the top-10 predictions, asking whether incorrect predictions preserve the relational structure of the prompt or instead reflect surface-level similarity. Section 4.3 addresses the mechanistic sub-question (SQ3): it reports the activation patching maps for each model and condition and asks which attention heads causally drive the prediction.

4.1 Behavioral Results

SQ1 asks how accurately the two models solve four-term verbal analogies and how that accuracy is shaped by prompting format, finetuning, relation type, and semantic distance. To answer it, I evaluate both models on the 60-item analogy set across all conditions and report rank-based accuracy as described in Chapter 3. The accuracies are reported per experiment: the single-prompt conditions in Table 4.1, the few-shot conditions in Table 4.2, and the GPT-2 finetuning conditions in Table 4.3. Because the models, and Llama in particular, can assign high probabilities to punctuation tokens before producing the correct lexical completion, the reported accuracies are lexical-filtered. For Llama the filter raises accuracy in the single-prompt conditions; for GPT-2 it changes the result only in the restricted forced-choice condition and leaves every other condition unchanged. All accuracies are reported as percentages with 95% Wilson score confidence intervals over the 60 items (240 in the restricted condition, and 20 or 30 in the held-out finetuning splits), as discussed in Chapter 3.

4.1.1 Open-vocabulary zero-shot results

The first experiment conducted under the behavioral research question is the open-vocabulary zero-shot experiment. In this experiment, both Llama and GPT-2 are evaluated using a prompt with an open end and no demonstration. The prompt is formatted like:

`"{A} is to {B} as {C} is to”`

The results in Table 4.1 show that Llama places the correct target within the top-10 for 36.67% of items, against 20.00% for GPT-2. As discussed in Chapter 3, a lexical filter was used to filter Llama's top-1 and top-10 predictions, since Llama often predicted punctuation. Therefore, the results shown in Table 4.1 show Llama's accuracy after this filter is applied. Llama's scores before applying the filter were 11.67% top-1 and 35.00% top-10 accuracy, indicating that punctuation tokens systematically displace the correct completion in approximately 5% of items. Applying the same filter to GPT-2 had no effect.

	GPT-2 Small		Llama 3.2 1B	
Condition	top-1_{lex} [CI]	top-10_{lex} [CI]	top-1_{lex} [CI]	top-10_{lex} [CI]
Open-vocabulary zero-shot (baseline)	**3.33 [0.9, 11.4]**	**20.00 [11.8, 31.8]**	**16.67 [9.3, 28.0]***	**36.67 [25.6, 49.3]***
Restricted forced-choice	2.92 [1.4, 5.9]	16.25 [12.1, 21.4]	0.83 [0.2, 3.0]	14.58 [10.7, 19.6]
Restricted forced-choice, correct-choice metric (chance = 50%):				
Correct-choice	64.17 [57.9, 70.0]		61.67 [55.4, 67.6]	

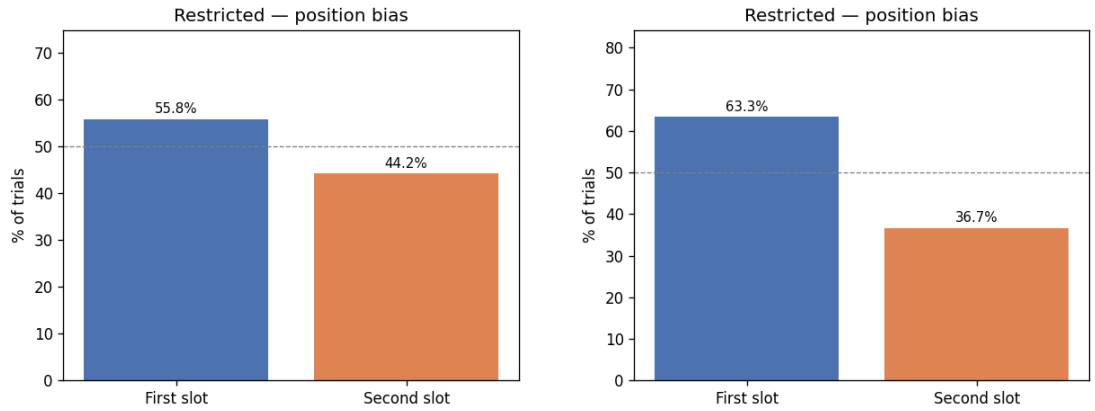
Table 4.1. Accuracy (%) on the single-prompt conditions (open-vocabulary zero-shot and restricted forced-choice) for GPT-2 Small and Llama 3.2 1B, with 95% Wilson score confidence intervals. All values are lexical-filtered. Bold marks the highest value in each column. An asterisk (*) marks a value whose 95% confidence interval does not overlap those of the other values in the same column, which corresponds to a statistically significant difference at the 0.05 level. This is not a formal significance test, but a significance conservatively inferred from non-overlapping 95% Wilson intervals.

4.1.2 Restricted forced-choice results

The second experiment conducted under the behavioral research question is the restricted forced-choice experiment. In this experiment, both Llama and GPT-2 are evaluated using a prompt that provides two explicit candidate words instead of an open end, similar to how the original experiment of Jones et al. (2022) was designed for humans. The prompt is formatted like:

```
"{A} is to {B} as {C} is to either {option 1} or {option 2}"
```

The results in Table 4.1 show that the rank-based accuracy in the restricted forced-choice condition is numerically lower than in the open-vocabulary condition, although the confidence intervals of the two conditions overlap and this difference is therefore not significant by the convention used here. Both conditions nevertheless fall far below 100%, suggesting that the models barely solve the task. This low performance stems from the autoregressive nature of both models, which output a probability distribution over their entire vocabulary rather than restricting their predictions to the two options provided in the prompt. Consequently, the rank-based top-1 accuracy, which evaluates whether the gold target is the absolute most probable token across the full vocabulary, remains remarkably low (2.92% for GPT-2 and 0.83% for Llama). To address this limitation, a correct-choice metric was introduced to measure whether the target is simply ranked higher than the distractor in the predicted probability distribution.



(a) GPT-2 Small **(b)** Llama 3.2 1B

Figure 4.1. Rate of selecting the first-position candidate in the restricted forced-choice condition, broken down by whether the gold target occupies the first or second slot, for (a) GPT-2 Small and (b) Llama 3.2 1B.

Two additional patterns characterize the restricted condition. First, both models show a positional bias, preferring whichever candidate appears first in the prompt regardless of whether that candidate is the gold target. Figure 4.1 shows this pattern for both models. Crucially, when the gold target occupies the first slot, GPT-2 selects the first-position candidate in 70.0% of trials; when the gold target occupies the second slot instead, GPT-2 still selects the first-position candidate (now the distractor) in 41.7% of trials. Llama shows the same pattern (75.0% vs. 51.7%), with a slightly smaller gap between the two slots (23.3%, against 28.3% for GPT-2). This indicates that prompt order itself influences the model's choice, suggesting that the restricted-choice format introduces a positional bias that should be acknowledged, when interpreting the correct-choice accuracy.

Second, distractor salience modulates performance. As described in the methodology, the analogy dataset of Jones et al. (2022) contains low- and high-salience distractors to test the influence of distractor type on the choice made by participants (or in this case models). Both models replicate the salience effect originally reported by Jones et al. (2022) in human participants: they are more easily misled when the distractor is semantically closer to the source pair as low-salience distractors are easier to reject than high-salience ones (GPT-2: 72.5% vs. 55.8%; Llama: 65.8% vs. 57.5%). This human-like sensitivity to distractor salience emerges in both models despite their weak rank-based accuracy. This suggests that both models track some notion of semantic plausibility, even when they fail to rank the correct target highest.

4.1.3 Few-shot results

The third experiment conducted under the behavioral research question is the few-shot experiment. In this experiment, both Llama and GPT-2 are evaluated using a prompt that includes one or two complete analogy demonstrations before the test prompt, in order to test the influence of in-context learning. The demonstrations are drawn either from the same relation type as the test item (within-relation) or from a different relation type (cross-relation), which yields four conditions in total.

Within-relation few-shot

In the within-relation conditions, the demonstrations share the relation type (categorical, compositional, or causal) of the test item. The prompt is formatted like:

```
"{A1} is to {B1} as {C1} is to {D1}
{A2} is to {B2} as {C2} is to {D2}
{A3} is to {B3} as {C3} is to"
```

The results in Table 4.2 show that within-relation demonstrations are associated with Llama's highest observed accuracy across all conditions in this thesis: the 1-shot within-relation condition reaches 23.33% top-1 accuracy (CI 14.4–35.4) and 50.00% top-10 accuracy (CI 37.7–62.3). This increase over the zero-shot baseline (16.67% top-1, CI 9.3–28.0) falls within the baseline's confidence interval, however, so it should be read as the strongest numerical pattern observed rather than a statistically confirmed effect. Adding a second within-relation demonstration produces no further gain, suggesting that one matching example is sufficient for Llama to extract the relevant relation type and that additional demonstrations beyond this point yield no measurable benefit.

Cross-relation few-shot

Cross-relation demonstrations, by contrast, are numerically lower than the zero-shot baseline (8.33% top-1 for 1-shot cross-relation; 6.67% for 2-shot cross-relation), though as Table 4.2 notes, these differences fall within the baseline's confidence interval and should be read as a suggestive rather than an established pattern. If this pattern holds, it would indicate that, for Llama, in-context examples improve analogical reasoning only when the demonstrated relation matches that of the test item, with a mismatched relation type misleading rather than simply failing to help. This is in line with the Structure-Mapping Theory (Gentner, 1983), according to which analogical reasoning proceeds by aligning the relational structure of a base example with that of the target. A cross-relation demonstration supplies a base structure (e.g., a causal relation) that does not match the target item's structure (e.g., a categorical relation). Under the Structure-Mapping Theory, the model is expected to carry this mismatched relational structure over into the target, producing a systematically wrong completion. This would be consistent with cross-relation demonstrations numerically depressing Llama's accuracy rather than simply leaving it unchanged, although, as noted above, this pattern does not reach significance given the overlapping confidence intervals.

GPT-2 Small, in contrast, shows no reliable benefit from demonstrations of either kind. Its highest few-shot score is a modest increase to 5.00% top-1 accuracy in the 1-shot cross-relation condition, up from a zero-shot baseline of 3.33%. This increase falls within the confidence interval of the baseline, so I cannot confidently attribute it to the demonstration itself. The within-relation demonstrations that most benefit Llama are, in fact, slightly harmful: they leave GPT-2 at 1.67% top-1 accuracy, below its own zero-shot baseline. The two models therefore diverge sharply under the same manipulation, as the within-relation demonstration that helps Llama produces no measurable effect in GPT-2. This contrast is consistent with the view that reliable in-context learning depends on model scale and on the induction-head machinery that emerges with it (Olsson et al., 2022), a capacity that GPT-2 Small, at 124M parameters, appears to lack for this task.

	GPT-2 Small		Llama 3.2 1B	
Condition	top-1 [CI]	top-10 [CI]	top-1 [CI]	top-10 [CI]
Baseline (open-vocabulary zero-shot)	3.33 [0.9, 11.4]	**20.00 [11.8, 31.8]**	16.67 [9.3, 28.0]	36.67 [25.6, 49.3]
1-shot within-relation	1.67 [0.3, 8.9]	16.67 [9.3, 28.0]	**23.33 [14.4, 35.4]**	**50.00 [37.7, 62.3]**
1-shot cross-relation	**5.00 [1.7, 13.7]**	11.67 [5.8, 22.2]	8.33 [3.6, 18.1]	36.67 [25.6, 49.3]
2-shot within-relation	1.67 [0.3, 8.9]	13.33 [6.9, 24.2]	20.00 [11.8, 31.8]	**50.00 [37.7, 62.3]**
2-shot cross-relation	3.33 [0.9, 11.4]	10.00 [4.7, 20.2]	6.67 [2.6, 15.9]	31.67 [21.3, 44.2]

Table 4.2. Accuracy (%) on the few-shot conditions for GPT-2 Small and Llama 3.2 1B, with 95% Wilson score confidence intervals. The results are not lexical-filtered as this changes neither model's accuracy. Bold marks the highest value in each column. An asterisk (*) would mark a value whose 95% confidence interval does not overlap those of the other values in the same column, corresponding to a statistically significant difference at the 0.05 level; in this table all intervals overlap, so no value is marked. This convention is not a formal significance test, but a significance conservatively inferred from non-overlapping 95% Wilson intervals.

4.1.4 Finetuning results

The fourth and final experiment conducted under the behavioral research question is the finetuning experiment, which is applied to GPT-2 Small only. Where the previous conditions probe what the pretrained model already does, this experiment asks whether the analogical task can be taught explicitly from a small number of examples. I finetune a separate copy of GPT-2 Small for each split with a single pass over the training data, using the AdamW optimizer (Loshchilov and Hutter, 2017).

The results in Table 4.3 show that finetuning does not teach GPT-2 the task but instead degrades it. Every split collapses top-1 accuracy to zero, and top-10 accuracy reaches at most 13.33% (CI 5.3–29.7) in the far → near split, with the remaining splits at or near zero. These scores fall below GPT-2's own zero-shot baseline of 3.33% top-1 and 20.00% top-10 (Table 4.1), which indicates that a single-pass finetune on 30 to 40 analogies overwrites useful pretrained structure rather than instilling a relational skill, a pattern consistent with catastrophic interference on such a small training set.

Split	top-1 [CI]	top-10 [CI]
Baseline (open-vocabulary zero-shot)	**3.33 [0.9, 11.4]**	**20.00 [11.8, 31.8]**
cross-Categorical	0.00 [0.0, 16.1]	0.00 [0.0, 16.1]
cross-Causal	0.00 [0.0, 16.1]	5.00 [0.9, 23.6]
cross-Compositional	0.00 [0.0, 16.1]	0.00 [0.0, 16.1]
near → far	0.00 [0.0, 11.3]	0.00 [0.0, 11.3]
far → near	0.00 [0.0, 11.3]	13.33 [5.3, 29.7]

Table 4.3. GPT-2 Small accuracy (%) on the finetuning conditions, with 95% Wilson score confidence intervals. Llama 3.2 1B was not finetuned. Bold marks the highest value in the top-10 column; the top-1 column is zero throughout. An asterisk (*) would mark a value whose 95% confidence interval does not overlap those of the other values in the same column, corresponding to a statistically significant difference at the 0.05 level; in this table all intervals overlap, so no value is marked. This convention is not a formal significance test, but a significance conservatively inferred from non-overlapping 95% Wilson intervals.

This outcome contrasts with the finetuning literature reviewed in Section 2.3, which reports that the relational pattern can be learned from relatively little data. Ushio

et al. (2021) finetuned RoBERTa with a contrastive objective to produce relation embeddings, and Petersen and van der Plas (2023) finetuned BERT-base to classify valid analogies, raising accuracy on the Jones et al. (2022) set from 0.53 to 0.69. Both studies, however, share two features that the present experiment does not: they finetune encoder models, and they frame analogy as representing or classifying the relation between a word pair. The experiment here instead finetunes a generative decoder on the analogy-completion task directly, optimizing the next-token logit of D rather than a relation representation. This divergence suggests that the small-data learnability reported in that literature is tied to the relation-embedding or classification framing, and does not transfer to a decoder-only model trained on the generative completion objective.

4.1.5 Accuracy across relation types and semantic distances

Beyond the overall accuracy reported above, SQ1 also asks whether analogical performance is uniform across the different properties of the analogies or varies with them. To examine this, I segment the open-vocabulary zero-shot predictions using the two categorical labels copied from Jones et al. (2022) during preprocessing: the relation type of each analogy (categorical, causal, or compositional, 20 items each) and its semantic-distance label (near or far, 30 items each). Because the near/far split was originally derived from LSA distances rather than from the models studied here, I first confirm that it also holds in each model's own embedding space (Figure 4.2) before turning to the accuracy breakdown. Accuracy is then recomputed within each subgroup using the same top-1 and top-10 metrics as before. As explained in Section 4.1.1, the GPT-2 values are unaffected by the lexical filter and the Llama values are lexical-filtered.

Near/far partition

In Section 3.1.1 I noted that the near/far partition was originally constructed from LSA distance scores by Jones et al. (2022), and asked whether this separation also holds when cosine distance is computed per model, a more modern and common technique for measuring semantic distance between word embeddings. To confirm this, I compare the cross-pair cosine distances (`dist_AB_CD`) of the two groups in both models. The partition holds: for GPT-2 the mean cosine distance between the summed pair embeddings is 0.61 for far analogies against 0.43 for near analogies, and for Llama the corresponding values are 0.88 and 0.71. Figure 4.2 shows the full distributions behind these means: in both models the far box sits clearly above the near box and the medians are well separated. The far group has a significantly higher cross-pair cosine distance than the near group in both models (Mann–Whitney U, $p < .001$). The LSA-based near/far partition therefore reflects a statistically reliable difference in the models' own embedding spaces, and the original labels are retained in all subsequent analyses.

Results

The results in Table 4.4 reveal two patterns. First, near analogies are consistently easier than far analogies for both models. For Llama, near accuracy is approximately twice the far accuracy (50.00% vs. 23.33% top-10), and for GPT-2 far analogies are barely solved within the top-10 at all (10.00%, against 30.00% for near). This near-over-far advantage is exactly what the literature predicts: models reliably recognize near

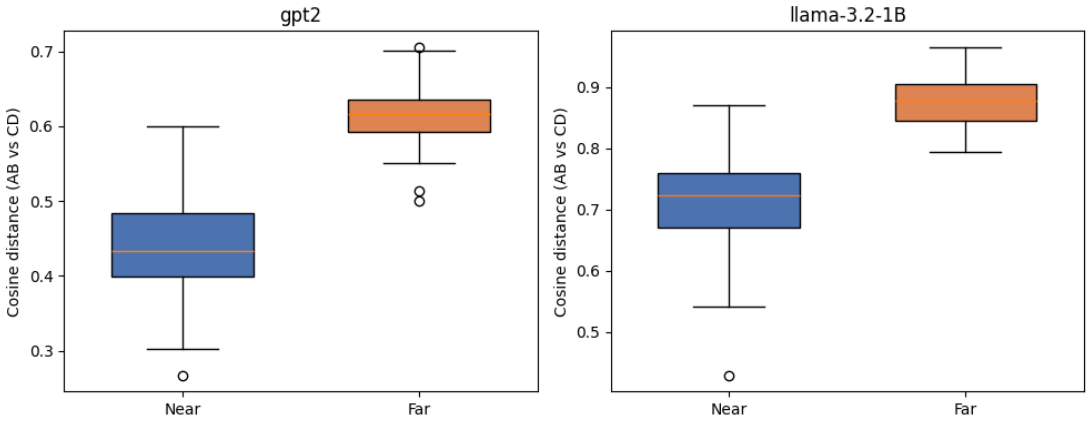


Figure 4.2. Boxplot for near and far analogies based on cosine distance

analogies but struggle with far ones that require relational transfer across dissimilar domains (Sourati et al., 2024; Jiayang et al., 2023). The same drop is reported for both humans (Jones et al., 2022) and GPT-3 (Webb et al., 2023) on this dataset. The pattern is consistent with a reliance on surface-level similarity: when the source and target pairs are drawn from similar domains, lexical and associative overlap can carry the prediction, whereas far analogies remove that overlap and expose whether the model has aligned the underlying relational structure, which both models largely fail to do.

Second, the relation-type pattern differs across models. GPT-2 shows a slight causal advantage at top-1 (10.00%, against 0.00% for the other two types), and a categorical advantage at top-10 (30.00%, against 15.00% for the other two types). Llama is most accurate on categorical analogies (top-10_{lex} 55.00%, against 25.00% causal and 30.00% compositional). Llama's categorical advantage is the cleaner of the two effects, and categorical is likely the easiest relation type as taxonomic co-occurrence in the training corpus most directly supports the completion. GPT-2's top-1 causal advantage, by contrast, rests on a single relation type reaching 10.00% while the others sit at zero, so given the 20-item subgroups it should not be over-interpreted. Taken together, the relation-type results indicate that neither model solves the three relations uniformly, but the small subgroup sizes preclude a strong claim about which relation is intrinsically hardest.

In conclusion, SQ1 is answered as follows. Llama 3.2 1B outperforms GPT-2 Small in the open-vocabulary zero-shot and few-shot conditions, consistent with the expectation that in-context analogical performance scales with model size (Webb et al., 2023); in the restricted forced-choice condition, however, this pattern reverses, with GPT-2 numerically ahead of Llama on every metric in Table 4.1 (though, as discussed above, not significantly). Neither model approaches ceiling in any condition. The rank-based accuracies remain low throughout, and the restricted forced-choice format does not raise them: under the more lenient correct-choice metric both models exceed chance, but even then they display a positional bias and a sensitivity to distractor salience. Of the remaining manipulations, the highest observed accuracy occurs under within-relation few-shot prompting for Llama, though given the overlapping confidence intervals reported in Table 4.2 this should be read as the strongest numerical pattern rather than a confirmed effect; cross-relation prompting, finetuning, and the restricted format all leave accuracy at or below the zero-shot baseline. The two structural properties of the

		GPT-2 Small		Llama 3.2 1B	
Split	Level	top-1_{lex} [CI]	top-10_{lex} [CI]	top-1_{lex} [CI]	top-10_{lex} [CI]
Baseline	(all 60 items)	3.33 [0.9, 11.4]	20.00 [11.8, 31.8]	16.67 [9.3, 28.0]	36.67 [25.6, 49.3]
Relation	Categorical	0.00 [0.0, 16.1]	**30.00 [14.5, 51.9]**	**25.00 [11.2, 46.9]**	**55.00 [34.2, 74.2]**
	Causal	**10.00 [2.8, 30.1]**	15.00 [5.2, 36.0]	15.00 [5.2, 36.0]	25.00 [11.2, 46.9]
	Compositional	0.00 [0.0, 16.1]	15.00 [5.2, 36.0]	10.00 [2.8, 30.1]	30.00 [14.5, 51.9]
Distance	Near	**6.67 [1.8, 21.3]**	**30.00 [16.7, 47.9]**	**23.33 [11.8, 40.9]**	**50.00 [33.2, 66.8]**
	Far	0.00 [0.0, 11.4]	10.00 [3.5, 25.6]	10.00 [3.5, 25.6]	23.33 [11.8, 40.9]

Table 4.4. Open-vocabulary zero-shot accuracy (%) over all 60 items (baseline) and broken down by relation type and by semantic distance, with 95% Wilson score confidence intervals. The subgroup accuracies decompose the baseline (each block averages to it). Bold marks the highest value in each column, within the relation and distance blocks separately (the baseline row is a reference and is not bolded). An asterisk (*) would mark a value whose 95% confidence interval does not overlap those of the other values in the same column and block, corresponding to a statistically significant difference at the 0.05 level; in this table all intervals overlap, so no value is marked. This convention is not a formal significance test, but a significance conservatively inferred from non-overlapping 95% Wilson intervals.

analogies modulate accuracy in opposite ways: near analogies are reliably easier than far analogies for both models, in line with the documented difficulty of cross-domain relational transfer, while the relation-type effect is smaller and model-specific. Taken together, both models solve the task only partially and appear to rely more on surface-level similarity than on the relational structure of the analogies, a tension that the representational analysis in the next section examines directly.

4.2 Error Analysis Results

SQ2 asks whether the models, when their top-ranked prediction is not the gold target, nevertheless preserve the relational structure of the prompt, which would indicate a partial representation of the A:B::C:D mapping rather than a random failure. The rank-based metrics in Section 4.1 report only whether the gold target lies in a model's top-1 or top-10; they do not reveal whether the competing predictions are random noise or whether they preserve the relational structure of the prompt. As explained in Section 3.4, I initially intended to answer this with an automatic structural criterion, labeling a prediction as structural when the offset ($\vec{v}_{\text{pred}} - \vec{v}_C$) is parallel to the source-pair offset ($\vec{v}_B - \vec{v}_A$). The validation reported there showed, however, that the analogies in this dataset are not parallel in either the models' embedding spaces (mean offset cosine 0.0425 for GPT-2 Small and 0.0418 for Llama 3.2 1B) or in Word2Vec (0.0561). This is consistent with prior work showing that the parallelogram structure holds only for a restricted set of relation types and fails for more abstract or associative relations (Peterson et al., 2020). The structural criterion could therefore not be applied, and the representational analysis instead rests on a manual inspection of the top-10 next-token predictions for each analogy in each model, classifying what the model produces when it does not return the target.

Table 4.5 reports the predictions for a representative subset of analogies that illustrate the recurring patterns; the full top-10 predictions for all 60 analogies are listed in Table 5 (Appendix D). The analogy indices match those of Table 2. Across both models, the dominant pattern is that the top predictions are similar at the surface level

to the analogy inputs rather than words that resolve the relational structure.

#	Analogy	GPT-2 Small top-10	Llama 3.2 1B top-10
2	calf : animal :: bull : mammal	dog, human, man, horse, sheep, bird, cow, cat, animal, cattle	–, what, ?, cow, ., ------, :, ., -----, (
5	physics : science :: ring : jewelry	physics, science, mathematics, astronomy, geometry, math, chemistry, the, music, magic	–, music, a, what, the, –, circle, . . . , . . . , mathematics
6	caffeine : drug :: rose : flower	wine, water, alcohol, poison, tea, caffeine, coffee, sugar, medicine, weed	–, what, **flower**, fruit, . . . , . . . , a, . . . , . . . , .
7	squirrel : rodent :: robin : bird	cat, squirrel, rat, **bird**, dog, monkey, rabbit, wolf, snake, rodent	**bird**, –, what, . . . , ?, ?, . . . , :, . . . , ------
11	diesel : fuel :: bed : furniture	bed, fuel, be, sleep, cook, feed, eat, fill, fire, make	–, bed, what, mattress, . . . , :, . . . , ?, __, .
17	wrench : tool :: sad : mood	be, work, use, make, do, die, find, tool, learn, live	–, what, . . . , ., happy, . . . , ------, tool, . . . , . . .
20	doll : toy :: turkey : poultry	toy, play, eat, be, chew, make, dog, go, fish, take	Christmas, Thanksgiving, bird, food, football, thank, holiday, meat, chicken, turkey
30	defeat : frown :: victory : smile	**smile**, laugh, frown, be, look, cry, stand, walk, sit, grin	**smile**, grin, laugh, a, raise, cheer, be, smirk, break, lift
34	sadness : crying :: genes : baldness	crying, being, feeling, laughing, thinking, the, screaming, eating, having, getting	genes, DNA, cells, dna, chromosomes, bacteria, the, blood, evolution, life
36	trout : fish :: lobster : crustacean	fish, eat, swim, crab, lobster, shrimp, fry, pick, be, fly	lobster, cooking, seafood, shell, crab, food, the, –, bait, –
40	period : cramps :: wind : erosion	the, rain, be, a, fall, blow, wind, snow, get, break	menstr, a, pregnancy, the, menstrual, childbirth, diarrhea, men, women, period
43	brass : trophy :: bronze : medal	silver, gold, bronze, trophy, platinum, diamond, be, crown, **medal**, win	**medal**, gold, –, sculpture, the, statue, medals, a, bronze, what

Table 4.5. Top-10 next-token predictions for a representative subset of analogies under the open-vocabulary zero-shot condition, for GPT-2 Small and Llama 3.2 1B. Rank 1 is the highest-probability prediction; the correct target is shown in bold where it appears. Empty-string tokens are shown as "–" and repeated punctuation as ". . . ". The full set for all 60 analogies is given in Table 5, Appendix D.

Repetition of analogy components

For GPT-2 Small, the most common pattern is the direct repetition of words already present in the prompt. In analogy 11 (*diesel : fuel :: bed : furniture*), the two highest-ranked predictions are *bed* and *fuel*, reproducing prompt words rather than completing the relation. This repetition is partly a consequence of the present evaluation setup: unlike the original parallelogram experiment of Mikolov et al. (2013), which excludes the analogy's own three terms from the candidate set, I score the full vocabulary, so the prompt words remain eligible completions and are frequently returned. This is the same failure mode that Linzen (2016) identified for the offset method, where the nearest vector is often simply one of the input words unless they are manually excluded. It is notable that a contextual model such as GPT-2 reproduces the behavior originally documented for static word embeddings, even though it has access to the full prompt rather than to isolated word vectors. In analogy 12 (*accountant : profession :: carrot : vegetable*), the top prediction is *carrot*, the input word C.

GPT-2 Small

Beyond repetition, two further patterns recur in GPT-2's predictions. First, the model frequently fixates on the semantic field of the source pair $A{:}B$ and never addresses the target pair at all. In analogy 5 (*physics : science :: ring : jewelry*), the top predictions are all sciences (*physics, science, mathematics, astronomy, geometry, math, chemistry*), and the target *jewelry* falls to rank 271. The same occurs in analogy 6 (*caffeine : drug :: rose : flower*), where the predictions (*wine, water, alcohol, poison, tea, caffeine, coffee*) elaborate the drink-and-drug field of the source pair while *rose* and *flower* are ignored. The top prediction *wine*, however, suggests that the model may have read *rose* in the sense of rosé wine rather than as a flower, which likely causes these predictions.

Second, on the categorical analogies the model often retrieves related hyponyms of the prompt words rather than the hypernym that the relation requires. In analogy 7 (*squirrel : rodent :: robin : bird*), GPT-2 ranks the small-animal neighbors *cat*, *squirrel*, and *rat* above the correct *bird*, which appears only at rank 4. In analogy 2 (*calf : animal :: bull : mammal*) it lists other mammals (*dog, human, horse, sheep, cow*) instead of the category label *mammal* (rank 42). Analogy 36 (*trout : fish :: lobster : crustacean*) combines both effects described: GPT-2 repeats the source hypernym *fish* at rank 1 and then lists related hyponyms of the target (*crab, lobster, shrimp*) without ever producing *crustacean*. These predictions show that the model captures the broad semantic field of the analogy but not the taxonomic level at which the relation operates.

The pattern becomes more pronounced in the far analogies, where the predictions collapse onto generic context words. In analogy 17 (*wrench : tool :: sad : mood*), the ten highest-ranked tokens (*be, work, use, make, do, die, find, tool, learn, live*) reflect either the tool-use context of the first pair or the sad context of the second pair. This breakdown on the far items mirrors the behavioral near-over-far gap reported in Section 4.1 and the broader finding that models recognize near analogies but fail on far ones that require relational transfer across dissimilar domains (Sourati et al., 2024; Jiayang et al., 2023). There are nevertheless cases where the model appears to capture the relational structure. In analogy 30 (*defeat : frown :: victory : smile*), the top prediction is *smile*, the correct target, and in analogy 31 (*motion : sickness :: drought : famine*), *famine* appears at rank 3. These correct predictions occur mostly in the near causal analogies, where the causal relation is lexically close to the surface form of the prompt, so it remains ambiguous whether they reflect relational mapping or close surface-level similarity.

Llama 3.2 1B

Llama 3.2 1B shows a qualitatively different failure mode. A large proportion of its top-ranked predictions are non-content tokens: empty strings, punctuation marks (., :, ?), repeated punctuation, and underscore tokens. In analogy 2 (*calf : animal :: bull : mammal*), eight of the ten top predictions are non-word tokens, which suggests that the model first assigns probability to formatting tokens before content words. This punctuation behavior, which the lexical filter in Section 4.1 was designed to absorb, is examined further in the Discussion.

Where Llama does produce content words, it often generates a coherent set from the correct domain that still misses the target. For analogy 19 (*apartment : shelter :: tea : beverage*), it produces *drink, refresh, comfort, tea, coffee, brew*, and for analogy 24 (*illness : treatment :: drug : relief*), the predictions *cure, medicine, disease, treatment,*

therapy, medication are all relevant to the domain, even though the target *relief* does not appear. More tellingly, Llama frequently fixates on a single prompt word and returns a tight cluster of its associates rather than the relation as a whole. In analogy 34 (*sadness : crying :: genes : baldness*), *genes* dominates and the predictions become a genetics cluster (*DNA, cells, chromosomes*) with no trace of *baldness* (rank 2662), and in analogy 40 (*period : cramps :: wind : erosion*), *period* triggers a menstruation cluster (*menstrual, pregnancy, childbirth*) while *wind* and *erosion* are ignored entirely. The clearest case is analogy 20 (*doll : toy :: turkey : poultry*), where the top predictions *Christmas*, *Thanksgiving*, *bird*, *food*, and *football* show that *turkey* activates cultural associations rather than the categorical relation. This single-word association effect mirrors the finding of Johnson et al. (2025), whose C-only prompt showed that models can solve up to 62% of analogies from the C-term alone, without access to the A:B relation. Llama does, however, resolve the relation more often than GPT-2: in analogy 43 (*brass : trophy :: bronze : medal*), GPT-2 lists metals (*silver, gold, bronze, platinum*) and reaches *medal* only at rank 9, whereas Llama returns *medal* at rank 1. Such cases, in which the larger model moves from surface-level association to the correct relational completion, are consistent with its higher behavioral accuracy in Section 4.1.

Taken together, the qualitative inspection answers SQ2: neither model reliably represents the relational structure of the analogy. Instead, both rely primarily on surface-level similarity, though in different ways. GPT-2 Small repeats prompt words, elaborates the semantic field of the source pair, and returns related hyponyms instead of the hypernym that the categorical relation requires. Llama 3.2 1B mixes domain-appropriate content words and clusters associated with single words together with non-content formatting tokens. This is consistent with the low open-vocabulary top-1 accuracy and with the offset-parallelism validation in Section 3.4, which found the source- and target-pair offsets to be nearly orthogonal rather than parallel for these analogies (mean cosine similarity 0.0425 for GPT-2 Small and 0.0418 for Llama 3.2 1B). The result speaks to the recurring tension in the analogy literature that strong or weak performance alone cannot reveal which strategy a model used. In this analysis, the top-10 predictions make the strategy visible, and for both models it is dominated by surface-level similarity rather than relational structure. This observational evidence cannot, however, establish which internal components produce the prediction, the question taken up by the mechanistic analysis in the next section.

4.3 Mechanistic Results

SQ3 asks which attention heads causally drive the analogical prediction in each model, and how stable that head-level structure is across conditions, finetuning, and resamplings of the corruption tokens. This section reports the activation patching results, as defined in Section 3.5. Section 4.3.2 reports the maps for the open-vocabulary zero-shot baseline. Section 4.3.3 compares them with the maps obtained under the best-performing condition for each model. Section 4.3.4 reports the robustness of the maps to the choice of corruption tokens. Section 4.3.5 aggregates across conditions and across the relation and distance breakdowns; the per-relation and per-distance heatmaps themselves are reported in the appendix (Figures 1 and 2).

4.3.1 Interpreting Recovery Scores

Before turning to the results, it is useful to be clear about what a recovery score can and cannot show. In activation patching, the model's activations are computed on both a clean and corrupted prompt. In patching, the activations of the corrupted prompt are taken as basis and one head at a time is restored to its clean-run value. The resulting logit for the target token is compared to the same token's logit under the corrupted and clean prompt to compute the logit recovery score (Section 3.5.2). A high recovery score for a head shows that restoring this single head, on its own, brings back a large part of the correct prediction (Heimersheim and Nanda, 2024).

A high score does not, however, show that this head is the only one involved. As (Wang et al., 2022; Yang et al., 2025; Eshuijs et al., 2025) show, a model's behavior is usually produced by several heads working together rather than by a single component. Because of this, even a head that plays a crucial role is unlikely to reach a recovery of 1.0 on its own. The reverse also holds: a head with a low score is not necessarily unimportant, since its contribution may only show up once it is combined with other heads, something that single-head patching cannot capture (Heimersheim and Nanda, 2024). The maps below should therefore be read as showing where a single head can most easily restore the model's behavior, not as proof that the heads found here are the complete set of components responsible for the prediction. Confirming that would require additional techniques, such as path patching, that trace how heads pass information to one another, which is beyond the scope of this analysis (Wang et al., 2022; Conmy et al., 2023).

Because the logit recovery score divides by the clean–corrupted logit gap (Section 3.5.2), it is only meaningful when that gap is clearly different from zero. If a corruption barely changes the target logit, the denominator approaches zero, and the recovery score no longer behaves as an interpretable fraction between roughly 0 and 1: dividing by a near-zero number instead inflates the score to extreme, uninterpretable values. To avoid this, a corruption is excluded from an analogy's average whenever its clean–corrupted logit gap falls below 1.0, and the analogy itself is dropped from the patching results once none of its five corruptions reaches this threshold, since no usable recovery score can then be computed for it at all. This filter is applied on top of the rank-based candidate selection described in the Methodology. Table 4.6 reports, for every condition discussed below, exactly how many analogies were candidates for patching under the `topk` filter and how many of these survive the logit-gap threshold.

4.3.2 Baseline (Open-vocabulary Zero-shot)

Figure 4.3 shows the GPT-2 Small recovery map under the open-vocabulary zero-shot condition and Table 4.7 reports the highest ranked attention heads. A single mid-layer head, L8H6, leads the ranking under both the `topk` and `top1` filters. The next-ranked heads have a progressively lower recovery. This drop-off indicates that, although several mid-layer heads contribute, one head plays a large role relative to the rest. Additionally, the by-layer view in Figure 4.3 demonstrates that recovery is concentrated in layers 5 through 8, while the earliest and latest layers contribute almost nothing. This suggests that the relevant computation for completing an analogy takes place roughly midway through the network. This confirms the hypothesis stated in Section 2.5 that the recovery signal would concentrate in mid-layer attention heads, consistent with the intermediate-layer symbolic-induction stage proposed by Yang et al.

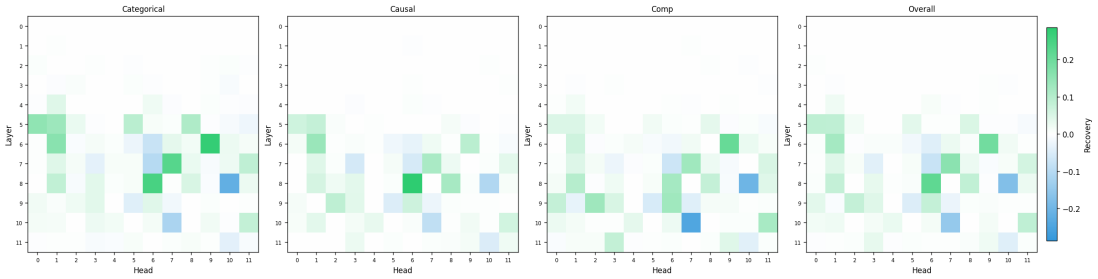
Condition	Model	Candidates	Retained	Dropped
Baseline (open-vocabulary zero-shot)	GPT-2 Small	12	12	0
Baseline (open-vocabulary zero-shot)	Llama 3.2 1B	21	21	0
1-shot within-relation	GPT-2 Small	10	7	3
1-shot within-relation	Llama 3.2 1B	30	29	1
1-shot cross-relation	GPT-2 Small	7	7	0
1-shot cross-relation	Llama 3.2 1B	22	22	0
2-shot within-relation	GPT-2 Small	8	7	1
2-shot within-relation	Llama 3.2 1B	30	29	1
2-shot cross-relation	GPT-2 Small	6	6	0
2-shot cross-relation	Llama 3.2 1B	19	18	1
Finetuning: cross-Categorical	GPT-2 Small	0	–	–
Finetuning: cross-Causal	GPT-2 Small	1	1	0
Finetuning: cross-Compositional	GPT-2 Small	0	–	–
Finetuning: near $\rightarrow$ far	GPT-2 Small	0	–	–
Finetuning: far $\rightarrow$ near	GPT-2 Small	4	3	1

Table 4.6. Number of analogies submitted to activation patching under the `topk` filter (Candidates), and how many survive the logit-gap threshold (Retained); Dropped is the difference. Candidates are analogies whose clean-run prediction already ranked the target within the top 10. A dash indicates that no analogies qualified as candidates in the first place (the model never ranked the target within its top 10), so no recovery map could be computed for that split. Under the stricter `top1` filter, this threshold drops nothing in any of the conditions above.

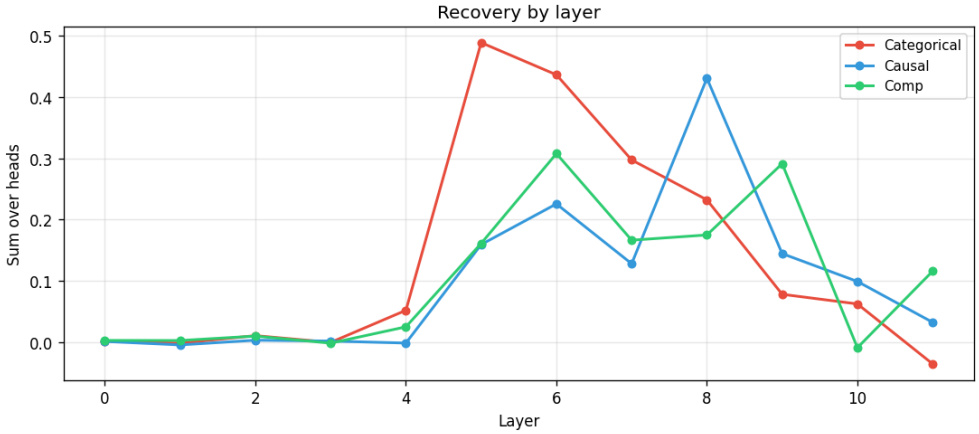
	GPT-2 Small				Llama 3.2 1B			
	`topk`		`top1`		`topk`		`top1`	
Rank	Head	Recovery	Head	Recovery	Head	Recovery	Head	Recovery
1	L8H6	0.227	L8H6	0.207	L7H5	0.272	L7H5	0.182
2	L6H9	0.214	L8H8	0.129	L11H9	0.107	L13H30	0.103
3	L7H7	0.176	L9H2	0.101	L9H16	0.105	L9H16	0.102
4	L6H1	0.131	L7H7	0.077	L9H31	0.101	L11H23	0.092
5	L5H0	0.105	L6H9	0.072	L13H30	0.093	L9H31	0.085

Table 4.7. The five attention heads with the highest mean logit recovery under the open-vocabulary zero-shot baseline, for GPT-2 Small and Llama 3.2 1B under the `topk` and `top1` candidate filters.

(2025). Moving to the stricter `top1` filter changes the picture in degree, not in kind: L8H6 remains the leading head, but with only two surviving analogies (Table 4.6) and a shift of by-layer mass toward the later layers. This version of the map should therefore be read as illustrative rather than as a reliable estimate. Figure 4.4 and



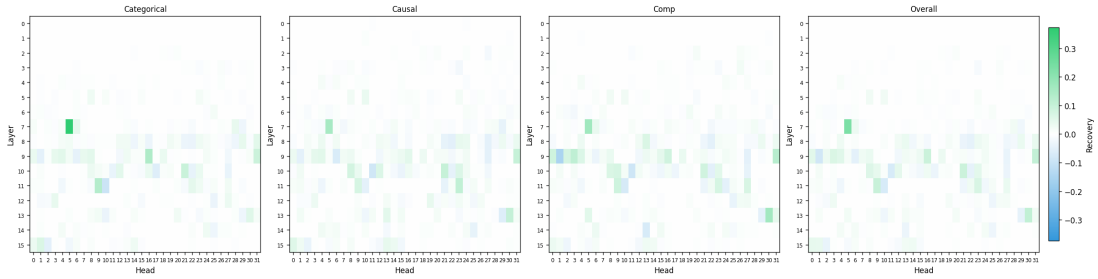
(a) Head-level recovery heatmap



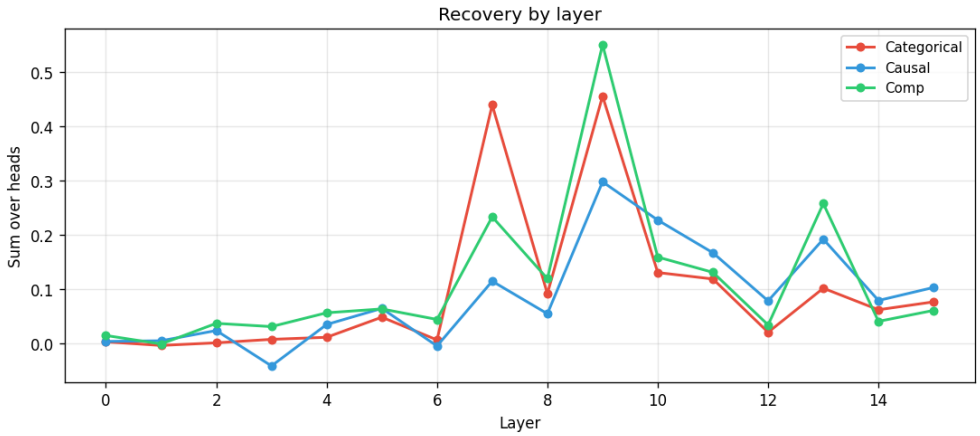
(b) Summed recovery by layer

Figure 4.3. GPT-2 Small activation patching under the open-vocabulary zero-shot condition, filter `topk` ($n = 12$).

Table 4.7 show the same pattern for Llama 3.2 1B: a single head, L7H5, leads the ranking under both filters, with the same drop-off and mid-to-late layer concentration observed for GPT-2 Small, again consistent with the hypothesis stated in Section 2.5. This concentration is particularly informative here, since Yang et al. (2025) identified an analogous intermediate-to-late symbolic-induction and retrieval stage in a much larger model from the same Llama family. This consistency across such a large difference in scale suggests that the mid-to-late localization of analogy completion is not simply a byproduct of model size, although actually confirming this would require comparing more than one model at each scale. Under the stricter `top1` filter, L7H5 remains the peak and the by-layer mass shifts further toward the later layers; with seven surviving analogies (Table 4.6), this map rests on a larger sample than GPT-2's and can be read with somewhat more confidence. Taken together, these maps show that GPT-2 Small and Llama 3.2 1B, despite their difference in scale, both concentrate the recovery signal in a small number of mid-layer heads rather than spreading it evenly across the network. Neither model's leading head, however, reaches full recovery: the corrupted prediction is only partially restored by any single component, which indicates that the lost information is reconstructed jointly by several heads rather than carried by one component alone. This kind of redundancy, where multiple heads each partly



(a) Head-level recovery heatmap



(b) Summed recovery by layer

Figure 4.4. Llama 3.2 1B activation patching under the open-vocabulary zero-shot condition, filter `topk` ($n = 21$).

compensate for the loss of relevant information, has also been documented for the indirect-object-identification circuit in GPT-2 small (Wang et al., 2022).

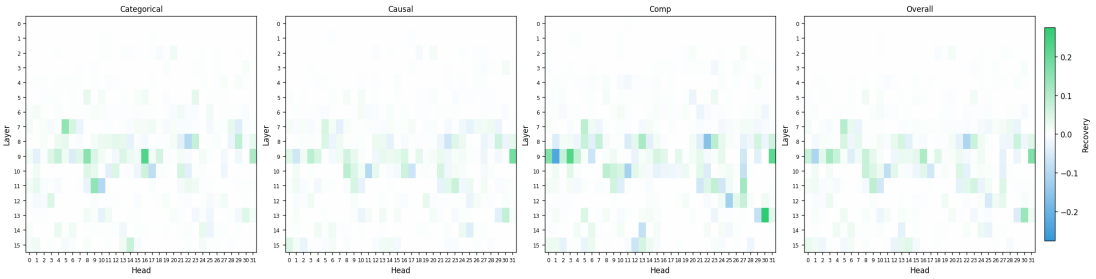
4.3.3 Best-performing Setup

The best-performing condition (by top-1 accuracy, see Tables 4.1 and 4.2) is the 1-shot within-relation condition for Llama, reported below. For GPT-2 the best-performing condition is identical to the baseline, so it is not repeated here.

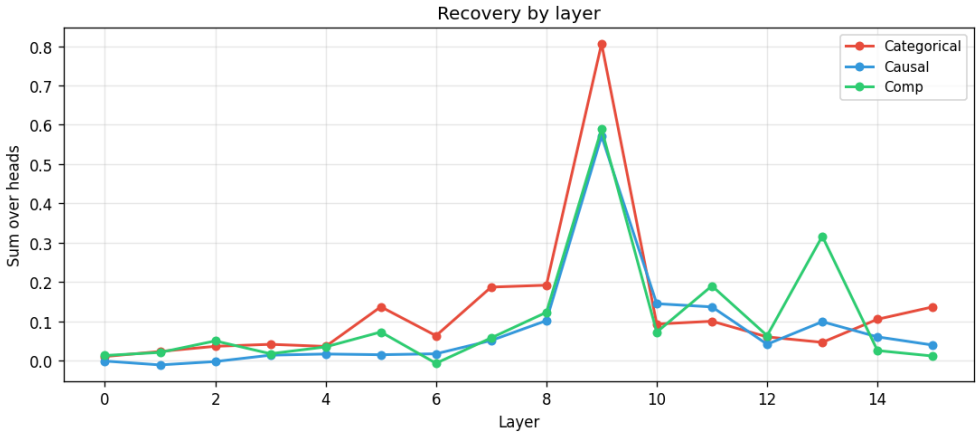
Table 4.8 reports the leading heads for Llama under 1-shot within-relation, and Table 4.6 reports the corresponding sample sizes. The recovery map remains highly correlated with the baseline ($r = 0.856$ under `topk`; $r = 0.851$ under `top1`), which indicates that the in-context examples recruit largely the same population of heads rather than a new circuit. The dominant head, however, shifts from L7H5 in the baseline to the layer-9 cluster. L7H5 drops to third place under `topk`, and falls out of the top five entirely under `top1`. The by-layer view in Figure 4.5 confirms this consolidation: recovery mass now concentrates almost entirely on layer 9. This indicates that the in-context examples that improve Llama's accuracy do not re-route the analogical circuit to a different head population, but instead shift its emphasis from layer 7 toward layer 9, to the point that the original baseline-dominant head becomes only a marginal contributor under the stricter filter.

	topk		top1	
Rank	Head	Recovery	Head	Recovery
1	L9H31	0.149	L9H16	0.145
2	L9H16	0.141	L9H31	0.121
3	L7H5	0.104	L9H8	0.106
4	L11H9	0.101	L11H9	0.093
5	L9H8	0.098	L13H30	0.078

Table 4.8. The five attention heads with the highest mean logit recovery for Llama 3.2 1B under the 1-shot within-relation condition (best-performing setup), for the `topk` and `top1` candidate filters.



(a) Head-level recovery heatmap



(b) Summed recovery by layer

Figure 4.5. Llama 3.2 1B activation patching under the 1-shot within-relation condition, filter `topk` ($n = 29$).

4.3.4 Corruption-Token Sensitivity

To assess how far the recovery maps depend on the specific corruption tokens used, I re-run the open-vocabulary baseline for both models under three independent seeds (42, 123, 456) for the corruption-token sample (Section 3.6.5). Table 4.9 reports the resulting correlations. For both models, the flattened recovery maps correlate almost perfectly across seeds, and the maximum mean absolute difference between any two seeds stays below 0.01. The peak head is also unchanged across all three seeds: L8H6 for GPT-2 Small (Figure 4.6) and L7H5 for Llama 3.2 1B (Figure 4.7). The recovery

maps are therefore not artifacts of the specific corruption tokens drawn under seed 42, but reflect head-level structure that is stable under resampling of the corruption distribution. This robustness shows that each head's recovery score is stable when the corruption tokens change.

Model	42 vs. 123	42 vs. 456	123 vs. 456
GPT-2 Small	0.992	0.992	0.992
Llama 3.2 1B	0.994	0.992	0.992

Table 4.9. Pairwise Pearson correlations (r) between the flattened recovery maps obtained under three independent corruption-token seeds (42, 123, 456) for the open-vocabulary zero-shot baseline.

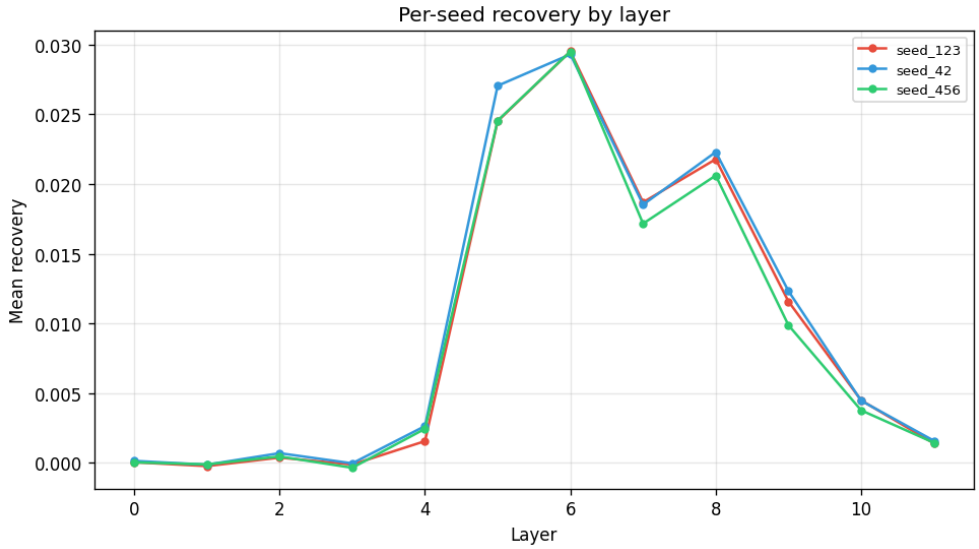


Figure 4.6. Per-layer summed recovery for GPT-2 Small under three independent corruption-token seeds (42, 123, 456) on the open-vocabulary zero-shot condition.

4.3.5 Across Experimental Conditions and Analogy Subsets

The recovery maps obtained under the few-shot conditions recruit largely the same attention heads as the open-vocabulary baseline. For GPT-2, the peak head across all four conditions remains L8H6, with recovery ranging from 0.24 to 0.33 (Table 4.6 reports the corresponding sample sizes), and the by-layer distribution is preserved. For Llama, consistent with the peak shift reported in Section 4.3.3, the few-shot peak is no longer L7H5 but the layer-9 cluster (L9H31/L9H16), with recovery ranging from 0.15 to 0.18. The finetuning maps tell a different story. Three of the five splits (cross-categorical, cross-compositional, near $\rightarrow$ far) retain no analogies at all once the logit-gap filter is applied, since the finetuned model never ranks the target within its top 10 for those splits in the first place. Of the remaining two splits, only one to three analogies survive, and the peak head is not preserved across them: L10H0 at recovery 0.233 in the far $\rightarrow$ near split, and L11H3 at recovery 0.406 in the cross-Causal split. Finetuning therefore reduces the number of items on which patching is interpretable to the point that no firm conclusion about head relocation can be drawn.

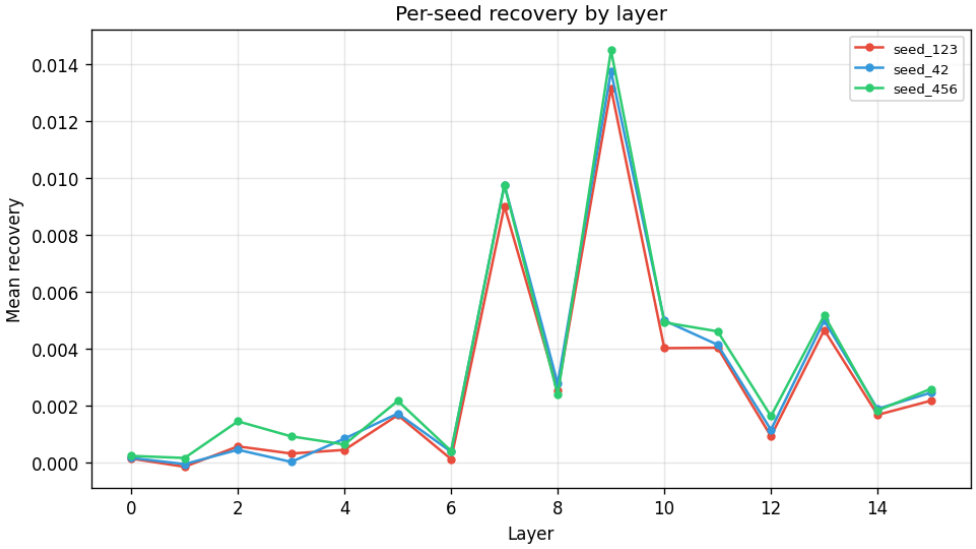


Figure 4.7. Per-layer summed recovery for Llama 3.2 1B under three independent corruption-token seeds (42, 123, 456) on the open-vocabulary zero-shot condition

Table 4.10 reports each model's baseline peak head broken down by relation type and semantic distance. For Llama, recovery is clearly highest on the categorical relation, the relation type the model also solves most accurately, which suggests that recovery scales with task difficulty rather than remaining constant across relation types. For GPT-2 the ordering is less clean, with causal sitting only marginally above categorical; given the small subgroup sizes, this difference should not be over-interpreted. The semantic-distance breakdown shows a far-bias for GPT-2's peak head, while for Llama the effect reverses but stays marginal, a difference that falls within the range expected from such small subset sizes. Together, these breakdowns indicate that the dominant heads identified in Section 4.3.2 are shared across relation types and distance categories, but that the strength of their contribution still varies somewhat with the difficulty of the underlying analogy.

Taken together, the activation patching results answer SQ3 as follows. In both models, the corrupted prediction is most readily restored by a small number of mid-layer attention heads: L8H6 for GPT-2 Small, and L7H5 together with the layer-9 cluster (L9H16, L9H31) for Llama 3.2 1B. As explained in Section 4.3.1, restoring one of these heads on its own recovers a substantial but partial share of the corrupted-to-clean logit gap (up to 0.227 for GPT-2 and 0.272 for Llama). This recovery score is highly stable under resampling of the corruption tokens. It is also largely preserved across prompting conditions, with one qualification: Llama's dominant head shifts from layer 7 to the layer-9 cluster once few-shot demonstrations are added. The finetuning splits, meanwhile, retain too few analogies to support any conclusion about head relocation. The recovery maps for the two models are broadly consistent with the qualitative pattern that Yang et al. (2025) describe for a much larger model: the heads that matter most sit past the network's midpoint, at two-thirds depth for GPT-2 (layer 8 of 12) and around the midpoint for Llama (layers 7–9 of 16), in the general region of their proposed symbolic-induction stage. However, the present analysis patches one head at a time. It does not trace how heads pass information to each other, the role played

Model	Subset	Peak head	Recovery (n)
GPT-2 Small	Categorical	L8H6	0.249 (6)
GPT-2 Small	Causal	L8H6	0.286 (3)
GPT-2 Small	Compositional	L8H6	0.122 (3)
GPT-2 Small	Near	L8H6	0.195 (9)
GPT-2 Small	Far	L8H6	0.322 (3)
Llama 3.2 1B	Categorical	L7H5	0.373 (11)
Llama 3.2 1B	Causal	L7H5	0.156 (5)
Llama 3.2 1B	Compositional	L7H5	0.168 (5)
Llama 3.2 1B	Near	L7H5	0.278 (14)
Llama 3.2 1B	Far	L7H5	0.261 (7)

Table 4.10. Mean logit recovery of each model's baseline peak head (L8H6 for GPT-2 Small, L7H5 for Llama 3.2 1B), broken down by relation type and semantic distance, under the open-vocabulary zero-shot condition (`topk` filter). n is the number of analogies retained in each subset after the logit-gap filter.

by path patching. It can therefore identify which heads matter, but not how they are connected. For this reason, it cannot confirm whether a staged circuit of the kind Yang et al. (2025) propose is also present in these smaller models (Wang et al., 2022; Conmy et al., 2023). Moreover, the representational analysis in Section 4.2 indicates that both models solve the task primarily through surface-level similarity rather than relational mapping. The heads localized here should therefore be read differently: as the components that causally support whatever surface-level computation produces the prediction, not as evidence for a dedicated analogical-reasoning circuit. This distinction is taken up further in the discussion.

Chapter 5

Discussion

This thesis aimed to address the gap that still exists in our understanding of how LLMs solve analogies. To that end, three sub-questions were formulated:

SQ1 (Behavioral). How accurately do GPT-2 Small and Llama 3.2 1B solve four-term verbal analogies, and how is this accuracy modulated by the prompting format, by finetuning on held-out splits (GPT-2 Small only), by relation type, and by semantic distance?

SQ2 (Representational/Error-analysis). When the model's top-ranked prediction is not the gold target, do its predictions nevertheless reflect the relational structure of the A:B::C:D mapping rather than random failure?

SQ3 (Mechanistic). Which attention heads causally drive the analogical prediction in each model, and how stable is this head-level structure across prompting conditions, finetuning, and resampling of the corruption tokens used for activation patching?

Chapter 4 reported the results and answered each sub-question in place, interpreting each finding as it was introduced. This chapter does not repeat that interpretation. Instead, it steps back to relate the findings to the broader literature, acknowledges the limitations of the present study, and identifies directions for future research.

5.1 Behavioral Discussion

The behavioral results, summarized in Section 4.1, show that both models perform substantially below the ceiling and that most of the manipulations intended to help do not improve performance. Two patterns are worth situating in the broader literature.

First, the positional bias observed in the restricted forced-choice condition, where both models favor whichever candidate appears first regardless of the relational structure. This is not found in the human study by Jones et al. (2022) on which this condition is modeled. This suggests that presenting two explicit candidates does not simply replicate the difficulty humans face when choosing between a target and a distractor. It introduces a separate, surface-level sensitivity to token position. This is consistent with the answer-order effects reported by Lewis and Mitchell (2024) for GPT-3.5 and GPT-4 on Webb's analogy domains, suggesting that this sensitivity is a more general

property of autoregressive LLMs on forced-choice analogy tasks rather than one specific to GPT-2 Small and Llama 3.2 1B. The behavioral results are nevertheless not purely surface-level: the salience effect, where both models are more easily misled by distractors that are semantically closer to the source pair, replicates the same direction of effect that Jones et al. (2022) reported for human participants. This suggests that GPT-2 Small and Llama 3.2 1B share at least some sensitivity to relational structure, even though their rank-based accuracy falls far below human performance on the same items.

Second, Llama 3.2 1B shows its highest observed accuracy under within-relation demonstrations and its lowest under cross-relation ones, while GPT-2 Small is indifferent to either; as noted in Section 4.1, neither shift reaches significance given the overlapping confidence intervals, so this pattern should be read as suggestive rather than established. If this pattern reflects a genuine effect, it would be consistent with the broader claim that few-shot in-context learning is an emergent capacity that scales with model size (Brown et al., 2020; Olsson et al., 2022). The asymmetry between within-relation and cross-relation examples would also be compatible with Structure-Mapping Theory (Gentner, 1983), which suggests that transfer depends on the relational alignment between the demonstration and the test item: a model could have the capacity to use in-context examples and still apply this capacity destructively if the examples and test items are relationally misaligned (i.e., in the cross-relation condition). Given the present data, however, this remains a plausible interpretation of a numerical trend rather than a confirmed extension of the theory to model behavior.

However, when attempting to teach these relations through fine-tuning rather than in-context demonstrations, the model's structured behavior breaks down entirely. Collapsing GPT-2 Small's top-1 accuracy to zero on every split after a single pass over 30 to 40 analogies is consistent with the well-documented vulnerability of neural networks to catastrophic forgetting when updated on a narrow distribution (Kirkpatrick et al., 2017), a phenomenon also observed specifically in fine-tuned LLMs across domain knowledge and reasoning tasks (Luo et al., 2023).

5.2 Error Analysis Discussion

SQ2 was initially intended as a representational question: when the model's top-ranked prediction is not the gold target, does it nevertheless preserve the relational structure defined by the source pair? As reported in Section 4.2, the automatic structural criterion planned for this purpose could not be applied, because the valid analogies in this dataset are themselves not parallel in either model's embedding space or in Word2Vec. The question therefore had to be addressed through manual inspection, which makes SQ2 in practice closer to an error analysis than to a representational probe.

The failure of the offset criterion itself is already an informative finding. The assumption that the analogical structure is linearly encoded as parallel offsets holds for the kinds of relations on which the parallelogram method was originally demonstrated (Mikolov et al., 2013), but it does not extend to the more abstract categorical, causal and compositional relations studied here. Offset parallelism therefore cannot serve as evidence of relational structure for this dataset, reflecting a genuine limitation of the parallelogram account.

The manual inspection of the predictions shows that the models' failures are not random: they are dominated by surface-level similarity. The two models combine

this surface-level similarity with different failure modes. GPT-2 Small's most common failure, repeating one of the analogy's own input tokens, is consistent with inductive copying performed by induction heads as described by Olsson et al. (2022): rather than constructing a novel completion, the model attends back to a token already present in the prompt and copies it forward.

Llama 3.2 1B's dominant failure mode is different: a large part of its top-ranked predictions are non-content tokens like empty strings, repeated punctuation, and underscores. The lexical filter applied throughout the experiments treats this as noise and removes it before scoring. This behavior is worth a moment of reflection. An explanation lies in the composition of the training data: Llama 3.2 1B is trained on a much larger corpus than GPT-2 Small, including substantial amounts of code and structured web text (Grattafiori et al., 2024). In this kind of training data, punctuation and formatting tokens occur frequently and are highly predictable continuations. A complementary explanation is architectural rather than data-driven: Transformer language models have been shown to allocate a disproportionate share of attention, and consequently probability mass, to tokens with limited semantic content, such as punctuation marks. This phenomenon is known as the attention sink effect (Xiao et al., 2024). Under this reading, Llama's preference for punctuation reflects a possible fusion of training data bias because of the large amount of punctuation present in the training data, and a structural bias due to the attention sink effect. Taken together, both accounts suggest that the punctuation preference is not a failure of analogical reasoning specifically, but a more general property of the model's internal mechanisms.

In terms of Structure-Mapping Theory (Gentner, 1983), neither model achieves successful mapping of relational structure: the predictions are better explained by surface-level similarity. This is consistent with the broader view, discussed in Section **??**, that Transformers can exhibit analogy-solving behavior as a consequence of the statistical regularities they learned during training.

5.3 Mechanistic Discussion

The activation patching maps localize the attention heads that are causally involved in producing the prediction. Because both models rely primarily on surface-level similarity rather than on relational structure, as concluded in the behavioral and error analysis, these maps cannot be read as evidence for an analogical-reasoning circuit. Instead, they localize where in the network the surface-level computation is carried out that leads to the observed predictions. This reinterpretation does not diminish the value of the activation patching results: it identifies a set of components on which future work could intervene to determine more precisely what kind of similarity these heads track.

The activation patching maps are highly stable for both models, yet accuracy varies considerably across conditions. This tension raises a question: if largely the same heads drive the prediction in both zero-shot and few-shot conditions, what does in-context learning actually change? For GPT-2 Small, the same head dominates throughout. For Llama, the same head population is recruited, but its emphasis shifts toward the layer-9 cluster under few-shot prompting, so the maps remain highly correlated without being identical. One interpretation is that the mid-layer heads reliably encode the relational direction, but that the bottleneck lies in the final projection from the residual stream to the vocabulary space (Wang et al., 2022). Under this interpretation, in-context examples sharpen that final step without rewiring the upstream circuit. An alternative

interpretation is that these mid-layer heads do not encode relational structure at all, but instead help predict tokens that are similar at the surface level to the context.

The relationship between near and far analogies in the patching results seems counterintuitive. Behaviorally, near analogies are consistently easier than far analogies for both models, yet GPT-2's dominant head shows higher logit recovery on far analogies than on near ones. For Llama, this near/far difference in logit recovery is much smaller, suggesting that the far-analogy bias is specific to GPT-2 rather than a general property of both models. One interpretation is that the mid-layer head is more causally necessary for far analogies, since solving them places greater demands on relational extraction, which would imply that the head encodes some form of relational structure rather than purely surface-level association. Near analogies, by contrast, may require less relational understanding to complete successfully. Given the small number of items in each subset, however, this difference may also reflect sampling variability rather than a systematic effect.

There are barely any patching results available for the finetuning condition, for a basic reason: three of the five finetuning splits leave no analogies eligible for patching at all, since the finetuned model never ranks the target within its top 10 for those splits in the first place. Only the cross-Causal and far,→,near splits retain any items, and there the dominant head is not preserved between them. The mechanistic data is therefore too sparse to support any claim about whether finetuning relocates the recovery signal.

5.4 Limitations and Future Work

In this section, several limitations of the present study are acknowledged, alongside recommendations for future work to improve and extend the current research.

The most fundamental limitation in this thesis is the size of the dataset. With only 60 analogies spanning three relation types and two distance categories, the subgroup analyses breakdowns contain as few as one to five items per cell. The Wilson score confidence intervals reflect this uncertainty: many of the subgroup-level differences should be treated as descriptive rather than statistically robust findings. A larger dataset, spanning more relation types and semantic distances than the 60 analogies used here, would allow these subgroup-level patterns to be evaluated with more than a handful of items per cell and would strengthen the conclusions that can be drawn from them.

A second limitation concerns the corruption procedure used in the activation patching experiments. Only the B-term was replaced with a corruption token, targeting the relational signal carried by the A:B pair. The contributions of other token positions, including the A-term, the C-term, and the structural connector tokens, were not examined. It is possible that attention heads at these positions also contribute to the analogical prediction but are not captured by the present patching configuration. Extending the corruption procedure to these other positions would reveal whether they carry additional heads relevant to the prediction that the present design cannot detect. In addition, applying path patching to trace the information flow between the identified heads would make it possible to evaluate whether the three-stage symbol-abstraction-induction-retrieval architecture that Yang et al. (2025) proposed for Llama 3.1 70B is also present in the much smaller models studied here. Alternatively, this might reveal a different circuit structure entirely, a question the present single-component patching cannot address.

Third, accuracy is evaluated by the rank of the first sub-word token of the target D. For analogy items where D tokenizes into multiple sub-word tokens, the metric may not capture the correct answer precisely: a first sub-word token that ranks highly could correspond to multiple possible completions, not exclusively the target word. In addition, a broader problem with probability-based evaluation has been identified: Holtzman et al. (2021) show that a model's probability mass is often distributed across multiple surface forms that express the same underlying concept. Therefore, synonyms of the target answer might be ranked in the top 10 while the exact target is not. Standard rank-based metrics fail to capture this nuance because they only look for exact matches. The same issue can occur in reverse within restricted-choice experiments: synonyms of a distractor might rank higher than the target. Even though the actual distractor is ranked lower, a rank-based metric would falsely classify the answer as correct, overlooking the fact that more probability mass shifted toward the distractor concept. This demonstrates that a low rank for the target does not necessarily reflect a failure to extract the relational structure. While an inspection of the top-10 predictions in Appendix D suggests this is not a major driver of errors in the current dataset, relying on a token-level rank metric remains a limitation.

Fourth, the corruption token selection depends on a specific threshold: these tokens are sampled from the 200 candidate nouns most distant from both the A-term and the C-term embeddings. Beyond this distance-based threshold, candidate tokens are also required to be nouns according to WordNet, be absent from the NLTK English stopword list, and to consist only of lowercase alphabetic characters at least three letters long. Although these settings were chosen based on established practice in the literature, it remains worthwhile to investigate the influence of alternative settings on the results. Changing these settings alters which tokens qualify as corruptions, and consequently might change the logit recovery scores I observed. Although I already investigate the influence of differences in the lexicality of corruption tokens, a systematic analysis of the influence of the other settings that determine the pool size would clarify how much the reported scores depend on this specific choice. Such an analysis would strengthen confidence in the robustness of the reported findings.

Finally, the error analysis rests on a manual, single-annotator classification of the top-10 predictions into qualitative patterns such as repetition, semantic-field fixation, and non-content-token preference. This classification was not independently verified by a second annotator, and the categories themselves were defined from the predictions of the two models studied here. Independent annotation, or replication on predictions from additional models, would establish whether the same categories generalize beyond this thesis, rather than describing only what was observed here.

Beyond addressing these specific limitations, the findings raise further questions that future work could pursue. On the behavioral side, the destructive effect of finetuning on GPT-2 Small's accuracy raises an open question: the present design cannot determine whether this destructive effect is a consequence of finetuning itself or an artifact of how finetuning was implemented here. Testing alternative finetuning techniques and settings would therefore be a natural direction for future work, to establish whether a different approach could avoid this collapse. On the mechanistic side, comparing the activation patching maps obtained for analogical reasoning against maps obtained when the model is prompted for similarity reasoning would help establish whether the mid-layer heads identified in this thesis encode relational structure or surface-level similarity, since the two accounts currently remain indistinguishable on the present data.

Chapter 6

Conclusion

This thesis investigated how GPT-2 Small and Llama 3.2 1B solve four-term verbal analogies, using a 60-item dataset of analogies and combining a behavioral evaluation across multiple prompting and finetuning conditions, an error analysis of the top-10 predictions, and head-level activation patching. The aim was to address the gap between behavioral analogy benchmarks and mechanistic interpretability research by examining not only whether the models produce the correct output, but also whether incorrect predictions preserve relational structure and which internal components drive that structure.

With respect to SQ1, both models showed limited and fragile analogical reasoning accuracy, with Llama 3.2 1B outperforming GPT-2 Small throughout. Top-1 accuracy collapsed under the restricted forced-choice condition, few-shot demonstrations and finetuning failed to reliably improve performance, and accuracy was further shaped by relational distance and type. A weak relational preference nonetheless persisted in forced-choice comparisons, indicating that the models encode some sensitivity to the relational structure that rank-based evaluation alone does not capture.

With respect to SQ2, the offset-cosine criterion could not be applied, as the analogies in this dataset are not parallel in either model's embedding space. Manual inspection of the top-10 predictions showed instead that incorrect predictions are not random: both models default to associative retrieval from the prompt's semantic neighborhood rather than the relational target. Additionally, probability-based behavior is observed as token-copying in GPT-2 Small and punctuation prediction in Llama 3.2 1B. Neither model achieves successful mapping of relational structure; their predictions are better explained by surface-level similarity.

With respect to SQ3, activation patching localized the attention heads that causally drive the prediction in both models to a small number of mid-layer heads, L8H6 in GPT-2 Small and L7H5 in Llama 3.2 1B, both within the intermediate-to-late stage of the network. These maps were stable under resampling of the corruption tokens and largely preserved across prompting conditions, even as accuracy varied considerably between them. However, given the behavioral and error-analysis findings, these heads are better characterized as encoding structural associative retrieval than as components of a deliberate analogical reasoning circuit.

Taken together, these findings suggest that both GPT-2 Small and Llama 3.2 1B solve four-term verbal analogies primarily by relying on surface-level similarity rather than genuine mapping of relational structure. Their failed predictions preserve the semantic neighborhood of the prompt rather than failing at random, and the mid-layer

heads identified through activation patching localize where this surface-level computation is carried out. The models therefore reproduce the surface behavior of analogy solving, returning relationally plausible words, without reliably identifying the underlying relation. This positions their behavior between random failure and genuine analogical reasoning: they exhibit a signature of surface-level similarity without consistently achieving the outcome of relational structure mapping. These results speak directly to the central debate over whether LLMs reason or simply reproduce statistical regularities in their training data: the models' behavior is better characterized as the product of surface-level associative retrieval than as deliberate analogical reasoning.

Appendix A: Per-relation and Per-distance Activation Patching

This appendix reports the per-relation and per-distance breakdowns of the open-vocabulary zero-shot activation patching maps. The aggregated numbers are reported inline in Section 4.3.5; the corresponding heatmaps are shown here.

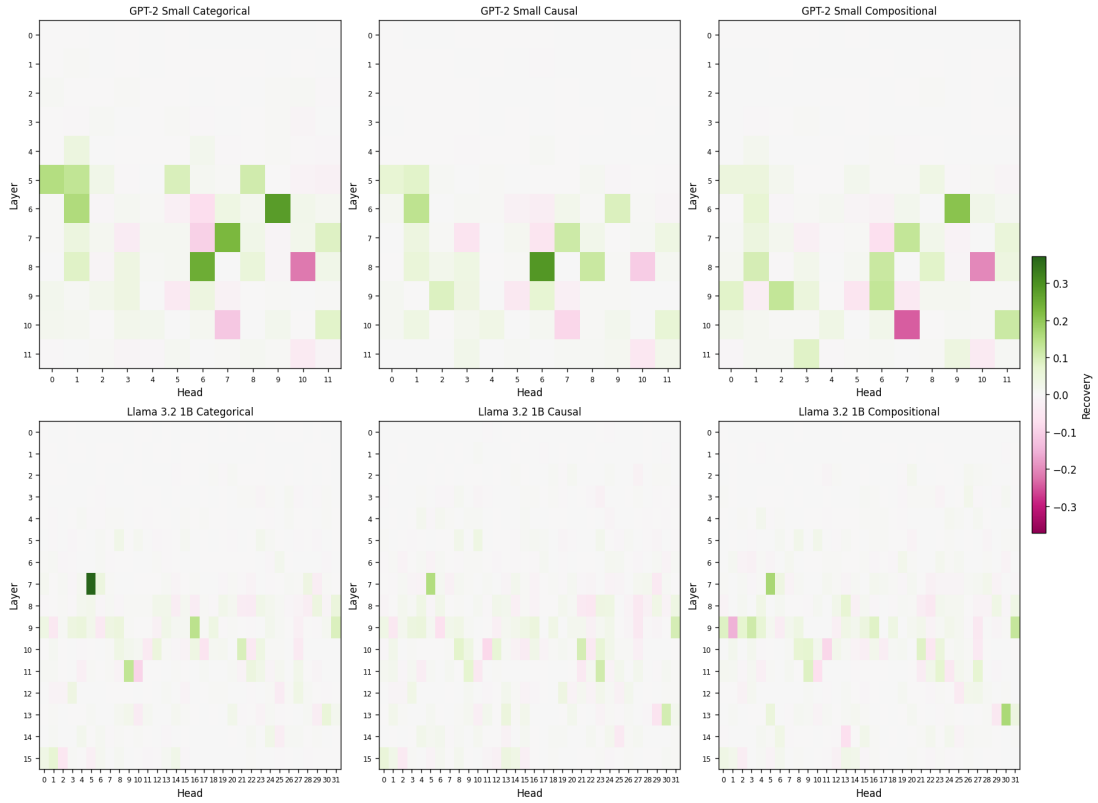


Figure 1. Mean head-level recovery by relation type under the open-vocabulary zero-shot condition (filter `topk`). Top row: GPT-2 Small ($n = 6$ Categorical, $n = 3$ Causal, $n = 3$ Compositional). Bottom row: Llama 3.2 1B ($n = 11$, $n = 5$, $n = 5$).

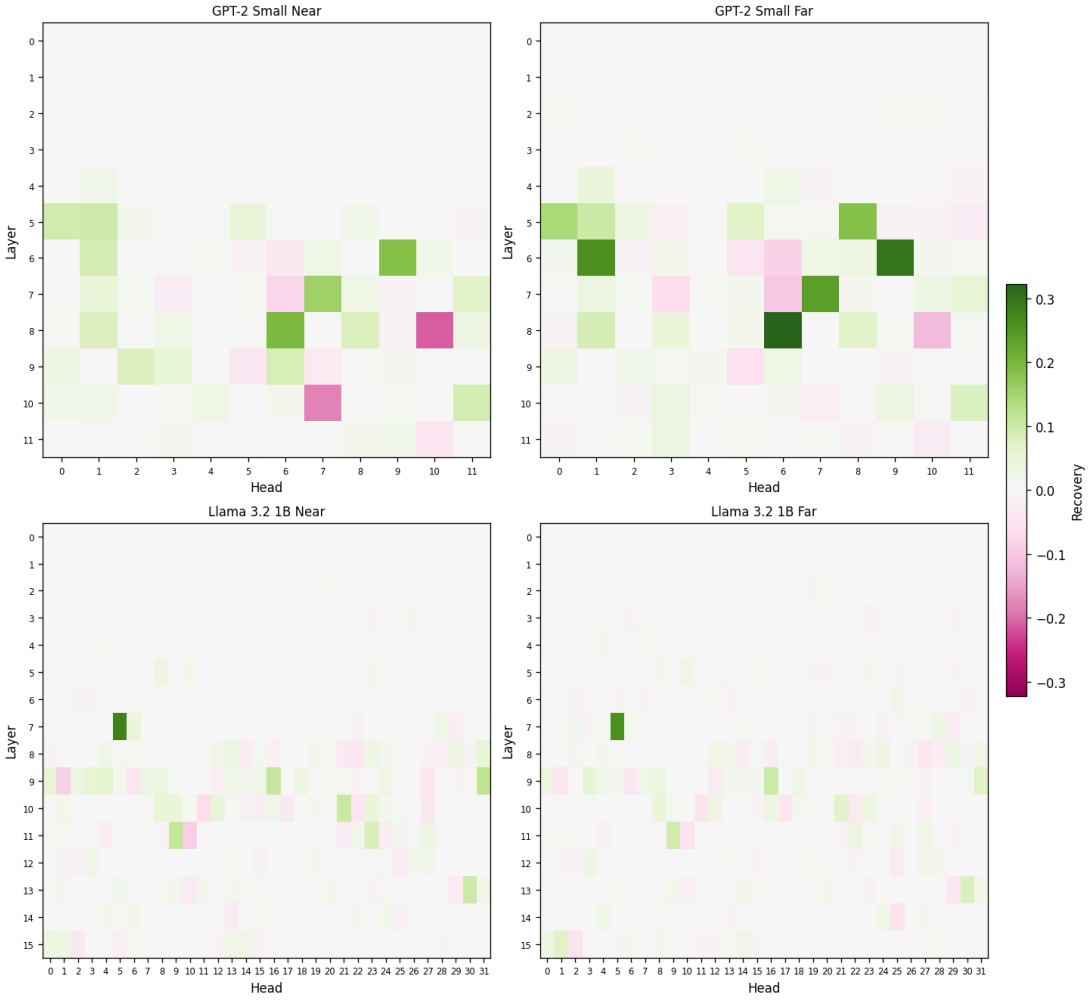


Figure 2. Mean head-level recovery by semantic distance category under the open-vocabulary zero-shot condition (filter `topk`). Top row: GPT-2 Small ($n = 9$ near, $n = 3$ far). Bottom row: Llama 3.2 1B ($n = 14$ near, $n = 7$ far).

Appendix B: Restricted Forced-choice Breakdown

This appendix reports the *correct-choice* metric of the restricted forced-choice condition, broken down by distractor salience and target position. The aggregate numbers are reported in Section 4.1.

Split	GPT-2 Small	Llama 3.2 1B
All prompts ($n = 240$)	64.17%	61.67%
High-salience distractor ($n = 120$)	55.83%	57.50%
Low-salience distractor ($n = 120$)	72.50%	65.83%
Target in position 1 ($n = 120$)	70.00%	75.00%
Target in position 2 ($n = 120$)	58.33%	48.33%

Table 1. Restricted forced-choice *correct-choice* accuracy (= target probability greater than distractor probability at the final token), broken down by distractor salience and by target position within the candidate list.

Both models show a position bias favoring the first candidate, and the salience effect originally reported in human participants by Jones et al. (2022) (low-salience distractors easier to reject than high-salience ones) is recovered for both models, with the effect notably larger for GPT-2.

Appendix C: Full Analogy Dataset

This appendix lists all 60 analogies used in the experiments. Table 2 provides the full analogy set with relation type and distance category. Table 3 reports the cosine distance scores per model using the analogy index from Table 2. Table 4 lists the five corruption words per analogy per model and their distance to the target word C.

#	Analogy	Relation	Distance
1	tarantula : spider :: bee : insect	Categorical	near
2	calf : animal :: bull : mammal	Categorical	near
3	botanist : biologist :: chemist : scientist	Categorical	near
4	swimming : sport :: golf : activity	Categorical	near
5	murder : crime :: gun : weapon	Categorical	near
6	trout : fish :: lobster : crustacean	Categorical	near
7	squirrel : rodent :: robin : bird	Categorical	near
8	jazz : music :: saxophone : instrument	Categorical	near
9	bowl : dish :: spoon : silverware	Categorical	near
10	january : month :: winter : season	Categorical	near
11	diesel : fuel :: bed : furniture	Categorical	far
12	accountant : profession :: carrot : vegetable	Categorical	far
13	west : direction :: cat : feline	Categorical	far
14	rectangle : shape :: lunch : meal	Categorical	far
15	physics : science :: ring : jewelry	Categorical	far
16	caffeine : drug :: rose : flower	Categorical	far
17	wrench : tool :: sad : mood	Categorical	far
18	oven : appliance :: sweden : country	Categorical	far
19	apartment : shelter :: tea : beverage	Categorical	far
20	doll : toy :: turkey : poultry	Categorical	far
21	trauma : coma :: accident : death	Causal	near
22	virus : flu :: bacteria : infection	Causal	near
23	traffic : delay :: crash : damage	Causal	near
24	illness : treatment :: drug : relief	Causal	near
25	training : fitness :: education : career	Causal	near
26	fracture : cast :: incision : scar	Causal	near
27	eclipse : darkness :: moon : tide	Causal	near
28	scratch : blood :: sprain : swell	Causal	near

continued on next page

Table 2 – *continued*

#	Analogy	Relation	Distance
29	failure : termination :: success : promotion	Causal	near
30	defeat : frown :: victory : smile	Causal	near
31	motion : sickness :: drought : famine	Causal	far
32	allergy : sneeze :: entertainment : laughter	Causal	far
33	loan : debt :: football : concussion	Causal	far
34	sadness : crying :: genes : baldness	Causal	far
35	crime : arrest :: mutation : cancer	Causal	far
36	speeding : ticket :: run : exhausted	Causal	far
37	order : delivery :: salt : thirst	Causal	far
38	hormones : mood :: smoke : cough	Causal	far
39	invitation : visit :: tornado : destruction	Causal	far
40	period : cramps :: wind : erosion	Causal	far
41	oatmeal : cookie :: banana : muffin	Compositional	near
42	marble : stairs :: brick : patio	Compositional	near
43	brass : trophy :: bronze : medal	Compositional	near
44	tomato : sauce :: cheese : soup	Compositional	near
45	strawberry : cake :: cherry : pie	Compositional	near
46	fudge : candy :: chocolate : brownie	Compositional	near
47	corduroy : pants :: denim : skirt	Compositional	near
48	feather : pillow :: foam : cushion	Compositional	near
49	fur : coat :: vinyl : gloves	Compositional	near
50	plastic : table :: wood : desk	Compositional	near
51	steel : scissors :: apple : cider	Compositional	far
52	paper : money :: beef : broth	Compositional	far
53	log : fort :: ceramic : jar	Compositional	far
54	satin : dress :: concrete : bridge	Compositional	far
55	crystal : vase :: cotton : socks	Compositional	far
56	leather : saddle :: gold : earring	Compositional	far
57	turkey : burger :: rubber : boots	Compositional	far
58	flour : tortilla :: silk : tie	Compositional	far
59	sugar : icing :: stone : house	Compositional	far
60	leaf : nest :: wool : scarf	Compositional	far

Table 2. All 60 analogies from the dataset, grouped by relation type and sorted by semantic distance category.

	GPT-2 Small			Llama 3.2 1B		
#	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$
1	0.609	0.462	0.408	0.905	0.714	0.716
2	0.619	0.652	0.336	0.839	0.949	0.724
3	0.591	0.331	0.267	0.902	0.788	0.430
4	0.647	0.799	0.450	0.865	0.973	0.709
5	0.445	0.421	0.529	0.645	0.643	0.808

continued on next page

Table 3 – *continued*

	GPT-2 Small			Llama 3.2 1B		
#	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$
6	0.426	0.560	0.403	0.573	0.899	0.674
7	0.423	0.650	0.440	0.893	0.818	0.704
8	0.489	0.561	0.434	0.720	0.803	0.724
9	0.595	0.600	0.484	0.739	0.816	0.762
10	0.637	0.489	0.461	0.861	0.717	0.725
11	0.468	0.642	0.628	0.679	0.896	0.886
12	0.616	0.535	0.592	0.848	0.721	0.840
13	0.691	0.553	0.669	0.923	0.875	0.965
14	0.614	0.395	0.638	0.837	0.638	0.934
15	0.414	0.625	0.705	0.622	0.838	0.892
16	0.581	0.541	0.618	0.834	0.760	0.860
17	0.625	0.646	0.643	0.828	0.884	0.917
18	0.552	0.677	0.613	0.784	0.956	0.879
19	0.616	0.527	0.640	0.857	0.739	0.869
20	0.491	0.426	0.595	0.663	0.618	0.818
21	0.628	0.664	0.484	0.860	0.888	0.741
22	0.576	0.448	0.377	0.833	0.715	0.634
23	0.717	0.678	0.500	0.948	0.878	0.791
24	0.587	0.767	0.453	0.912	0.935	0.782
25	0.584	0.625	0.486	0.827	0.891	0.723
26	0.745	0.660	0.541	0.929	0.916	0.858
27	0.657	0.671	0.486	0.850	0.853	0.744
28	0.773	0.650	0.600	0.955	0.895	0.871
29	0.595	0.647	0.443	0.851	0.909	0.722
30	0.740	0.647	0.302	0.988	0.920	0.669
31	0.721	0.424	0.579	0.942	0.569	0.831
32	0.648	0.605	0.554	0.940	0.801	0.867
33	0.469	0.653	0.622	0.650	0.904	0.876
34	0.456	0.649	0.501	0.729	0.906	0.834
35	0.598	0.580	0.595	0.833	0.871	0.798
36	0.639	0.785	0.644	0.875	0.960	0.905
37	0.714	0.682	0.701	0.907	0.937	0.945
38	0.657	0.657	0.595	0.880	0.841	0.868
39	0.616	0.641	0.637	0.878	0.873	0.865
40	0.686	0.655	0.615	0.920	0.906	0.909
41	0.615	0.631	0.422	0.825	0.916	0.755
42	0.577	0.668	0.417	0.851	0.872	0.665
43	0.726	0.515	0.398	0.902	0.686	0.542
44	0.518	0.625	0.333	0.761	0.838	0.593
45	0.581	0.643	0.403	0.835	0.864	0.634
46	0.675	0.572	0.378	0.927	0.847	0.710
47	0.682	0.581	0.379	0.966	0.814	0.695

continued on next page

Table 3 – *continued*

	GPT-2 Small			Llama 3.2 1B		
#	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$	$d(AB)$	$d(CD)$	$d(A{\to}B, C{\to}D)$
48	0.591	0.582	0.428	0.836	0.813	0.629
49	0.626	0.689	0.510	0.863	0.876	0.845
50	0.780	0.719	0.433	0.967	0.967	0.763
51	0.695	0.451	0.576	0.895	0.640	0.835
52	0.665	0.591	0.628	0.882	0.833	0.904
53	0.726	0.685	0.610	0.932	0.871	0.904
54	0.725	0.693	0.624	0.896	0.909	0.900
55	0.736	0.603	0.631	0.911	0.832	0.840
56	0.557	0.722	0.513	0.818	0.976	0.889
57	0.552	0.633	0.591	0.777	0.882	0.794
58	0.639	0.667	0.577	0.876	0.914	0.889
59	0.640	0.703	0.632	0.842	0.899	0.909
60	0.723	0.542	0.551	0.907	0.783	0.874

Table 3. Cosine distance scores for all 60 analogies by model. Analogy numbers correspond to Table 2. $d(AB)$ and $d(CD)$ are the within-pair cosine distances between the source terms (A, B) and the target terms (C, D), respectively. $d(AB, CD)$ is the cosine distance between the summed source-pair and target-pair embeddings, $(\vec{v}_A + \vec{v}_B)$ and $(\vec{v}_C + \vec{v}_D)$, i.e. the `dist_AB_CD` measure used for the near/far split (Section 3.1.1).

		GPT-2 Small		Llama 3.2 1B	
#		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	1	data	0.848	total	0.989
	2	host	0.786	caption	0.980
1	3	navigation	0.770	part	0.995
	4	whole	0.802	official	0.989
	5	sport	0.808	analysis	1.012
	1	vac	0.824	con	0.974
	2	worlds	0.842	present	1.004
2	3	practice	0.836	model	1.001
	4	fusion	0.822	former	0.984
	5	rooms	0.816	mod	1.011
	1	peripheral	0.796	nada	1.017
	2	encoding	0.762	licenses	1.005
3	3	repeat	0.805	auxiliary	1.006
	4	boots	0.750	transmitting	0.999
	5	entitlement	0.773	ordinary	1.005
	1	first	0.875	name	0.982
	2	individual	0.811	lab	0.981
4	3	inc	0.861	ens	0.986
	4	system	0.850	system	0.998
	5	independent	0.828	least	0.982

continued on next page

Table 4 – *continued*

#		GPT-2 Small		Llama 3.2 1B	
		Corruption	$d(w, C)$	Corruption	$d(w, C)$
5	1	contingent	0.830	trough	0.980
	2	format	0.840	sites	0.986
	3	qualifiers	0.809	applications	0.992
	4	touches	0.807	due	0.981
	5	domain	0.804	information	0.981
6	1	space	0.804	version	1.004
	2	two	0.813	central	1.008
	3	future	0.804	low	0.997
	4	means	0.814	making	1.009
	5	second	0.798	overall	0.990
7	1	default	0.782	plus	1.008
	2	placement	0.788	found	1.000
	3	temporary	0.791	content	1.000
	4	class	0.794	three	1.008
	5	author	0.779	main	1.005
8	1	action	0.833	number	1.007
	2	item	0.787	sets	0.994
	3	base	0.821	group	1.017
	4	season	0.793	six	0.986
	5	link	0.821	taking	0.998
9	1	rep	0.840	operations	1.008
	2	far	0.827	points	0.987
	3	low	0.827	still	0.983
	4	access	0.810	call	1.012
	5	exports	0.810	variant	0.982
10	1	clear	0.815	imp	0.984
	2	free	0.785	person	0.993
	3	spot	0.801	acc	0.976
	4	tone	0.808	prior	0.986
	5	reason	0.798	system	0.976
11	1	value	0.821	tang	0.996
	2	ceremony	0.811	con	0.973
	3	start	0.795	prior	0.972
	4	outside	0.837	gains	0.974
	5	rep	0.810	features	0.983
12	1	units	0.817	times	1.005
	2	cell	0.803	large	1.005
	3	match	0.800	must	0.988
	4	time	0.841	first	1.047
	5	stage	0.801	program	0.985
13	1	figure	0.817	neurons	0.995
	2	alert	0.827	bullying	0.981

continued on next page

Table 4 – *continued*

#		GPT-2 Small		Llama 3.2 1B	
		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	3	royalty	0.834	annoying	0.995
	4	sounding	0.839	buttons	0.980
	5	liquid	0.838	substitution	0.983
	1	left	0.808	cons	0.979
	2	moves	0.800	particular	0.973
14	3	heavy	0.789	gen	0.972
	4	professional	0.807	experience	0.974
	5	tune	0.802	dos	0.977
	1	forth	0.834	cere	0.981
	2	haw	0.812	using	0.972
15	3	status	0.807	alb	0.989
	4	replacement	0.804	need	0.980
	5	right	0.810	liquid	0.972
	1	operations	0.798	entry	0.975
	2	record	0.822	trad	0.980
16	3	test	0.821	former	0.990
	4	current	0.808	taking	0.978
	5	hand	0.825	exit	0.990
	1	transfer	0.801	classification	1.000
	2	favour	0.817	superior	0.990
17	3	fire	0.821	supports	0.998
	4	associate	0.809	course	0.993
	5	pipeline	0.820	control	0.997
	1	stable	0.776	classical	0.983
	2	part	0.806	drop	0.978
18	3	line	0.770	effects	0.992
	4	branch	0.782	plus	0.982
	5	place	0.789	may	0.996
	1	evolution	0.805	contact	0.971
	2	load	0.858	outcomes	0.971
19	3	release	0.825	point	0.985
	4	disadvantage	0.799	multiple	0.986
	5	human	0.827	best	0.992
	1	meet	0.830	change	0.988
	2	like	0.808	team	0.977
20	3	order	0.808	functions	0.977
	4	sources	0.820	ins	1.034
	5	network	0.807	rem	0.977
	1	bid	0.819	words	0.974
	2	base	0.823	gain	0.992
21	3	part	0.822	classic	1.017
	4	stores	0.818	main	0.979

continued on next page

Table 4 – *continued*

#		GPT-2 Small		Llama 3.2 1B	
		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	5	bracket	0.798	library	0.974
	1	art	0.793	mean	0.997
	2	stable	0.808	shot	0.985
22	3	comp	0.795	use	1.017
	4	square	0.799	lead	0.982
	5	trad	0.793	basic	0.983
	1	tongue	0.817	manifestation	0.989
	2	allies	0.838	ill	0.981
23	3	novel	0.812	urn	0.992
	4	bleach	0.817	sin	0.979
	5	item	0.825	allen	0.991
	1	chrome	0.795	profile	0.998
	2	vis	0.818	ratio	0.970
24	3	welcome	0.832	comes	1.000
	4	buck	0.828	mit	0.984
	5	preview	0.845	ping	0.991
	1	result	0.821	suck	0.982
	2	fold	0.862	verb	1.003
25	3	pinch	0.839	plague	0.983
	4	creeps	0.833	lambda	0.991
	5	vines	0.821	lint	0.985
	1	track	0.781	top	0.986
	2	last	0.783	better	0.979
26	3	crowd	0.779	may	0.979
	4	running	0.790	well	0.999
	5	example	0.784	bounty	0.975
	1	store	0.847	firm	0.981
	2	safe	0.834	general	0.992
27	3	normal	0.837	den	0.998
	4	right	0.840	present	0.996
	5	nos	0.820	reports	0.976
	1	right	0.795	evidence	0.995
	2	achievement	0.792	functioning	0.981
28	3	mission	0.798	formation	0.996
	4	audience	0.816	news	0.995
	5	ethic	0.802	case	0.995
	1	tit	0.819	ants	0.996
	2	mat	0.848	dryer	0.987
29	3	ling	0.822	artisans	0.989
	4	hugs	0.828	sparkling	0.993
	5	ether	0.815	revolver	0.994
	1	models	0.831	local	0.992

continued on next page

Table 4 – *continued*

		GPT-2 Small		Llama 3.2 1B	
#		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	2	beds	0.816	ins	1.005
	3	plug	0.841	experts	0.976
	4	hip	0.819	current	0.995
	5	short	0.807	inc	0.981
	1	phantom	0.828	even	1.013
	2	professional	0.823	hex	0.972
31	3	trainers	0.824	comb	0.989
	4	outside	0.842	anti	0.981
	5	complement	0.804	wee	0.971
	1	red	0.828	force	0.989
	2	number	0.809	last	1.010
32	3	process	0.838	put	1.016
	4	block	0.827	order	0.982
	5	chapter	0.811	string	0.983
	1	vets	0.812	gap	0.973
	2	med	0.817	pos	1.005
33	3	superior	0.833	current	0.974
	4	true	0.819	implements	0.978
	5	session	0.805	des	1.009
	1	launch	0.822	product	0.996
	2	third	0.819	light	1.009
34	3	means	0.782	transport	0.988
	4	mobile	0.831	time	1.027
	5	prep	0.780	space	0.985
	1	stern	0.823	simple	1.003
	2	seal	0.818	als	0.985
35	3	audible	0.798	much	0.989
	4	surf	0.805	para	0.987
	5	empty	0.828	ins	0.988
	1	selection	0.819	values	0.967
	2	symbol	0.846	https	0.977
36	3	pal	0.826	organ	0.966
	4	origin	0.864	features	0.969
	5	elements	0.835	community	0.974
	1	rival	0.811	blooms	1.004
	2	behind	0.852	capabilities	1.009
37	3	wounding	0.815	lem	1.015
	4	breakthrough	0.818	glitches	1.004
	5	ward	0.822	leans	1.019
	1	creation	0.812	instance	0.978
	2	office	0.812	enough	0.976
38	3	station	0.836	mini	0.982

continued on next page

Table 4 – *continued*

#		GPT-2 Small		Llama 3.2 1B	
		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	4	dot	0.816	independent	0.993
	5	return	0.824	group	0.985
	1	human	0.830	plus	1.020
	2	daily	0.843	groups	1.000
39	3	native	0.803	unit	0.984
	4	focus	0.810	quality	0.981
	5	body	0.808	lower	0.999
	1	mem	0.855	acronym	0.992
	2	transplant	0.830	obesity	1.013
40	3	behind	0.829	nonsense	0.995
	4	ops	0.834	leveraging	0.992
	5	brethren	0.837	subsets	1.002
	1	space	0.816	start	0.991
	2	setting	0.827	working	0.987
41	3	rep	0.783	experience	0.980
	4	left	0.789	method	1.008
	5	hand	0.763	leading	0.981
	1	talking	0.808	type	0.975
	2	rev	0.851	review	0.984
42	3	least	0.819	sup	1.014
	4	classes	0.800	min	0.990
	5	shift	0.804	advice	0.986
	1	identity	0.818	degree	0.985
	2	school	0.813	mon	0.978
43	3	process	0.821	experience	0.981
	4	transport	0.821	cross	1.002
	5	outside	0.827	filter	0.991
	1	cycle	0.850	support	0.986
	2	charge	0.805	proof	0.982
44	3	close	0.852	major	1.014
	4	age	0.825	ref	0.994
	5	aug	0.808	record	0.977
	1	past	0.810	former	0.982
	2	intent	0.806	version	0.987
45	3	condition	0.855	name	0.994
	4	control	0.854	sent	0.982
	5	moves	0.809	www	0.986
	1	starting	0.779	says	0.993
	2	sheet	0.801	disorder	0.974
46	3	reception	0.812	review	0.986
	4	audition	0.779	partition	0.992
	5	appearance	0.809	gas	0.974

continued on next page

Table 4 – *continued*

		GPT-2 Small		Llama 3.2 1B	
#		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	1	class	0.861	med	1.000
	2	records	0.780	part	1.058
47	3	main	0.853	using	1.052
	4	first	0.894	functions	0.987
	5	manifest	0.780	offer	0.994
	1	services	0.814	even	0.991
	2	average	0.823	parts	1.002
48	3	people	0.828	dis	0.985
	4	fore	0.808	law	0.979
	5	general	0.847	plain	0.989
	1	exit	0.807	alternative	0.994
	2	range	0.819	presentation	0.991
49	3	suite	0.800	stopping	0.974
	4	honor	0.805	names	0.974
	5	forward	0.808	shop	0.976
	1	respect	0.809	rover	0.982
	2	command	0.809	dangling	0.980
50	3	tweets	0.810	conclusion	0.979
	4	record	0.806	thus	0.984
	5	forward	0.826	due	0.994
	1	lit	0.842	vac	0.989
	2	practice	0.831	neglect	0.976
51	3	home	0.809	mandate	0.982
	4	minors	0.824	uses	0.977
	5	possible	0.826	resources	1.009
	1	alias	0.874	directories	0.982
	2	enrichment	0.832	compounds	0.988
52	3	mechanics	0.841	totals	0.997
	4	tent	0.818	interaction	0.990
	5	escape	0.881	void	0.982
	1	major	0.810	comes	0.984
	2	fever	0.790	disease	0.969
53	3	borders	0.795	father	0.970
	4	yawn	0.791	issues	0.969
	5	eyes	0.803	existence	0.975
	1	sweep	0.805	user	0.987
	2	understatement	0.792	water	0.978
54	3	crews	0.800	trust	0.974
	4	amateur	0.822	row	0.979
	5	disclaimer	0.793	item	0.981
	1	staff	0.813	extent	0.978
	2	management	0.812	appearance	0.977
55					

continued on next page

Table 4 – *continued*

#		GPT-2 Small		Llama 3.2 1B	
		Corruption	$d(w, C)$	Corruption	$d(w, C)$
	3	discipline	0.821	median	1.002
	4	use	0.822	following	1.016
	5	saves	0.814	alert	1.010
	1	wake	0.854	biases	0.981
	2	cause	0.815	intern	0.979
56	3	critic	0.823	analysis	0.978
	4	probes	0.805	idea	0.972
	5	episode	0.821	comments	0.975
	1	grad	0.806	ant	0.975
	2	cos	0.810	default	0.985
57	3	time	0.807	set	0.996
	4	counter	0.830	ten	0.995
	5	world	0.829	uses	0.984
	1	individual	0.798	approach	0.982
	2	home	0.832	super	1.009
58	3	head	0.811	third	0.999
	4	show	0.814	make	0.998
	5	like	0.806	review	0.997
	1	rep	0.812	fatigue	0.985
	2	bidding	0.809	net	0.983
59	3	exec	0.843	mit	1.003
	4	bond	0.842	dealing	0.977
	5	future	0.816	person	0.982
	1	face	0.813	augmentation	0.972
	2	due	0.810	mere	1.009
60	3	prom	0.808	consistency	0.984
	4	still	0.843	breaking	0.973
	5	convention	0.818	respect	0.974

Table 4. Corruption words and their cosine distance to the C-term, per model. Analogy numbers correspond to Table 2. Each analogy has five corruption words, sampled independently per model. The corruption words are selected to be distant from both the A-term and the C-term (Section 3.5.1); the column $d(w, C)$ reports the distance to the C-term.

Appendix D: Open-vocabulary Top-10 Predictions

Table 5 lists the top-10 next-token predictions for every analogy under the open-vocabulary zero-shot condition (Experiment 1), for GPT-2 Small and Llama 3.2 1B. Predictions are ordered by probability (rank 1 is highest). Analogy indices correspond to Table 2. Rows are grouped by relation type and distance category, separated by horizontal rules. Llama sometimes assigns highest probability to whitespace or punctuation tokens; empty-string tokens are shown as –.

#	Analogy	GPT-2 Small top-10	Llama 3.2 1B top-10
1	tarantula : spider :: bee : insect	bee, spider, fly, cat, snake, butterfly, bat, bird, swallow, fish	was, what, honey, insect, bee, ant, –, fly, butterfly, flower
2	calf : animal :: bull : mammal	dog, human, man, horse, sheep, bird, cow, cat, animal, cattle	–, what, ?, cow, ., ------, :, ., -----, (
3	botanist : biologist :: chemist : scientist	chemist, philosopher, physicist, biologist, scientist, bot, mathematician, ge, zo, engineer	chem, physicist, –, biologist, engineer, bio, ge, scientist, phys, what
4	swimming : sport :: golf : activity	swim, play, sport, dance, be, swimming, golf, compete, run, sports	–, what, ------, ., . . . , sport, a, exercise, the, . . .
5	murder : crime :: gun : weapon	murder, crime, gun, robbery, homicide, drug, sex, life, weapon, rape	–, crime, weapon, violence, murder, gun, killing, what, weapons, the
6	trout : fish :: lobster : crustacean	fish, eat, swim, crab, lobster, shrimp, fry, pick, be, fly	lobster, cooking, seafood, shell, crab, food, the, –, bait, –
7	squirrel : rodent :: robin : bird	cat, squirrel, rat, bird, dog, monkey, rabbit, wolf, snake, rodent	bird, –, what, ..., ?, ?, . . . , :, . . . , ------
8	jazz : music :: saxophone : instrument	music, dance, piano, guitar, the, jazz, song, writing, rock, a	music, what, sax, the, jazz, sound, –, a, wind, instrument
9	bowl : dish :: spoon : silverware	dish, spoon, eat, cook, pour, drink, serve, make, stir, scoop	fork, –, cup, knife, plate, spoon, ., hand, lad, what
10	january : month :: winter : season	month, fall, summer, year, winter, come, spring, season, date, months	season, –, month, day, time, year, week, what, :, .
11	diesel : fuel :: bed : furniture	bed, fuel, be, sleep, cook, feed, eat, fill, fire, make	–, bed, what, mattress, ..., :, . . . , ?, __, .
12	accountant : profession :: carrot : vegetable	carrot, apple, man, stick, fruit, tree, bread, cake, pie, vegetable	vegetable, food, –, what, fruit, ..., vegetables, . . . , ., root
13	west : direction :: cat : feline	direction, move, go, be, walk, tail, run, leave, cat, dog	–, what, animal, dog, ., cat, the, direction, which, .
14	rectangle : shape :: lunch : meal	eat, be, dinner, lunch, go, bake, sit, drink, take, serve	food, meal, –, what, ..., ., dinner, ..., eating, .
15	physics : science :: ring : jewelry	physics, science, mathematics, astronomy, geometry, math, chemistry, the, music, magic	–, music, a, what, the, –, circle, ..., ..., mathematics

continued on next page

Table 5 – *continued*

#	Analogy	GPT-2 Small top-10	Llama 3.2 1B top-10
16	caffeine : drug :: rose : flower	wine, water, alcohol, poison, tea, caffeine, coffee, sugar, medicine, weed	–, what, flower, fruit, . . . , ..., a, . . . , ..., .
17	wrench : tool :: sad : mood	be, work, use, make, do, die, find, tool, learn, live	–, what, ..., ., happy, . . . , _____, tool, ..., ...
18	oven : appliance :: sweden : country	the, a, country, furniture, house, home, appliance, kitchen, Swedish, food	–, country, what, ?, ..., the, ., ..., a, .
19	apartment : shelter :: tea : beverage	be, shelter, drink, provide, sleep, house, eat, cook, make, wash	drink, refresh, comfort, tea, the, a, nour, keep, coffee, brew
20	doll : toy :: turkey : poultry	toy, play, eat, be, chew, make, dog, go, fish, take	Christmas, Thanksgiving, bird, food, football, thank, holiday, meat, chicken, turkey
21	trauma : coma :: accident : death	death, life, coma, die, recovery, be, brain, fall, a, paralysis	–, death, injury, accident, ., fracture, a, :, ?, ...
22	virus : flu :: bacteria : infection	bacteria, viruses, a, yeast, mosquitoes, chicken, virus, the, fungus, human	virus, cancer, viruses, bacteria, –, fungus, pneumonia, stre, st, tuberculosis
23	traffic : delay :: crash : damage	be, increase, stop, continue, go, take, reduce, make, avoid, become	delay, stop, slow, kill, prevent, death, –, avoid, damage, speed
24	illness : treatment :: drug : relief	treatment, cure, therapy, prevention, drug, the, medicine, punishment, disease, medication	cure, medicine, disease, treatment, therapy, the, illness, medication, dose, remedy
25	training : fitness :: education : career	education, health, training, exercise, science, the, fitness, learning, sport, physical	learning, knowledge, health, life, business, nutrition, intelligence, the, –, training
26	fracture : cast :: incision : scar	cast, cut, remove, fracture, make, fix, draw, break, bend, strike	surgery, knife, fracture, the, a, inc, sut, cut, wound, operation
27	eclipse : darkness :: moon : tide	light, dawn, sunlight, sunrise, sun, bright, darkness, the, night, daylight	day, light, night, sun, –, the, eclipse, sky, darkness, stars
28	scratch : blood :: sprain : swell	blood, bleed, tear, bone, tears, sp, skin, spit, be, vomit	–, fracture, bone, break, bru, injury, wound, what, a, _____
29	failure : termination :: success : promotion	failure, success, fail, be, the, completion, termination, a, succeed, terminate	birth, creation, failure, the, completion, achievement, beginning, a, success, non
30	defeat : frown :: victory : smile	smile, laugh, frown, be, look, cry, stand, walk, sit, grin	smile, grin, laugh, a, raise, cheer, be, smirk, break, lift
31	motion : sickness :: drought : famine	drought, sickness, famine, death, the, disease, life, rain, hunger, climate	famine, the, food, hunger, death, water, rain, a, crop, health
32	allergy : sneeze :: entertainment : laughter	sne, eat, be, laugh, breathe, get, vomit, feel, have, smell	a, the, laughter, watching, watch, children, music, an, television, laugh
33	loan : debt :: football : concussion	football, money, life, the, sport, debt, play, a, sports, business	soccer, –, what, football, baseball, basketball, the, _____, cricket, a
34	sadness : crying :: genes : baldness	crying, being, feeling, laughing, thinking, the, screaming, eating, having, getting	genes, DNA, cells, dna, chromosomes, bacteria, the, blood, evolution, life
35	crime : arrest :: mutation : cancer	kill, be, cure, destroy, die, commit, make, grow, become, remove	breed, reproduce, create, cure, evolve, evolution, be, fertil, change, mutate
36	speeding : ticket :: run : exhausted	run, ticket, pass, be, go, drive, ride, win, walk, stop	–, run, stop, ..., drive, walk, arrest, ., ..., speed

continued on next page

Table 5 – *continued*

#	Analogy	GPT-2 Small top-10	Llama 3.2 1B top-10
37	order : delivery :: salt : thirst	be, receive, ship, deliver, order, the, take, salt, buy, get	the, salt, water, a, life, sugar, pepper, blood, food, sea
38	hormones : mood :: smoke : cough	sleep, smell, smoke, breath, feel, be, drink, taste, breathe, body	fire, –, the, a, ..., flame, food, . . . , air, flavor
39	invitation : visit :: tornado : destruction	be, visit, go, come, fly, enter, travel, arrive, leave, get	a, the, visit, tornado, be, thunder, an, hurricane, hurricanes, lightning
40	period : cramps :: wind : erosion	the, rain, be, a, fall, blow, wind, snow, get, break	menstr, a, pregnancy, the, menstrual, childbirth, diarrhea, men, women, period
41	oatmeal : cookie :: banana : muffin	chocolate, cake, cookie, banana, peanut, egg, bread, pizza, cookies, apple	–, cake, ..., ..., ?, what, bread, chocolate, cookie, banana
42	marble : stairs :: brick : patio	stairs, the, walls, bricks, doors, floors, a, windows, floor, stair	–, walls, windows, floors, ..., a, the, ., what, . . .
43	brass : trophy :: bronze : medal	silver, gold, bronze, trophy, platinum, diamond, be, crown, medal, win	medal, gold, –, sculpture, the, statue, medals, a, bronze, what
44	tomato : sauce :: cheese : soup	cheese, sauce, pepper, tomato, bread, butter, cream, mustard, toast, wine	–, pizza, pasta, bread, what, sauce, salad, cheese, soup, . . .
45	strawberry : cake :: cherry : pie	cake, cherry, strawberry, chocolate, apple, be, pie, b, ice, fruit	pie, cake, –, what, ice, fruit, jam, dessert, ..., :
46	fudge : candy :: chocolate : brownie	chocolate, candy, ice, peanut, cake, sugar, water, coffee, a, cookies	what, –, chocolate, ice, cake, dessert, ..., ..., candy, . . .
47	corduroy : pants :: denim : skirt	shoes, jeans, pants, denim, shirts, socks, boots, sneakers, leather, underwear	jeans, shirts, shorts, denim, jackets, what, trousers, shirt, t, skirts
48	feather : pillow :: foam : cushion	pillow, bed, pill, fabric, cushion, blanket, foam, carpet, t, cover	pillow, mattress, –, cushion, bed, feather, ..., what, ..., the
49	fur : coat :: vinyl : gloves	paint, be, wear, coat, cover, make, knit, dye, hide, stain	–, plastic, rubber, fabric, what, leather, paint, the, carpet, a
50	plastic : table :: wood : desk	table, be, paper, wood, plate, sheet, stone, roll, glass, floor	–, table, ?, ..., ., what, chair, the, :, . . .
51	steel : scissors :: apple : cider	apple, a, knife, apples, bread, the, pick, scissors, pie, butter	–, ..., apple, ?, . . . , ..., ., a, orange, banana
52	paper : money :: beef : broth	money, food, cheese, meat, the, butter, water, beef, beer, bread	butter, chicken, bread, cheese, the, –, pork, milk, a, wheat
53	log : fort :: ceramic : jar	be, stone, fort, make, clay, solid, iron, fire, die, build	–, what, wood, metal, steel, stone, glass, __, ?, concrete
54	satin : dress :: concrete : bridge	dress, wear, hide, make, be, use, put, look, take, paint	architecture, build, the, building, a, buildings, construction, –, paint, be
55	crystal : vase :: cotton : socks	flower, c, cotton, v, t, fabric, b, wool, silk, paint	–, ..., what, ., ?, flower, glass, ..., . . . , vase
56	leather : saddle :: gold : earring	saddle, leather, be, wear, lace, ride, buckle, make, hide, flee	horse, the, a, money, gold, horses, coin, jewelry, silver, saddle
57	turkey : burger :: rubber : boots	cheese, chicken, steak, burger, hamb, pizza, bread, beer, pork, meat	car, –, ..., . . . , tire, . . . , ..., –, what, rubber
58	flour : tortilla :: silk : tie	tort, cake, bread, flour, butter, chocolate, pastry, potato, sugar, cheese	–, ..., _____, ______, what, . . . , cotton, __, ____, silk
59	sugar : icing :: stone : house	icing, frost, the, stone, sugar, ice, cake, snow, baking, water	–, cake, fruit, a, ice, ..., what, rock, water, . . .

continued on next page

Table 5 – *continued*

#	Analogy	GPT-2 Small top-10	Llama 3.2 1B top-10
60	leaf : nest :: wool : scarf	nest, be, hide, lay, grow, weave, knit, flee, make, sleep	sheep, –, a, knit, wool, the, fleece, hair, clothing, be

Table 5. Top-10 next-token predictions per analogy under the open-vocabulary zero-shot condition, for GPT-2 Small and Llama 3.2 1B. Rank 1 is the highest-probability prediction. Analogy indices correspond to Table 2.

References

E. M. Bender and A. Koller. Climbing towards nlu: On meaning, form, and understanding in the age of data. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 5185–5198, 2020.

S. Bills, N. Cammarata, D. Mossing, H. Tillman, L. Gao, G. Goh, I. Sutskever, J. Leike, J. Wu, and W. Saunders. Language models can explain neurons in language models. `https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html`, 2023. OpenAI.

E. Biran, D. Gottesman, S. Yang, M. Geva, and A. Globerson. Hopping too late: Exploring the limitations of large language models on multi-hop queries. *arXiv preprint arXiv:2406.12775*, 2024.

L. D. Brown, T. T. Cai, and A. DasGupta. Interval estimation for a binomial proportion. *Statistical Science*, 16(2):101–133, 2001.

T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language models are few-shot learners. *CoRR*, abs/2005.14165, 2020. URL `https://arxiv.org/abs/2005.14165`.

A. Conmy, A. Mavor-Parker, A. Lynch, S. Heimersheim, and A. Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems*, 36:16318–16352, 2023.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

L. Eshuijs, S. Wang, and A. Fokkens. Short-circuiting shortcuts: Mechanistic investigation of shortcuts in text classification. *arXiv preprint arXiv:2505.06032*, 2025.

K. Ethayarajh, D. Duvenaud, and G. Hirst. Towards understanding linear word analogies. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3253–3262, Florence, Italy, 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1315.

D. Gentner. Structure-mapping: A theoretical framework for analogy. *Cognitive science*, 7(2):155–170, 1983.

D. Gentner and F. Maravilla. Analogical reasoning. In *International handbook of thinking and reasoning*, pages 186–203. Routledge, 2017.

Google. LLMs: What's a Large Language Model?, 2026. URL `https://developers.google.com/machine-learning/crash-course/llm/transformers`. Machine Learning Crash Course, Google for Developers.

A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

W. Gurnee, N. Nanda, M. Pauly, K. Harvey, D. Troitskii, and D. Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *arXiv preprint arXiv:2305.01610*, 2023.

S. Heimersheim and N. Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

M. Hersche, G. Camposampiero, R. Wattenhofer, A. Sebastian, and A. Rahimi. Towards learning to reason: Comparing llms with neuro-symbolic on arithmetic relations in abstract reasoning. *arXiv preprint arXiv:2412.05586*, 2024.

A. Holtzman, P. West, V. Shwartz, Y. Choi, and L. Zettlemoyer. Surface form competition: Why the highest probability answer isn't always right. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Y. Hou, J. Li, Y. Fei, A. Stolfo, W. Zhou, G. Zeng, A. Bosselut, and M. Sachan. Towards a mechanistic interpretation of multi-step reasoning capabilities of language models. *arXiv preprint arXiv:2310.14491*, 2023.

X. Hu, S. Storks, R. L. Lewis, and J. Chai. In-context analogical reasoning with pretrained language models. *arXiv preprint arXiv:2305.17626*, 2023.

C. Jiayang, L. Qiu, T. Chan, T. Fang, W. Wang, C. Chan, D. Ru, Q. Guo, H. Zhang, Y. Song, et al. Storyanalogy: Deriving story-level analogies from large language models to unlock analogical understanding. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 11518–11537, 2023.

T. Johnson, M. ter Veen, R. Choenni, H. van der Maas, E. Shutova, and C. E. Stevenson. Do large language models solve verbal analogies like children do? In *Proceedings of the 29th Conference on Computational Natural Language Learning*, pages 627–639, 2025.

L. L. Jones, M. J. Kmiecik, J. L. Irwin, and R. G. Morrison. Differential effects of semantic distance, distractor salience, and relations in verbal analogy. *Psychonomic bulletin & review*, 29(4):1480–1491, 2022.

J. Kim, A. Podlasek, K. Shidara, F. Liu, A. Alaa, and D. Bernardo. Limitations of large language models in clinical problem-solving arising from inflexible reasoning. *Scientific reports*, 15(1):39426, 2025.

J. H. Kirchner. Image from: Serendipitous connections: applying explanations from ai to the brain. `https://universalprior.substack.com/p/serendipitous-connections-applying`, May 2022. Accessed: 2026-03-30.

J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114 (13):3521–3526, 2017.

M. Lewis and M. Mitchell. Evaluating the robustness of analogical reasoning in large language models. *arXiv preprint arXiv:2411.14215*, 2024.

T. Linzen. Issues in evaluating semantic spaces using word analogies. In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 13–18, Berlin, Germany, 2016. Association for Computational Linguistics. doi: 10.18653/v1/W16-2503.

I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *arXiv*, 2017. doi: 10.48550/arxiv.1711.05101.

Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*, 2023.

K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372, 2022.

T. Mikolov, W.-t. Yih, and G. Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, 2013.

P. Mondorf and B. Plank. Beyond accuracy: Evaluating the reasoning behavior of large language models – a survey, 2024. URL `https://arxiv.org/abs/2404.01869`.

N. Nanda and J. Bloom. Transformerlens, 2022. URL `https://github.com/TransformerLensOrg/TransformerLens`.

C. Olsson, N. Elhage, N. Nanda, N. Joseph, N. DasSarma, T. Henighan, B. Mann, A. Askell, Y. Bai, A. Chen, et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022.

OpenAI. Introducing ChatGPT, Mar. 2024. URL `https://openai.com/index/chatgpt/`.

L. Pan, J. Liang, J. Ye, M. Yang, X. Lu, and F. Zhu. Opening the black box: A survey on the mechanisms of multi-step reasoning in large language models, 2026. URL `https://arxiv.org/abs/2601.14270`.

M. Petersen and L. van der Plas. Can language models learn analogical reasoning? investigating training objectives and comparisons to human performance. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 16414–16425, 2023.

J. C. Peterson, D. Chen, and T. L. Griffiths. Parallelograms revisited: Exploring the limitations of vector space models for simple analogies. *Cognition*, 205:104440, 2020. doi: 10.1016/j.cognition.2020.104440.

A. Prabhakar, T. L. Griffiths, and R. T. McCoy. Deciphering the factors influencing the efficacy of chain-of-thought: Probability, memorization, and noisy reasoning. *arXiv preprint arXiv:2407.01687*, 2024.

A. Radford, K. Narasimhan, T. Salimans, I. Sutskever, et al. Improving language understanding by generative pre-training. 2018.

A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.

A. Rogers, A. Drozd, and B. Li. The (too many) problems of analogical reasoning with word vectors. In *Proceedings of the 6th Joint Conference on Lexical and Computational Semantics (*SEM 2017)*, pages 135–148, Vancouver, Canada, 2017. Association for Computational Linguistics. doi: 10.18653/v1/S17-1017.

Z. Sourati, F. Ilievski, P. Sommerauer, and Y. Jiang. Arn: Analogical reasoning on narratives. *Transactions of the Association for Computational Linguistics*, 12:1063–1086, 2024.

F. Stahlberg. Neural machine translation: A review. *Journal of Artificial Intelligence Research*, 69:343–418, 2020.

R. J. Sternberg and G. Nigro. Developmental patterns in the solution of verbal analogies. *Child Development*, 51(1):27–38, 1980.

B. Stollnitz. The transformer architecture of gpt models, July 2023. URL `https://bea.stollnitz.com/blog/gpt-transformer/`. Accessed: 2026-03-18.

P. D. Turney, M. L. Littman, J. Bigham, and V. Shnayder. Combining independent modules to solve multiple-choice synonym and analogy problems. In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP-03)*, pages 482–489, Borovets, Bulgaria, 2003.

A. Ushio, J. Camacho-Collados, and S. Schockaert. Distilling relation embeddings from pretrained language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9044–9062, 2021.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv preprint arXiv:2211.00593*, 2022.

T. Webb, K. J. Holyoak, and H. Lu. Emergent analogical reasoning in large language models. *Nature Human Behaviour*, 7(9):1526–1541, 2023.

P. Wicke, L. Hirlimann, and J. M. Cunha. Using analogical reasoning to prompt llms for their intuitions of abstract spatial schemas. In *First Workshop on Analogical Abstraction in Cognition, Perception, and Language at IJCAI*, 2024.

T. Wijesiriwardene, R. Wickramarachchi, B. G. Gajera, S. M. Gowaikar, C. Gupta, A. Chadha, A. N. Reganti, A. Sheth, and A. Das. Analogical–a novel benchmark for long text analogy evaluation in large language models. *arXiv preprint arXiv:2305.05050*, 2023.

E. B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

Z. Xu, S. Jain, and M. Kankanhalli. Hallucination is inevitable: An innate limitation of large language models. *arXiv preprint arXiv:2401.11817*, 2024.

Y. Yang, D. Campbell, K. Huang, M. Wang, J. Cohen, and T. Webb. Emergent symbolic mechanisms support abstract reasoning in large language models, 2025. URL `https://arxiv.org/abs/2502.20332`.

S. Yuan, J. Chen, X. Ge, Y. Xiao, and D. Yang. Beneath surface similarity: Large language models make reasonable scientific analogies after structure abduction. *arXiv preprint arXiv:2305.12660*, 2023.

F. Zhang and N. Nanda. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023.

Z. Zheng, Y. Wang, Y. Huang, S. Song, M. Yang, B. Tang, F. Xiong, and Z. Li. Attention heads of large language models: A survey. *arXiv preprint arXiv:2409.03752*, 2024.